

Conceptual Modelling of Legal Compliance Requirements - The cases of eIDAS2 and NIS2

Paulo Alexandre Pereira Bolinhas

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Supervisor: José Luís Brinquete Borbinha

Examination Committee

Chairperson: Manuel Fernando Cabido Peres Lopes

Supervisor: José Luís Brinquete Borbinha

Member of the Committee: André Ferreira Ferrão Couto e Vasconcelos

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Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the *Universidade de Lisboa*.

Acknowledgments

Conceptual Modelling of Legal Compliance Requirements - The cases of eIDAS2 and NIS2 is the result of the work I did during the second year of my Master's degree. It represents months of effort, perseverance, and learning, and it is with great dedication and satisfaction that I now bring it to completion.

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"I can do all things through Christ who strengthens me."

- *Philippians 4:13*

Abstract

This work explores the hypothesis that conceptual models can simplify the understanding of complex legal compliance requirements. There is a recognized absence of complete and well-defined models for eIDAS2 (Electronic Identification, Authentication, and Trust Services) and NIS2 (Network and Information Systems), two critical pieces of European legislation addressing digital identity and cybersecurity, respectively. Through conceptual modelling, this research fills both gaps by developing two detailed architecture descriptions, aiming to improve eIDAS2 and NIS2 legal clarity and observance. ArchiMate is used as the modelling language since it provides a structured enterprise architecture framework that visually represents legal requisites. The final architecture descriptions include multiple documented views offering insights for entities seeking to understand and comply with these two legislative instruments, along with practical examples demonstrating their real-world application. Furthermore, work with Artificial Intelligence is explored for validation and documentation of these views within ArchiMate.

Keywords

Legal Compliance; Conceptual Modelling; ArchiMate; Architecture Description; eIDAS2 Regulation; NIS2 Directive

Resumo

Este trabalho explora a hipótese de que os modelos conceituais podem simplificar a compreensão de requisitos complexos de conformidade legal. Existe uma ausência reconhecida de modelos completos e corretamente definidos para a eIDAS2 (*Electronic Identification, Authentication and Trust Services*) e a NIS2 (*Network and Information Systems*), duas legislações europeias importantes que abordam a identidade digital e a cibersegurança, respetivamente. Através da modelação conceitual, esta investigação preenche ambas as lacunas através do desenvolvimento de duas descrições de arquitetura detalhadas, visando melhorar a clareza e conformidade legal da NIS2 e da eIDAS2. O ArchiMate é usado como linguagem de modelação pois fornece uma *framework* de arquitetura empresarial que consegue representar requisitos legais visualmente. As descrições de arquitetura finais incluem múltiplas vistas documentadas que oferecem *insights* para entidades que procuram entender e cumprir com ambas as legislações, bem como cenários práticos que demonstram a aplicação destas em casos reais. Além disso, trabalho com Inteligência Artificial é explorado para a validação e documentação das vistas em ArchiMate.

Palavras-Chave

Conformidade Legal; Modelação Conceitual; ArchiMate; Descrição de Arquitetura; Regulamento eIDAS2; Diretiva NIS2

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Acronyms

AD - Architecture Description

AI - Artificial Intelligence

AdES - Advanced Electronic Signature

ADL - Architecture Description Language

ADM - Architecture Development Method

AMA - *Agência para a Modernização Administrativa*

ANS - *Autoridade Nacional de Segurança*

ARF - Architecture Reference Framework

CDI - Corporate Digital Identity

CEN - European Committee for Standardization

CERT - Computer Emergency Response Team

CNCS - *Centro Nacional de Cibersegurança*

CMAI - Conceptual Modelling & Artificial Intelligence

CMAF - Cybersecurity Maturity Assessment Framework

CMD - *Chave Móvel Digital*

CPC - Citizens for Cybersecurity

CSA - Cybersecurity Security Act

CSIRT - Computer Security Incident Response Team

EA - Enterprise Architecture

EAA - Electronic Attestation of Attributes

eID - Electronic Identification

eIDAS2 - Electronic Identification, Authentication, and Trust Services Regulation (version 2)

EIF - European Interoperability Framework

EIRA - European Interoperability Reference Architecture

EN - European Norm

ENISA - European Union Agency for Cybersecurity

EU - European Union

EU-CyCLONe - European cyber crisis liaison organization network

ETSI - European Telecommunications Standards Institute

EUDI Wallet - EU Digital Identity Wallet

FIMSM - Federated Identity Management Service Model

GDPR - General Data Protection Regulation

GNS - *Gabinete Nacional de Segurança*

GPFI - Global Partnership for Financial Inclusion

IAM - Identity and Access Management

ID - Identification

IMSMF - Identity Management Service Model Framework

ISO - International Organization for Standardization

ISMS - Information Security Management System

IST - *Instituto Superior Técnico*

IT - Information Technology

KPI - Key Performance Indicator

LLM - Large Language Model

LoA - Level of Assurance

LOTL - List of Trusted Lists

NAB - National Accreditation Body

NATO - North Atlantic Treaty Organization

NCS - National Cybersecurity Strategy

NIST - National Institute of Standards and Technology

NOBID - Nordic-Baltic eID Project

MFA - Multi-Factor Authentication

MS - Member State

NIS2 - Network and Information Systems Directive

OECD - Organization for Economic Co-operation and Development

OOP - Once-Only Principle

PID - Person Identification Data

QES - Qualified Electronic Signature

QTSP - Qualified Trust Service Provider

RP - Relying Party

SBB - Solution Building Block

SCAP - *Sistema de Certificação de Atributos Profissionais*

SME - Small and Medium-sized Enterprise

SPC - Single Point of Contact

SSI - Self-Sovereign Identity

TOGAF - The Open Group Architecture Framework

TSP - Trust Service Provider

UIDAI - Unique Identification Authority of India

1 Introduction

Nowadays, it's getting more challenging for organizations and individuals to keep up with all the "rules". The legal framework is a significant constraint for all businesses, notably when dealing with an increasing number of regulations and directives [1]. Legislation and the process of their creation must "reflect the needs and reality of society", so stakeholders, as business-concerned actors, should have appropriate mechanisms to ensure their compliance [2]. However, a persistent gap remains between legal texts and the business environments required to implement them. Legislation is often complex, abstract, and difficult to interpret consistently, making compliance hard. What is needed is a structured enterprise framework that translates legal obligations into clear, business-relevant representations. Such a framework would allow stakeholders to understand requirements more effectively and ensure legal compliance [3].

This work focuses on two key European legal frameworks: the Electronic Identification, Authentication and Trust Services (eIDAS2) regulation and the Network and Information Systems (NIS2) directive. Both play a central role in digital trust and cybersecurity across the European Union (EU). By applying conceptual modelling (in this case with ArchiMate), this work aims to create formal architecture descriptions (ADs) that align legal requirements with Enterprise Architecture (EA) practice. The following sub-sections introduce the problem, objective, contributions, and structure of this project, establishing an introductory overview.

1.1 Problem and Motivation

eIDAS2 and NIS2 are key to the digital strategy within the EU. eIDAS2 introduces complex identity and trust service provisions but doesn't deliver a unified architecture for stakeholders to follow, while NIS2 expands cybersecurity obligations, but its implementation guidance remains fragmented. Without formalized models of eIDAS2 and NIS2, compliance with their requirements turns out being inconsistent. For eIDAS2, a complete structured model is absent in its ecosystem [4], leading to misunderstandings and compliance challenges. As the regulation pushes toward decentralized identity architectures, the need for well-defined models becomes increasingly urgent [5]. In parallel, NIS2 expands to include more stringent requirements, compelling organizations to manage complex cybersecurity needs. This calls for detailed architectural representations to ensure the comprehensive implementation of cybersecurity measures [6]. Moreover, research mentions a notable gap in structured architectural models for NIS2 implementation, particularly from an EA perspective [7].

Optimistically, previous initiatives already show the value of EA modelling (3.7), but no equivalent effort exists for eIDAS2 or NIS2. This creates an opportunity to pioneer structured models (in this case based on ArchiMate) that ultimately facilitate legal compliance. For instance, the eArchiving initiative aligns closely with this, as "at the heart of the eArchiving Reference Architecture (eRA) is a series of ArchiMate elements and views" [8], where "the real-world cases are represented as ArchiMate eRA viewpoints" [9]. Similarly, the Once-Only Principle (OOP) Project has adopted architectural modelling

based on ArchiMate and BPMN, allowing public administrations to design interoperable data reuse processes and ensure legal and technical compliance across Member States (MSs), illustrating how EA languages support cross-border digital government initiatives [10]. The CEF Building Blocks project has used The Open Group Architecture Framework (TOGAF) and ArchiMate to model reusable services like electronic identification (eID), eDelivery, and eSignature, helping organizations clearly document integration patterns, trust frameworks, and governance models that are critical for cross-EU digital service provision ¹. Furthermore, the European Interoperability Framework (EIF) has also been mapped to ArchiMate in various national implementations (e.g., in the Netherlands and Finland), showing how semantic, organizational, and technical interoperability layers can be formally aligned using EA languages [11].

On top of that, this work proposed to bridge the gap between eIDAS2 and NIS2 using the ArchiMate language, introducing a pioneering approach that translates both laws into structured ArchiMate ADs. ArchiMate was the EA language chosen since it can be applied to a wide range of architectural contexts. For instance, in the regulatory-based architectural context, research illustrates how ArchiMate facilitates modelling General Data Protection Regulation (GDPR) compliance by translating legal principles into architectural fragments, showing its adaptability for legal architectures [1].

Nevertheless, the motivation of this work is not limited to creating these two legal representations, it also aims to demonstrate how can they be applied in real-world cases and how Artificial Intelligence (AI) can support their refinement.

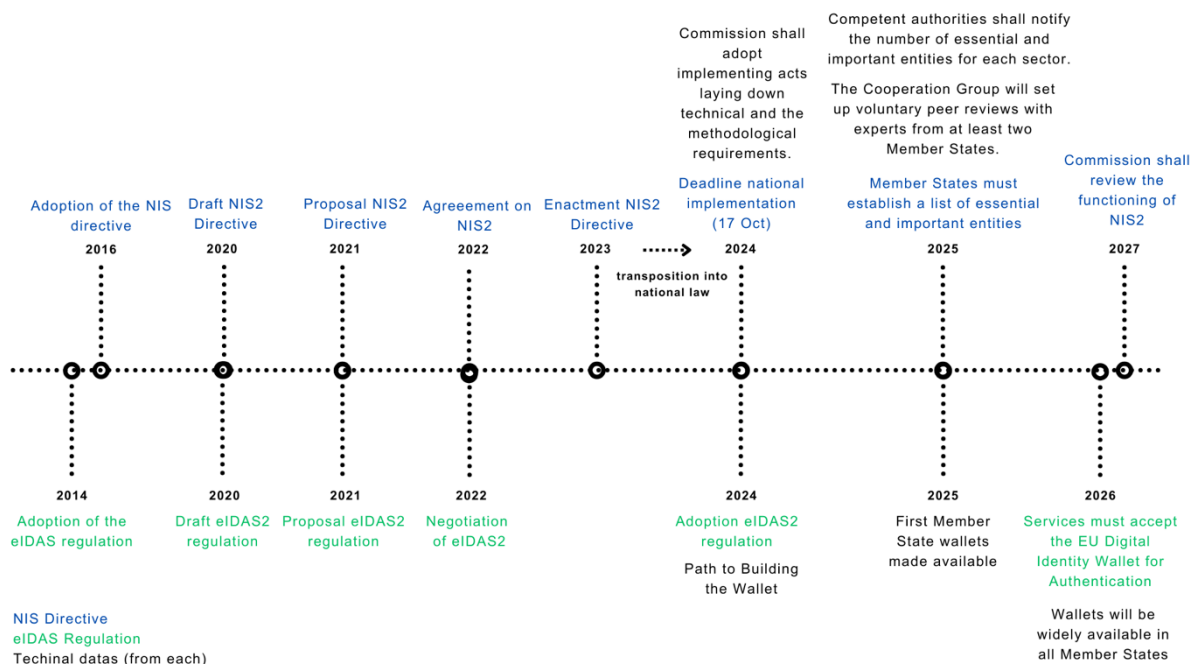


Figure 1: Timeline of events for NIS2 and eIDAS2

The timeline illustrated in Figure 1, shows significant milestones of these legal frameworks, presenting their evolution from inception to future implementation phases. This historical context highlights how

¹ CRYPTOGRAPHIC. EU's CEF Building Blocks: eID, eSignature & eDelivery.

eIDAS2 and NIS2 have contributed and will continue to contribute to enhancing digital development across the EU, underscoring their significance ^{2 3}.

1.2 Objective

The main objective of this work was to develop two ArchiMate ADs:

- one for eIDAS2, and
- one for NIS2.

The goal was to create a structured conceptual model that clearly represents eIDAS2 and NIS2, facilitating their understanding and compliance. Ultimately, the final ADs must capture legal requirements in structured conceptual models, enabling:

- legal coverage of NIS2 and eIDAS2, ensuring a clear understanding of the requirements,
- traceability, between legal clauses and architectural elements, and
- extraction of independent and documented views.

To support that, practical scenarios, implemented in viewpoints and views as extensions to the initial ADs, demonstrate how the models can be applied in real-world contexts. Exploratory validation and documentation with emerging AI were also tested to assess the potential of "AI-assisted" verification and refinement of large-scale legal models, thereby addressing a well-known limitation of conceptual modelling and EA approaches to complex models.

Since this is pioneering work that was done in a constrained interval of time, conventional validation methods could not have been applied. Direct validation with stakeholders would require an entirely new investigation process. Therefore, the focus is on demonstrating applicability through practical scenarios, and exploring AI-assisted techniques for assessment, rather than targeting a regular validation.

1.3 Contributions

The main contributions of this work are:

- **Methodical traceability approach:** It decomposes legislation into identifiable legal clauses, organizes them through a traceability matrix, associates them to ArchiMate views, and ensures consistency with a numbering system.
- **Two detailed ArchiMate ADs for eIDAS2 and NIS2:** Each systematically derived from the legislative text. The resulting ADs provide more than one hundred views and over a thousand uniquely identified elements, offering good coverage of both eIDAS2 and NIS2, following the same color coding and architecture principles.
- **Practical demonstration views:** These take the form of practical scenarios, designed in accordance with ISO/IEC/IEEE 42010 and associated with hypothetical stakeholders such as national authorities, agencies, and students. Each scenario was developed by first identifying the concerns of the stakeholders, grouping them into relevant viewpoints, and then modelling

² European Commission. About the EUDI wallet initiative.

³ NIS2 Directive. NIS2 release date. NIS2 Directive.

the legal environment through selected elements of the AD. The views were built using a “high-level to detailed” approach, gradually refining the models to address specific compliance questions. In this way, the project not only clarifies and structures each legislation but also illustrates their relevance in real-world contexts, showing how stakeholders can extend them into their own “legal scenarios”, adapting the legislation to their business environment and representing it clearly in a view.

- **Exploratory work on validation and documentation with AI:** While a comprehensive validation framework is not yet feasible for this type of modelling, this work tests preliminary approaches using AI. Two complementary approaches were tested: a unit-based review with ChatGPT, which allowed small, traceable segments of the AD to be validated iteratively, and an AI-enhanced JArchi integration, which provided automated scripts for analysis, critique, and documentation within the Archi tool, using LLaMA and Gemini. Together, these approaches explored how AI could support validation and documentation of large-scale ADs, not as substitutes for formal validation but as experimental methods that suggest possible future directions.

1.4 Structure

This project is divided into ten sections, organized to provide a clear and logical progression:

- **Section 1: Introduction** - Context, motivation, objectives, and structure of the work.
- **Section 2: Legal Foundations** - EU legislation, privacy, cybersecurity, digital identity, and standardization bodies.
- **Section 3: Conceptual Modelling** - EA, ISO/IEC/IEEE 42010, ArchiMate, and related work.
- **Sections 4-5:** In-depth study of NIS2 and eIDAS2.
- **Section 6: Method** - Traceability matrix, numbering system, development of ArchiMate views.
- **Section 7: Results** - Presentation of views from the NIS2 and eIDAS2 ADs.
- **Section 8: Demonstration** - Practical scenarios applying the ADs in hypothetical real-world cases.
- **Section 9: Validation** - Exploratory AI-based validation and documentation.
- **Section 10: Conclusion** - Outcomes, considerations, and future work.

2 Legal Foundations

European legislation provides the legal frameworks for digital identity and cybersecurity. Legislation such as NIS2 and eIDAS2 play a central role in shaping the European digital ecosystem by promoting trust, privacy, security, and resilience across all MSs, while ensuring that citizens, businesses, and public administrations can securely interact and exchange information in the digital world. Moreover, standards are essential in this context, as they provide common technical and procedural benchmarks that support legal compliance and the consistent implementation of legislative requirements across MSs.

Understanding this requires examining the legal foundations addressed in this project, including privacy in the digital environment, cybersecurity frameworks, the digital identity paradigm, and the most relevant standardization bodies.

2.1 European Legislation

The EU is based on the rule of law. This means that every action the EU takes is founded on treaties its members have approved democratically. EU laws help achieve the objectives of EU treaties and put EU policies into practice. There are two main types of EU legislation, primary (“both the treaties and general principles of EU law”) and secondary (“including regulations, directives, decisions, recommendations, and opinions”) [12]. Secondary EU legislation impacts nearly every aspect of governance by creating binding rules that ensure uniformity and coherence across MSs, addressing many areas, such as digital transformation and security.

Delving into secondary legislation, a regulation is a binding legislative act that must be applied in its entirety throughout the EU, while a directive is a legislative act that outlines an objective that EU countries must achieve [13]. Therefore, eIDAS2, as a regulation, has a binding legal force that applies uniformly across all MSs. Since its goal is to establish a common framework for eID, authentication, and trust services in a way that supports cross-border interoperability, it requires a consistent approach across the entire EU. In contrast, NIS2, as a directive, sets certain cybersecurity goals that MSs must achieve but allows each country to determine how to implement these measures through national laws. Both are required to achieve privacy in a digital world.

2.2 Privacy in the Digital World

The concern for privacy has existed since the dawn of civilization, although the formalized concept of privacy is a relatively recent development. Traditionally, “privacy interests were implicit in legal or social protection of personal property and space, intimate settings, or personal effects” [14]. In the digital world, privacy extends to “personal data”. Since the development of cookies in the 1990s, web users have been “concerned with online privacy, i.e., how to protect their personal information and data while enjoying the benefits of a connected digital life” [15]. The European understanding of personal data is the “information that permits to identify a person directly or indirectly, in particular by reference to an

identification number or to one or more factors specific to his physical, physiological, mental, economic, cultural or social identity” [16].

GDPR stands as one of the most impactful privacy and data protection laws in many years. While its origins lie in the EU, its scope extends globally, applying to any organization that collects or processes the personal data of EU citizens, regardless of the geographic location of the organization [17]. This regulation guarantees EU citizens and residents “rights over the ownership of their own data and requires their permission (opt-in) for its commercial or other usage with substantial fines for intended or unintended use, release, or even theft/hacking by third parties” [18].

Aside from safeguarding personal data, GDPR addresses modern challenges in the digital world, such as data portability, consent management, automated decision-making, cross-border data transfers, breach notifications, and the right to be forgotten [19]. Thus, solid cybersecurity is a must for securing GDPR compliance. This intersection creates a safe environment for protecting digital data. Without these security measures, personal information becomes vulnerable to attacks and misuse. Essentially, cybersecurity and data protection go hand in hand to ensure our digital data remains safe and secure [20].

2.3 Cybersecurity

Cybersecurity is a “security matter which spans from the security of the individual against cyber threats to all of society” [21]. To clarify the importance of cybersecurity for individuals, it is essential to underline how safeguarding personal data and digital interactions protects individual privacy and contributes to the broader resilience of society against cyber threats. Moreover, cybersecurity is equally critical for organizations. Nowadays, companies are prime targets for cyberattacks, and a breach can lead to financial losses, reputational damage, and legal reverberations [22]. Consequently, understanding the legal foundations of cybersecurity is necessary in every business [23].

EU enhances cybersecurity by providing legislation such as NIS2, improving the critical security of infrastructure and resilience across MSs ⁴. NIS2 emphasizes that cybersecurity strategies must be considered, recognizing that safeguarding digital data mitigates individual vulnerabilities and fortifies the collective defense against growing cyber vulnerabilities [24]. Adopting a risk management approach is crucial to further strengthen cybersecurity, as it helps balance the costs of security measures with the potential consequences of cyber threats, ensuring that investments are directed toward mitigating the most significant risks, both for individuals and businesses at large [25].

In addition to these measures, applying principles like “secure by design” is key at the business and compliance layers. This proactive approach focuses on integrating security throughout the lifecycle of products and services, from their initial design to deployment, ensuring that vulnerabilities are minimized from the outset. Accordingly, addressing security concerns early in the development process reduces the risk of cyber incidents and better protects user data [26].

⁴ NIS2 Directive. <https://nis2directive.eu/>

It is important to mention that, to establish these strategies, measures, and approaches, cybersecurity needs to always consider and follow three principal requirements, as in general computer networks [27]:

- Confidentiality (privacy): “The data should be available for access only by those who are authorized to access it”;
- Integrity: “The information of various organizations and sources should not be modified or altered by any unauthorized entity (i.e., an attacker) under any circumstances in a network”;
- Authentication: “Authentication is a mechanism by which the identity of a user is verified. Thus, it is expected that only the authorized account holder should be able to access his/her account after successful mutual authentication with the system”.

Adequate cybersecurity protection, which includes all data security solutions, can only be achieved “by adopting a cybersecurity framework that provides a structure and methodology for protecting critical digital assets”. Moreover, security experts advise adopting cybersecurity standards that comprise a set of best practices designed to safeguard organizations against cyber threats. However, “many organizations, companies, and governments lack experienced personnel in the cybersecurity domain, so they have difficulty adopting a standard approach or cybersecurity framework” [28].

A brief overview of the cybersecurity framework from Europe will be presented alongside the national frameworks from the United States and Portugal.

2.3.1 Europe

The European Union Agency for Cybersecurity (ENISA) is the agency of the EU dedicated to achieving a high common level of cybersecurity across Europe, being the primary agency specifically focused on enhancing cybersecurity at the EU level [29]. ENISA is the major entity supporting the implementation of NIS2, designed to ensure that MSs take proper cybersecurity measures and collaborate on a common level to face evolving cyber threats [30].

According to the NIS2 directive, cybersecurity maturity refers to the ability of an organization to effectively manage and respond to cyber risks, with maturity levels ranging from initial to optimized practices. Maturity models are used to evaluate the maturity levels. For instance, the Cybersecurity Maturity Assessment Framework (CMAF) helps organizations assess their cybersecurity posture, ensuring compliance with NIS2 while improving resilience against cyber threats. Higher maturity levels lead to better protection of critical infrastructure and continuity of essential services [31].

International standards like ISO 27001 align with CMAF. While NIS2 sets the legal requirements for cybersecurity within the EU, ISO 27001 provides a structured approach to achieving these requirements, offering a set of best practices and controls for managing information security [32]. Organizations assessing their cybersecurity maturity using tools like CMAF can integrate ISO 27001 standards to enhance their compliance with NIS2, ensuring comprehensive cybersecurity management.

Further reinforcing cybersecurity within the EU, the Cybersecurity Act (Regulation (EU) 2019/881) was introduced. This regulation “strengthens the ENISA and establishes a cybersecurity certification framework for products and services” [33].

Additionally, the EU Cybersecurity Act (CSA), a proposed regulation, aims to strengthen cybersecurity rules to ensure more secure hardware and software products [34]. This regulation is particularly focused on “safeguarding consumers and businesses buying or using products or software with a digital component,” [33] ensuring that these products adhere to stringent cybersecurity requirements.

2.3.2 United States

The National Institute of Standards and Technology (NIST) is a United States federal agency under the Department of Commerce. The mission of NIST is to “promote United States innovation and industrial competitiveness by advancing measurement science, standards, and technology in ways that enhance economic security and improve quality of life” [35]. They develop cybersecurity standards, guidelines, best practices, and other resources to meet the needs of the United States.

NIST is recognized as the leading authority on cybersecurity in the United States. Its Cybersecurity Framework (CSF) serves as the primary cybersecurity foundation of the nation. This voluntary framework provides organizations with a structured approach to managing and reducing cybersecurity risks. The recent NIST CSF 2.0, in Figure 2, is built on six core functions - **Govern, Identify, Protect, Detect, Respond, and Recover** - which guide organizations in establishing a robust cybersecurity program [36].

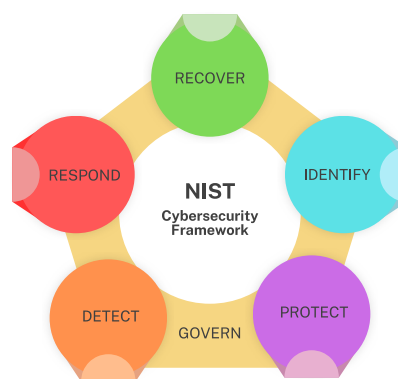


Figure 2: CSF 2.0 core functions ⁵

The CSF consists of three key components: the **Core**, **Profiles**, and **Tiers**. The Core, as mentioned already, consists of six functions that are further divided into categories and subcategories, providing a guide to risk management. Profiles allow organizations to align the Core outcomes with their current (Current Profile) or desired (Target Profile) cybersecurity postures, helping adapt risk management strategies to specific needs. Meanwhile, Tiers describe the maturity of the cybersecurity practices of an organization, enabling them to benchmark their cybersecurity efforts effectively [37]. That said, Tiers reflect the “practices of an organization for managing cybersecurity risk as Partial (Tier 1), Risk Informed (Tier 2), Repeatable (Tier 3), and Adaptive (Tier 4). They describe a progression from informal, ad hoc responses to approaches that are agile, risk-informed, and

⁵ NIST. The NIST Cybersecurity Framework (CSF) 2.0.

continuously improving. Selecting Tiers helps set the overall tone for how an organization will manage its cybersecurity risks” [36].

Beyond its development of the CSF, NIST is also responsible for the Risk Management Framework, the Privacy Framework, and the NICE Framework (Workforce Framework for Cybersecurity). These frameworks are used by federal agencies, private-sector organizations, and international entities, demonstrating the cyber influence of NIST [35].

2.3.3 Portugal

As a member of the EU, the cybersecurity framework of Portugal follows the framework set forth by the EU. The *Centro Nacional de Cibersegurança* (CNCS) and the *Gabinete Nacional de Segurança* (GNS) are the two key entities overseeing cybersecurity governance in Portugal.

CNCS represents both the NIS2 Competent Authority and NIS2 Single Point of Contact (SPC) for Portugal and focuses on improving national cybersecurity and supporting governmental strategies. It allows “organizations to reduce the risk associated with cyber threats, providing the basis for any entity to comply with the minimum-security requirements for networks and information systems” [38]. GNS takes a broader approach, securing the entire national security infrastructure while overseeing classified information and cryptography.

Regarding the hierarchy, in Figure 3, CNCS operates under GNS, being both under the *Autoridade Nacional de Segurança* (ANS). GNS primarily focuses on the protection of classified information. By defining security measures across public and private sectors, the GNS ensures alignment with international security legislation, including that of the North Atlantic Treaty Organization (NATO) and the EU. The CNCS is dedicated to enhancing national cybersecurity. It serves as the national cybersecurity authority, supporting compliance with cybersecurity legislative frameworks (e.g., NIS2) and engaging with entities like ENISA. CNCS also integrates a Computer Emergency Response Team (CERT), named CERT.PT, the Portuguese Computer Security Incident Response Team (CSIRT), a service that coordinates responses to incidents involving public administration entities [39].

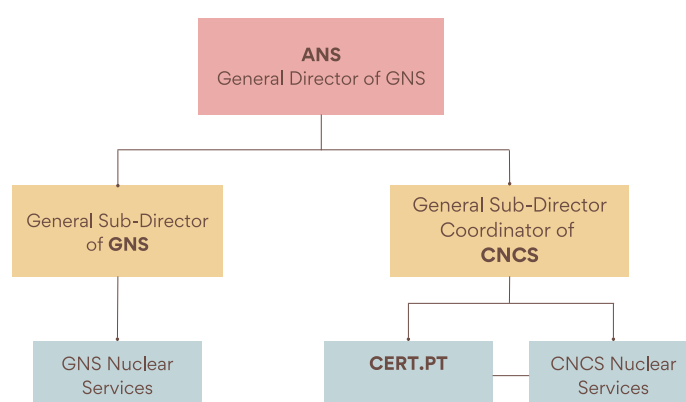


Figure 3: GNS organizational chart ⁶

⁶ Gabinete Nacional de Segurança. Organograma.

2.4 Digital Identity

Digital identity refers to the “unique characteristics representing an individual or entity online. It encompasses the distinct traits and attributes that define an individual or entity in the online environment” [40]. There are no equal identities. Every identity corresponds to a distinct set of characteristics [41].

Digital identity is built through a combination of personal identification data (e.g., a set of data issued under EU or national law that establishes the identity of an individual, legal entity, or a representative), online behavior, and digital footprints that individuals and organizations generate as they interact with digital platforms [42]. Regarding its purpose, as highlighted in discussions around eIDAS2, the primary regulation for digital identity, it plays a crucial role in enabling secure and interoperable access to online services across MSs [43].

As people increasingly rely on digital systems for everyday activities, scaling digital identity has become a critical concern for both individuals and organizations. Tackling this, the EU introduced functionalities to enhance digital identity systems. For instance, the EU Digital Identity Wallet (EUDI Wallet) “allows users to securely store, manage, and present their credentials in both online and offline cross-border use cases, extending identity verification to physical services” [44]. In addition to these functionalities, the EU emphasizes the importance of measures to protect digital identity systems from potential threats [45]. Despite advancements in identity technologies, cyberspace still introduces significant challenges, including opportunities for identity theft. “One inherent property of digital media is that it can be duplicated perfectly and easily” [41].

Furthermore, digital identity plays a crucial role in “fostering a secure, inclusive, and efficient digital transformation by enabling individuals and businesses to prove their identity and share verified information through digital means” [46]. It is completely established now, within different contexts, and with a lot of impact on the future. However, it has changed a lot in recent times.

A historical background on digital identity, some contexts from different countries, and a vision for its future will be conferred.

2.4.1 Historical Evolution

Identity issues have existed throughout human history [47]. The evolution of digital identity reflects a journey from rudimentary identification systems to sophisticated digital frameworks. Historically, identity verification relied on physical documents and personal interactions. Early approaches to digital identity often revolved around basic username-password combinations or personal identification numbers.

The digital revolution brought a whole new panorama. For instance, with social media, digital forms of identification (ID) gained popularity through social media logins, and with biometrics, the way we authenticate our personal identities has completely changed [48]. Modern innovations include blockchain and decentralized identity frameworks prioritizing user-centric security and privacy [49].

Nowadays, digital identity systems, frameworks that securely verify and manage the identities of individuals in digital environments, are being implemented globally. This rapid evolution brings fresh legal challenges.

2.4.2 Country Context

Digital identity systems enable access to public and private services while verifying security. The adoption and development of digital identity systems have changed significantly across countries.

For example, Estonia is renowned for its e-Residency program, a government-issued digital identity that allows citizens to “digitally sign documents and contracts, verify the authenticity of signed documents, encrypt and transmit documents securely, establish and administer a company in Estonia online, conduct e-banking and remote money transfers, access online payment service providers, and declare Estonian taxes online” [50]. In addition, citizens can start a company 100% online from anywhere. Estonia was the first country to offer e-Residency in 2014.

Similarly, the Indian Aadhaar system, managed by the Unique Identification Authority of India (UIDAI), is one of the largest digital identity systems globally. It is designed to provide every resident of India with a unique 12-digit identification number. Aadhaar is linked to the demographic and biometric data of an individual, including fingerprints and iris scans, ensuring robust identity verification [51].

Additionally, the Nordic countries are leaders in cross-border digital identity integration through the Nordic-Baltic eID project (NOBID), which is focused on facilitating the use of national electronic identification systems across the Nordic and Baltic regions [52]. NOBID addresses two issues in particular [53]:

- “Online users should be able to utilize an eID from their home country to identify also for digital services in other countries”;
- “Access to civil registration numbers for citizens of other countries when appropriate for digital cross-border service provision”.

However, challenges persist in regions with limited infrastructure or fragmented governance. Countries in Africa, for instance, are exploring mobile-first solutions to overcome traditional identity systems [54]. Initiatives such as M-Pesa-linked digital IDs in Kenya are driving innovation where traditional identity frameworks are lacking. M-Pesa is a “mobile service that allows money to be exchanged between mobile phone users” [55]. It is the preferred way to make payments across Africa. “Using the mobile number, a sender can transfer mobile money to a recipient using mobile messaging. Users who register with the service can, therefore, send or receive digital cash through their mobile number.” This service is particularly beneficial for people who own mobile phones but lack traditional bank accounts or have limited access to banking facilities.

Governments are collaborating globally to establish equitable digital identity systems. The G20 Digital Identity Framework, as outlined by the Global Partnership for Financial Inclusion (GPFI), emphasizes the critical role of digital identity systems in enhancing financial inclusion, providing digital equality, and promoting sustainable development [56].

2.4.3 For the Future

As digital identity systems evolve, their governance brings important responsibilities. Frameworks should be careful with critical issues, such as privacy protection, while mitigating potential risks. “Responsible innovation” in this domain requires balancing user control with security and compliance.

For example, frameworks like Self-Sovereign Identity (SSI) provide an alternate paradigm for identification where the individual creates and controls their digital credentials, enabling users to dictate how their information is shared and stored. The ethical principles behind these innovations are needed to build trust in digital identity systems [57]. G7 nations envision a future where interoperable digital identity systems improve access to services and promote trust, data privacy, and equity in the digital world. This outlook requires collaboration between nations to create frameworks that support innovation while safeguarding individual rights [58].

In addition to supporting broad digital transformation, digital identity plays a foundational role in the secure operation of government functions, particularly in tax administration. As noted by the Organization for Economic Co-operation and Development (OECD), secure digital identities allow taxpayers to access a range of digital services, such as filing returns and making payments, ensuring transparent interactions between individuals, businesses, and tax administrations [59]. This demonstrates the potential of digital identities not just as a tool for secure access but as a bridge for broader governmental innovation in public service delivery.

Looking to the future, the OECD emphasizes three pillars [60]:

- “Developing user-centered digital identity systems” that are accessible and portable across locations, technologies, and sectors;
- “Strengthening governance of digital identity” through clear policies, collaborative frameworks, and privacy safeguards;
- “Enabling cross-border interoperability” to support seamless access globally.

These principles aim for digital identity systems to respect democratic values, drive innovation, and maximize social and economic benefits for all.

2.5 Relevant Standardization Bodies

Standards represent documented knowledge that provides an opportunity to save time and material resources by using pre-developed common situations, technologies, and objects. They enhance the reliability of products and services, distributed in both printed and digital formats, as the technical solutions applied have been tested repeatedly in practice. In nearly every area of life, there is typically at least one standard that plays a crucial role in guiding practices [61].

Standardizing bodies such as International Organization for Standardization (ISO), European Telecommunications Standards Institute (ETSI), and European Committee for Standardization (CEN) play a huge role in shaping the frameworks that guide cybersecurity and digital identity [62]:

- ISO is an independent, non-governmental international organization that develops standards to ensure quality, safety, and efficiency across industries [62]. ISO is crucial for digital identity and cybersecurity as it provides frameworks that help organizations establish robust systems for protecting sensitive information. As an example, for cybersecurity, the ISO 27000 series is central, with ISO 27001 offering a comprehensive framework for an Information Security Management System (ISMS) to protect confidentiality, integrity, and availability of information [63]. For digital identity, standards like ISO/IEC 24760 define a framework for managing identity

information securely [64], and ISO/IEC 29115 provides assurance levels for identity authentication ⁷;

- ETSI is an European standards organization that specializes in developing globally applicable standards for telecommunications, broadcasting, and other electronic communications networks and services [65]. It is one of the European Standards Bodies (CEN, CENELEC, and ETSI) that define a standard as a document ⁸. For cybersecurity, ETSI has created standards like European Norm (EN) 303 645 for IoT device security and frameworks for securing AI and quantum-safe cryptography [67]. In digital identity, ETSI supports the EU Digital Identity Framework through digital wallets and secure authentication standards, incorporating technologies like zero-knowledge proofs to enhance privacy and user control ⁹. ETSI aligns with regulations such as eIDAS 2.0, enabling identity solutions across Europe [68];
- CEN develops European standards to harmonize technical specifications across Europe, ensuring interoperability and compliance with EU policies [62]. It is also part of the European Standards Bodies. Efforts from CEN include aligning with existing frameworks such as ISO/IEC standards and contributing to certification schemes under EU regulations. CEN is critical in defining standards for cybersecurity and digital identity in Europe. It collaborates with bodies like the European Network and Information Security Agency (ENISA) to create frameworks supporting secure and interoperable systems, aligning closely with NIS2. CEN has developed standards such as EN 17529, which focus on data protection and privacy by design and default ¹⁰. For digital identity, it plays a role in initiatives like the European Digital Wallet, providing guidelines for secure identity management and interoperability across borders.

Every field is affected by standards, even when designing and modelling architecture, especially if it involves the law.

⁷ ISO Standards. ISO/IEC 29115:2013.

⁸ CEN-CENELEC. European standards. <https://www.cencenelec.eu/european-standardization/european-standards/>

⁹ CEN-CENELEC. ETSI & CEN Workshop on EU Digital Identity Framework Standards.

¹⁰ CENELEC. EN 17529.

3 Conceptual Modelling

Conceptual modelling involves describing the semantics of systems at a high level of abstraction. Specifically, conceptual modelers describe [69]:

- “Structure models in terms of entities, relationships, and constraints”;
- “Behavior or functional models in terms of states, transitions among states, and actions performed in states and transitions”;
- “Interactions and user interfaces in terms of messages sent and received and information exchanged” ¹¹.

In their common usage, conceptual models are “high-level abstractions that enable clients and analysts to understand one another, enable analysts to communicate successfully with application programmers, and in some cases automatically generate (parts of) the software application”. They help us to “understand the structure and operations of organizations, the flow and the nature of data through business processes, and how the various software elements (objects, entities, functions, tasks, etc.) interact with elements inside and outside of the final system” [69].

Conceptual modelling ensures clarity in every system design, being essential for creating clear ADs (the objective of this project), which formally represent system architectures. Per the ISO/IEC/IEEE 42010 standard, which is “based upon a conceptual model of the terms and concepts pertaining to architecture description” [70], an AD documents the architecture of a system, addressing stakeholder concerns through views (representing the system from the perspective of a specific set of concerns) and viewpoints (rules for creating views).

This section introduces EA, ArchiMate, and ISO 42010 in the context of conceptual modelling, and examines related work in the current state of the art.

3.1 Enterprise Architecture and ArchiMate

Over recent decades, architectural quality has been recognized as a determinant of social, cultural, and economic development across Europe [71]. In the digital world, this focus translates into the strategic discipline of EA, which enables organizational systems to achieve overarching enterprise goals [72]. Moreover, EA “allows organizations to align their business goals and processes to Information Technology (IT). This alignment is delineated using modelling languages, one of which is ArchiMate” [73].

Guiding EA, there are frameworks like TOGAF that provide a structured methodology for developing and managing EAs, emphasizing the importance of aligning business and IT strategies. Central to TOGAF is the Architecture Development Method (ADM), a cyclical process that guides architects through the phases of architecture development and ensures alignment with business goals. ArchiMate is a powerful tool for effectively visualizing these architectures. It enables enterprise architects to create comprehensive representations of business processes, applications, and their interrelationships [74].

¹¹ Conceptual Modelling. Conceptual modelling. <https://conceptualmodelling.org/>

The ArchiMate specification emphasizes modelling with concepts, “an important part of information systems development and use that involves identifying and representing relevant aspects of reality” [75], offering a comprehensive way to represent abstract structures and their relationships, helping organizations to systematically capture the complexities of their EA systems. As an Architecture Description Language (ADL), ArchiMate is “any form of expression for use in architecture descriptions” and, through conceptual modelling, is used to describe the architecture of this project [76].

3.2 ISO/IEC/IEEE 42010

ISO/IEC/IEEE 42010 is an international standard developed to provide processes applicable to many types of architectural domains (such as companies, systems services, and software domains) and can be utilized by software architecture developers and users of these practices [77]. It enters conceptual modelling as the standard that governs how conceptual models are defined, justified, related to concerns, and integrated into an AD. It focuses on the development, evaluation, and maintenance of system architectures using detailed ADs. Well-known EA frameworks like TOGAF must employ it, as it provides a foundation for describing architectures that emphasize stakeholder concerns [78].

An architecture framework fundamentally “establishes a common practice for creating, interpreting, analyzing and using ADs within a particular domain of application or stakeholder community” [70]. Furthermore, “every architecture description needs a framework” [79]. According to ISO/IEC/IEEE 42010, an AD would consist of its: “Architecture framework (i.e., stakeholders, concerns, viewpoints, and correspondence rules)”, “Set of views and correspondences (satisfying the architecture framework)”, and “Rationale and required administrative information” [79]. Moreover, ISO/IEC/IEEE 42010 defines an AD as a structured work product that documents the architecture of a system (e.g., an entity of interest), ensuring clarity and alignment with the concerns of the stakeholders. Within an AD, a view represents the entity of interest from the perspective of a specific set of concerns, while a viewpoint defines the conventions and rules for constructing such views. Stakeholders are individuals or organizations with interests in the entity of interest, and their concerns are met through appropriate views and viewpoints, facilitating effective communication and architectural understanding of the entity of interest.

The second and latest edition of ISO/IEC/IEEE 42010 introduces stakeholder perspectives as “a means to group concerns and therefore to organize viewpoints framing those concerns” and architecture aspects, “characteristics of the entity of interest that are reflected in architecture views. Among other uses, aspects can be used to refine the traceability between concerns and views” [70].

Considering a small online business (an **entity of interest**) in a country from the EU that sells products and must comply with cybersecurity legislation, including NIS2, to exemplify the ISO 42010 definition of a conceptual model of an AD (Figure 4). To document its system **architecture**, the business develops an **AD** that defines how its website, payment system, and customer data storage are structured. The **stakeholders** include customers who want to ensure their data is secure and the business owner who needs to comply with NIS2 requirements. To effectively **address these concerns**, the business identifies **architectural aspects**, such as data security and system compliance, which are

essential to meeting the **perspective of the stakeholder**. To represent this, the business creates two views: a customer view showing how their data is protected during online transactions and a compliance view for the owner detailing how the business meets cybersecurity standards. Each view follows specific rules or **viewpoints** to ensure alignment with the concerns of the stakeholders and compliance with the NIS2 directive.

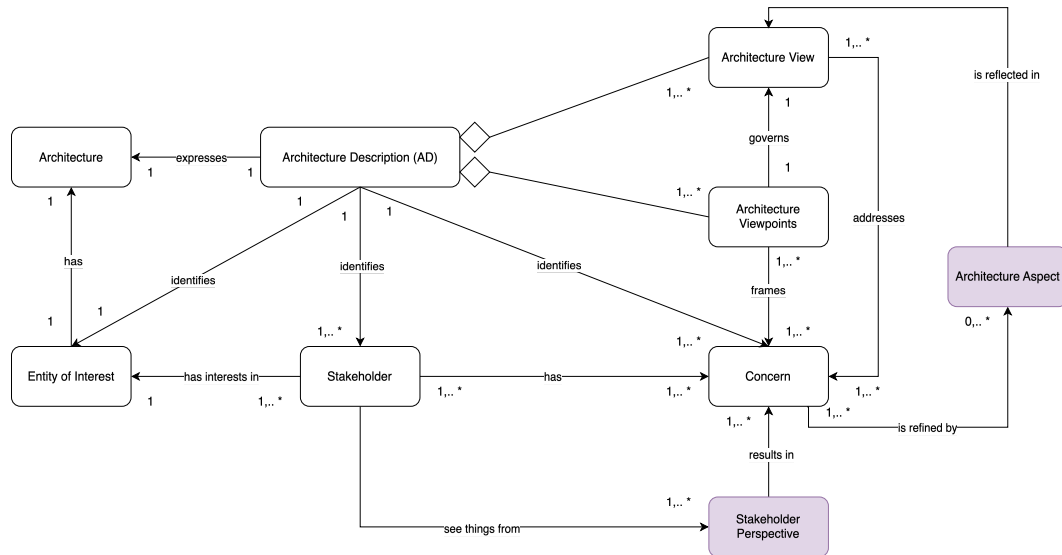


Figure 4: Example of legal representation of AD according to ISO 42010

Furthermore, views and viewpoints are extremely important when following this standard. In detail, a viewpoint provides a structured way to express specific architectural concerns, allowing architects to meet with stakeholder interests effectively. Each viewpoint is defined by the concerns it frames, the relevant stakeholders, and the model kinds it employs, which establish conventions for modelling. In the AD, each viewpoint corresponds to one architecture view (its governing viewpoint), which adheres to its conventions. This view includes identifying information, architecture models addressing the concerns, and records of any known issues related to the viewpoint.

All these definitions highlight the importance of following this standard when building ADs, having a huge impact when modelling the solution of this project within the ArchiMate language. The adaptability of ISO/IEC/IEEE 42020 allows it to be applied in technical domains, while also extending to regulatory frameworks. In the context of EU legislation, ISO/IEC/IEEE 42010 can provide a structured way to model and analyze frameworks like eIDAS2 and NIS2 [80], as illustrated in Figure 5.

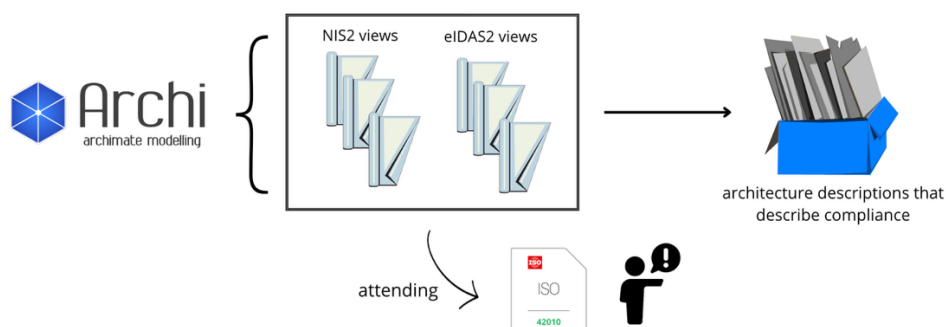


Figure 5: ISO 42010 describing legal ADs representation example

3.3 ArchiMate Framework: Layers and Aspects

The ArchiMate Core Framework consists of nine cells used to classify elements of its core language. It is structured around three layers and three aspects. In ArchiMate, layers and aspects provide a structured way to model and analyze EAs. Layers represent different levels of abstraction within an enterprise: the Business Layer focuses on business operations and services, the Application Layer defines software applications and their services, and the Technology Layer deals with IT infrastructure and physical systems. The underlying structure of models in the different layers is consistent.

Aspects classify elements based on layer-independent characteristics. The Active Structure Aspect represents subjects that perform actions (e.g., business actors and roles), the Behavior Aspect captures events, interactions, processes, and services executed by those subjects, representing the dynamic aspects of the enterprise, and the Passive Structure Aspect defines objects or data acted upon (e.g., business objects and products), that can be accessed by behavior elements.

The Full Framework, in Figure 6, extends this by adding more layers and an additional aspect, enabling richer modelling. It introduces the Strategy Layer to capture high-level business goals and strategic decisions, as well as physical elements in the Technology Layer for modelling facilities, equipment, and materials. The Motivation Aspect addresses stakeholder intentions, goals, and rationale, while the Implementation and Migration Layer supports modelling the transition and transformation of systems.

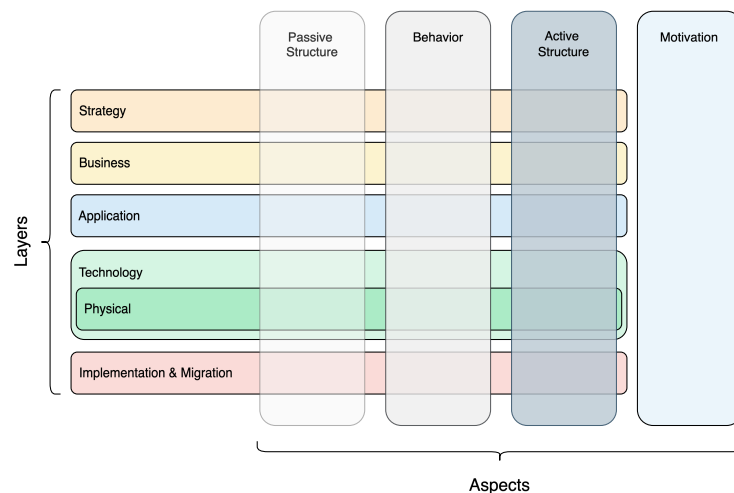


Figure 6: ArchiMate full framework

The layered approach of ArchiMate provides stakeholders with tools to map motivations, behaviors, and structures, linking traditional architectural excellence to modern enterprise transformation [72].

3.4 Elements, Relations, and Colors in ArchiMate

ArchiMate allows for conceptual modelling. In ArchiMate, “a model is a collection of concepts - a concept is either an element or a relationship” [76].

Elements are classified into four categories:

- Behavior Elements, which describe actions or processes performed;
- Structure Elements, which define the entities that perform or are acted upon;
- Motivation Elements, which capture goals, drivers, and rationales;
- Composite Elements, which combine multiple aspects into a single representation.

These elements are the fundamental building blocks used to represent the components of an EA, and they are key to creating coherent models that align layers. Aside from elements, ArchiMate “defines a core set of generic relationships, each of which can connect a predefined set of source and target concepts (in most cases elements, but in a few cases also other relationships). Many of these relationships are “overloaded”; i.e., their exact meaning differs depending on the source and destination concepts that they connect.” In terms of major relationships, there are:

- Structural relationships (e.g., association, composition, aggregation) which define the static structure and dependencies between elements;
- Dependency relationships (e.g., access, influence) which indicate how elements depend on or are affected by one another;
- Dynamic relationships (e.g., triggering, flow) which capture behavioral interactions over time;
- Other relationships (e.g., specialization) that define hierarchies and refinements.

Each relationship has exactly one “from” and one “to” concept (element, relationship, or relationship connector) as endpoints, subject to certain restrictions. Relations connect elements within and across layers and aspects. The relationships between these layers are “primarily built on serving relationships, where elements in one layer provide services to those in another (e.g., technology services enabling application services)” [76].

Together, elements and relations form the foundation of ArchiMate models, enabling architects to describe complex systems in a structured way. Relations, such as “association,” “realization,” or “serving,” connect elements to show how they collaborate, depend on, or influence each other. This ensures that the model comprehensively captures both the components and their interconnections.

ArchiMate also offers colors. Colors are used to “distinguish concepts belonging to the different aspects of the ArchiMate Framework” [76]. For instance, in the metamodel, green represents the Passive Structure Aspect, yellow represents the Behavior Aspect, and blue represents the Active Structure Aspect. Similarly, example models often use yellow for the Business Layer, blue for the Application Layer, and green for the Technology Layer to visually differentiate between layers. While these conventions enhance readability, the colors in ArchiMate lack formal semantics, leaving their use flexible for modelers to highlight specific elements or emphasize certain aspects as needed.

3.5 Views and Viewpoints in ArchiMate

In ArchiMate, a viewpoint can be understood as a set of guidelines or instructions that define how to present a system. It serves as the framework for determining which aspects of the architecture are relevant to specific stakeholders and what concerns need to be addressed. A view, on the other hand, is the actual representation created using that framework. It offers a visual description of the architecture from the perspective of a defined set of concerns, addressing the needs of different stakeholders. The

ArchiMate language facilitates this process by allowing architects and stakeholders to develop their own views based on the provided viewpoints. That's why defining views and viewpoints is one of the first steps when modelling with ArchiMate.

As shown in Figure 7, ArchiMate classifies viewpoints based on two dimensions: purpose (designing, deciding, informing) and content/ abstraction level (detail, coherence, overview). This helps identify suitable perspectives for specific architectural tasks. "The goal of this classification is to assist architects and others find suitable viewpoints given their task at hand; i.e., the purpose that a view must serve and the content it should display" [76].

Classification	Viewpoint	Description	Typical Stakeholders	Examples
Purpose	Designing	Supports architects and designers in the design process from initial sketch to detailed design. Typically consists of diagrams like UML.	Architects, Designers	UML diagrams
	Deciding	Assists managers in decision-making by offering insights into cross-domain architecture relationships using models, tables, maps, lists, and reports.	Managers	Cross-reference tables, landscape maps, lists, reports
	Informing	Helps inform any stakeholder about enterprise architecture to achieve understanding, commitment, and persuasion. Examples include illustrations and animations.	Any stakeholder	Illustrations, animations, cartoons, flyers
Abstraction Level	Details	Focuses on one layer and one aspect. Stakeholders include software engineers and process owners. Examples: BPMN process diagrams, UML class diagrams.	Software engineers, Process owners	BPMN process diagrams, UML class diagrams
	Coherence	Spans multiple layers or aspects, showing architecture relationships. Stakeholders are operational managers responsible for IT services or business processes.	Operational managers	Views showing process-uses-system, application-uses-object
	Overview	Addresses multiple layers and aspects, targeted at enterprise architects and decision-makers like CEOs and CIOs.	Enterprise architects, CEOs, CIOs	High-level architecture overviews



Figure 7: ArchiMate viewpoint classification dimensions

To exemplify (Figure 8), consider a citizen using an EUDI Wallet to access some digital public service. From the perspective of this user, key concerns include ease of use, security, and accessibility. To address these concerns, an architect can apply a "User experience" viewpoint, which defines the relevant elements and relationships to focus on, such as the roles of the user and the services they interact with. This viewpoint would specify which layers (business, application, technology) and which elements (e.g., business actor, service) should be included in the view to effectively represent the

interaction of the user with the EUDI Wallet. The resulting view then visually presents this information, using an EA language like ArchiMate, tailored to the concerns of the stakeholders.

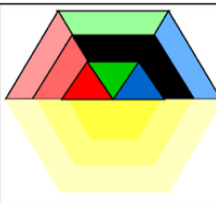
Stakeholders	1. Citizen (User) 2. EUDI Wallet Provider		
Concerns	1. ease of use 2. security 3. accessibility		
Purpose	Designing, Informing		
Abstraction Level	Coherence		
Layer	Business, Application		
Aspects	Active structure, Behavior, Motivation		

Figure 8: Example of a possible user experience viewpoint

3.6 Conceptual Modelling with Artificial Intelligence

The intersection of conceptual modelling and AI has given rise to a new interdisciplinary research domain known as Conceptual Modelling & Artificial Intelligence (CMAI). While conceptual modelling focuses on creating structured and human-understandable abstractions for communication and analysis, AI offers powerful capabilities for processing large, unstructured datasets, often through non-transparent "black-box" models. A recent study highlights how conceptual modelling can address the limitations of AI in explainability and control, while AI can enhance conceptual modelling through automation and pattern discovery. The study identifies mutual benefits from this integration, classifying contributions into categories such as conceptual modelling-driven, AI-driven, and co-designed approaches. This synergy is expected to drive innovation in domains like machine learning, natural language processing, and multi-agent systems, where modelling languages such as UML and DSMLs are combined with AI techniques to support system representation, analysis, and even code generation [81].

Furthermore, research reports initial experiments showing how ChatGPT (based on GPT-4) can generate and interpret conceptual models, including ER diagrams, UML class diagrams, business process models, and Heraklit models, based solely on structured and natural language prompts. The results suggest that Large Language Models (LLMs) can effectively assist in model synthesis tasks by using pretrained knowledge representations and autoregressive generation techniques. While limitations in precision and domain control remain, it presents the feasibility of integrating LLMs into early-phase modelling workflows to enhance productivity, reduce manual modelling effort, and support model-driven engineering in complex domains [82].

Illustrating the value of ArchiMate in structuring complex systems integrating AI and blockchain, an approach for Dry Port Digital Twins. While the architecture is preliminary, it envisions embedding AI models to optimize port operations, demonstrating how the layered framework of ArchiMate supports the incorporation of intelligent automation and data-driven decision-making. The proposed TO-BE

architecture models seven layers, from stakeholders and business services to physical infrastructure and IoT data sources, integrating blockchain to ensure data trustworthiness and AI-driven analytics to optimize container flows and detect damages. This approach demonstrates the possibility of using both ArchiMate and AI together but does not illustrate their direct integration [83]. For instance, a study shows how AI components, such as machine learning models, chatbots, or decision-support systems, can be accurately represented within the layered framework of ArchiMate. In the context of AI system development, ArchiMate is used to visualize and analyze the impact of AI components across the EA, including how AI algorithms interact with services, data, and infrastructure [84].

On the other hand, AI accelerates ArchiMate modelling by automatically extracting architectural elements from text using natural language processing, enabling faster creation and updating of models. It also helps validate consistency and suggests improvements, reducing manual effort and improving accuracy in EA ¹². To address this, more research explores the feasibility of generating ArchiMate diagrams with ChatGPT. Initially, it experimented by prompting ChatGPT to create an ArchiMate business view diagram for a restaurant business. The AI-generated XML output was close to the desired format but required adjustments to align with the ArchiMate 3.0 schema. After refining the prompts to ensure correct XML structure and schema references, the generated file was successfully imported into Archi, a popular ArchiMate modelling tool. The resulting diagram displayed elements neatly organized in folders, with relationships clearly depicted. This experiment demonstrated that ChatGPT can effectively assist in creating basic ArchiMate diagrams, streamlining the process for enterprise architects. The research suggests that integrating such AI capabilities directly into modelling tools could further enhance workflow efficiency ¹³.

In conclusion, the integration of AI with conceptual modelling, especially through tools like ArchiMate, is an emerging area that holds significant potential for future research. It could form the basis of an entire dedicated study, exploring how AI can enhance modelling, analysis, and automation within EA practices.

3.7 Related work

Models that enrich the understanding of legal compliance are lacking. This area is intricate due to its reliance on constantly evolving legislation and the need to balance legal requirements with business goals. The complexity and importance of eIDAS2 and NIS2 call for frameworks that simplify their interpretation.

Despite the absence of complete models representing these European laws, work has been done in related fields to address legal compliance through conceptual modelling. For example, frameworks such as TOGAF have been applied to align EA with legal compliance. A case study demonstrated how applying TOGAF can facilitate the implementation of the GDPR in small and medium-sized enterprises (SMEs) [86]. Additionally, research introduces an EA viewpoint to assist organizations in implementing

¹² Method Analytics. Accelerate Enterprise Architecture Modelling with AI.

¹³ Annala, R. (2023). Creating ArchiMate diagrams using ChatGPT. Medium. <https://medium.com/@roxblend/creating-archimate-diagrams-using-chatgpt-aed94bb67704>

the CSF by mapping its seven-step process onto the ADM. This combined process enables organizations to adopt the CSF within a single architecture development cycle [87].

Below, the state of the art will be further conferred by presenting existing research and frameworks relevant to this project.

3.7.1 EIRA

The European Interoperability Reference Architecture (EIRA) is a reference architecture for implementing interoperable European digital services modelled using ArchiMate. “It represents a metamodel defining the Architectural Building Blocks (ABBs) to build interoperable eGovernment systems” [87]. An ABB is an abstract component that encapsulates architectural requirements and directs the development of Solution Building Blocks (SBBs), which are concrete elements that define implementation and meet the necessary requirements.

Within EIRA, the Legal Interoperability Specification ABB plays a crucial role in ensuring that digital solutions comply with legal requirements. The e-procurement SAT is an example of how this specific ABB can be instantiated as a SBB, providing a framework for developing interoperable e-Procurement solutions that align with the Directive 2014/24/EU on public procurement. For instance, the “e-Procurement Submitting” phase focuses on the electronic submission of tenders, ensuring that the process is secure and adheres to the standards set by the directive. By using the e-Procurement SAT, public administrations can develop solutions that are not only legally compliant but also interoperable across different regions, demonstrating how EIRA facilitates legal compliance [88].

Important to note that EIRA is closely aligned with the EIF, which provides a comprehensive set of principles to enhance interoperability across European digital services. EIRA needs to ensure that its ABBs and SBBs adhere to the interoperability principles outlined in the EIF. While the EIF sets the strategic direction and high-level principles for interoperability, the EIRA translates these into practical architectural structures [87].

3.7.2 Modelling Identity with ArchiMate

ArchiMate itself provides a structured approach to modelling identity, supported by studies exploring how digital identity can be effectively represented within the language. It involves representing individuals (Actors) and their associated digital identities (Accounts) within an EA framework ¹⁴.

In this context, research investigates example frameworks for creating ArchiMate architecture models tailored to digital identity systems. For example, it explores how to integrate Identity and Access Management (IAM) into NIS2 by “mapping NIS2 directive stages” ¹⁵ with ArchiMate, providing a structured approach to visualizing compliance.

ArchiMate identity management-related research also includes:

¹⁴ ArchiMate Tool. (2018). Long read: Modelling identity in enterprise architecture with ArchiMate.

¹⁵ Archok Baokc, K. (2022). Integrating identity and access management with the NIS2 directive. LinkedIn.

- A proposed identity management service model framework (IMSMF) that consists of “a meta-model and several models for various protocols and implementations” [89]. By employing ArchiMate, the authors provide a structured approach to represent IAM services, enabling a clear visualization of complex IAM structures and promoting better understanding among stakeholders involved in identity management processes;
- A proposed universal federated identity management service model (FIMSM), applied to “describe various federated identity management scenarios in a generic service-oriented way” [90], using ArchiMate to represent key elements of their proposed service model. It emphasizes the need for a unified conceptualization of FIM services to deal with complexities in cross-organizational identity management.

3.7.3 More research

Representations of eIDAS2 and NIS2 exist, but these tend to be disorganized, difficult to follow, or decentralized, often focusing only on specific parts of the legislation. For instance, the European Digital Identity Architecture Reference Framework (ARF) summarizes the understanding, from eIDAS Expert Group, of the EUDI Wallet concept within a dedicated GitHub repository, which provides detailed architectural models and processes of the EUDI wallet [91]. Another example focuses on a specific area of implementation, such as eID for universities, which applies the eIDAS framework in a university context by building two cross-border eIDAS-enabled academic services based on the eIDAS infrastructure: “the eRegistration in the Erasmus student exchange program and the Login facility with national eIDs on the university portal” [92].

For NIS2, modelling approaches, such as an ontological approach to compliance verification, have been explored to achieve two fundamental objectives: “domain modelling and resource interrogation” [93]. This approach aims to be “helpful for any organization that is through the hard task of compliance verification” [94]. However, these approaches do not capture the broader legal landscape or the diverse stakeholder perspectives. Additionally, there is an analysis on how NIS2 can be translated into user-centered requirements to guide real-world security evaluations, presenting a structured, persona-based approach to interpret NIS2, facilitating compliance through clear stakeholder-focused requirements rather than abstract legal clauses. This relates to this work, as it supports the use of structured, stakeholder-driven modelling in ArchiMate to translate NIS2 legal requirements into clear architecture views.

As with eIDAS2, the full scope of NIS2 is often not modelled, and ADs remain limited. Even outside the EU, in the United States, NIST also offers some digital identity models that reflect technologies and architectures already available in the market, such as an “Identity Proofing and Enrollment” model [95]. Reference models often address specific areas like digital identity, digital wallet, or identity management, which align with both legal frameworks but do not fully cover them.

4 Network and Information Security

Directive 2016/1148 on the security of network and information systems (known as NIS) was the “first horizontal legislation undertaken at EU level for the protection of network and information systems across the Union” [96]. The NIS directive “was an important historical part in the evolution of the overall cybersecurity framework of the EU” [97]. On 16 January 2023, the Directive (EU) 2022/2555 (known as NIS2) entered into force, replacing NIS. ENISA is the entity that supports the implementation of the NIS2 directive as part of its mandate and work program [5].

As stated in the renewed version, “the review of the first version of NIS revealed inherent shortcomings that prevent it from addressing effectively emerging cybersecurity challenges” [98]. Therefore, NIS2 came with an updated list of sectors and activities subject to cybersecurity obligations.

The NIS2 directive is “the most comprehensive European cybersecurity directive yet. With stricter requirements for risk management and incident reporting, wider coverage of sectors, and more hard-hitting penalties for non-compliance, hundreds of thousands of EU organizations will need to reassess their cybersecurity posture” ¹⁶.

4.1 National Cybersecurity Strategy

Under NIS2, each MS is required to develop and maintain a National Cybersecurity Strategy (NCS) that outlines strategic objectives and concrete policy measures to achieve a high level of cybersecurity across its territory.

The NCS must define national priorities, particularly in essential and important sectors, set out governance structures, assign responsibilities across stakeholders (including competent authorities, CSIRTs, and SPCs), and establish mechanisms for risk assessment, incident preparedness, and public-private cooperation. It must also include supply chain security policies, vulnerability management, awareness-raising plans, support for education and research, and tools for information sharing. The strategy must be regularly assessed and updated at least every five years, with the possibility of assistance from ENISA to ensure alignment with EU-wide goals [6].

4.2 Cybersecurity Measures

NIS2 impacts all entities delivering essential or significant services to the European economy and society, including businesses and their suppliers [99]. Therefore, it requires these organizations to implement cybersecurity measures [100].

To efficiently deal with cybersecurity risks, there is a need to introduce measures in various domains. Rules for preserving the security of digital products and services, steps to protect the security of the digital supply chain, criminalization of cyberattacks, and actions to manage a cyber crisis - all these measures comprise a system for compliance with cybersecurity within the EU [101].

¹⁶ NIS2 Directive. <https://nis2directive.eu/>

Key cybersecurity measures of NIS2 define that organizations must implement strong “**Risk Management**” measures, including incident management, supply chain security, network defenses, access control, and encryption. “**Corporate Accountability**” requires senior management to oversee cybersecurity efforts, with potential penalties for breaches, including liability or bans from management roles. “**Reporting Obligations**” mandate timely reporting of significant security incidents, with a 24-hour “early warning” deadline. Lastly, a solid “**Business Continuity**” plan is required, covering system recovery, emergency procedures, and the formation of a crisis response team to ensure resilience during cyber incidents. Formally, the European Commission has come up with ten measures that (at least) need to be considered for organizations within the NIS2 scope, shown in Table 1 [6] [100].

Measure Number	Measure
1	Establishing or revising policies aligned with risk analyses and the security of information systems, as outlined in NIS2 requirements.
2	Implementing robust incident-handling procedures to address cybersecurity threats effectively.
3	Ensuring business continuity through measures such as backup management, disaster recovery plans, and crisis management strategies.
4	Securing the supply chain by addressing cybersecurity risks in relationships with suppliers or service providers, in accordance with NIS2 standards.
5	Integrating security in the acquisition, development, and maintenance of network and information systems, including managing vulnerabilities and disclosures (EUR-Lex NIS2 Directive).
6	Establishing policies and procedures to regularly evaluate the effectiveness of cybersecurity risk management measures, as mandated by the directive.
7	Promoting basic cyber hygiene practices and conducting cybersecurity awareness training programs (NIS2 compliance requirements).
8	Developing policies for cryptographic practices, including encryption where necessary, in line with NIS2 regulations.
9	Human resources security, access control policies and asset management have to be in place.
10	Implementing Multi-Factor Authentication (MFA) or continuous authentication solutions. Preparing secured communication systems (voice, video, and text) and emergency systems when applicable.

Table 1: Ten new measures from NIS2 (European Parliament, 2022)

4.3 Incident Reporting and Sector Coverage

Of all the measures NIS2 must consider, incident reporting is among the most crucial. Under NIS2, incident reporting obligations are needed to sustain cybersecurity resilience, requiring important entities to notify authorities of significant incidents without undue delay [102]. The evolution from the original NIS to NIS2 broadens the scope of reporting, extending mandatory obligations to include incidents that may not have caused immediate harm but still pose a potential threat. This change aims to provide a more comprehensive picture of the threat environment, achieving better preparedness and response

across the EU while promoting cooperation between national authorities. However, determining which incidents qualify for reporting remains challenging, particularly when balancing the harm thresholds against the need for timely information sharing to prevent incident spread [47].

In fact, Article 23 of the NIS2 Directive outlines a structured notification process: affected entities must first submit an early warning within 24 hours of becoming aware of a significant incident, followed by an incident notification within 72 hours that includes an initial assessment of severity and impact. A final report is then due no later than one month after the initial notification, providing a detailed root cause analysis, response measures, and cross-border implications if applicable. The directive also allows for intermediary updates and even public communication, should the incident affect the public interest.

Building on the expanded incident reporting obligations under NIS2, the directive also extends its reach to a broader range of sectors. It now covers essential services across industries such as energy, healthcare, transport, and digital infrastructure. This wider sector coverage ensures that all critical services are held to high cybersecurity standards [6]. NIS2 specifically outlines the importance of digital infrastructure, recognizing it as the “backbone” of modern society. The digital infrastructure sector includes key services such as telecom, data centers, cloud services, and trust services, all of which are now classified as essential entities under the directive. With increased regulatory oversight, organizations in the digital infrastructure sector must implement stronger physical and cybersecurity measures, improve incident response plans, and ensure compliance with stricter standards. This approach focuses on protecting the core digital systems upon which the EU relies, maintaining great resilience, and reducing vulnerabilities [103].

4.4 Cross-Border Cooperation

Cyber threats often originate abroad, affect multiple countries, and target companies providing cross-border services, complicating traditional notions of jurisdiction tied to physical territory. The transnational nature of cyberattacks makes it difficult to pinpoint the location of activities or actors, creating complex jurisdictional challenges [104]. To address this, NIS2 brings cooperation and information exchange between EU MSs and relevant authorities. NIS2 promotes a coordinated approach to cybersecurity incidents that may have cross-border implications. This collaboration, as depicted in Figure 9, improves response times and overall success in mitigating cyberattacks [105].

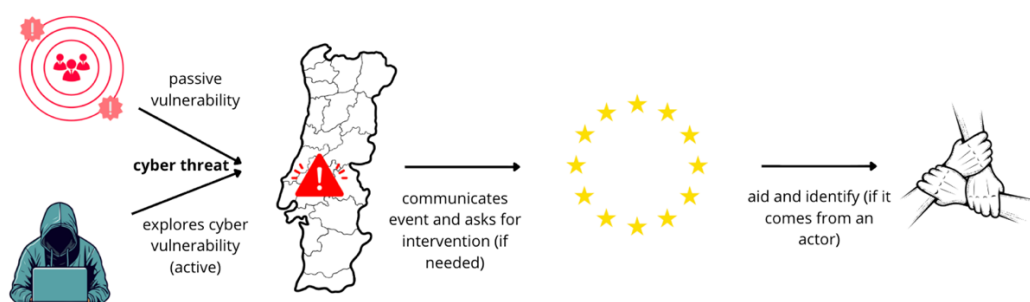


Figure 9: Cross-border cooperation representation example

There are several ways to illustrate cross-border cooperation. For example, providers of public electronic communications networks, trust service providers (TSPs), and top-level domain name registries play key roles in the security of electronic services. Under eIDAS2, which establishes trust frameworks for electronic transactions, disruptions to these services could create significant systemic risks, particularly in sectors where such interruptions can have cross-border effects, aligning with the emphasis of NIS2 on managing cybersecurity incidents with transnational impacts [106]. For instance, in the event of a cybersecurity incident with cross-border implications, such as a ransomware attack affecting multiple hospitals, the general hospital engages in cross-border cooperation in accordance with the NIS2 framework. It reports the incident to the national competent authority and collaborates with other hospitals, ENISA, and EU MSs to share threat intelligence, coordinate response efforts, and limit the impact of the attack across borders [105].

Additionally, NIS2 introduces formal mutual assistance mechanisms among MSs. This guarantees that, during significant incidents, competent authorities, CSIRTs, and ENISA can support one another by sharing expertise, resources, and operational help. When a MS faces a large-scale cyberattack, others can step in with technical or logistical assistance [6].

Ultimately, cross-border compliance is accomplished by simplifying legal compliance for cross-border service providers. That's where NIS2 comes into play, increasing the harmonization of security and reporting standards to support legal compliance for entities offering cross-border services.

4.5 CSIRTs and EU-CyCLONe

CSIRTs are specialized entities responsible for managing and responding to cybersecurity incidents within specific organizations, sectors, or nations. “Each MS shall designate or establish one or more CSIRTs” [6]. CSIRTs monitor and analyze cyber threats, provide warnings, respond to incidents, and assist organizations in managing vulnerabilities. They perform forensic analysis, support secure information-sharing, collaborate with other CSIRTs, and help coordinate vulnerability disclosures. MSs must “ensure that their CSIRTs jointly have the technical capabilities necessary to carry out these tasks.” CSIRTs network, composed of national CSIRTs and CERT-EU, facilitates their EU-wide collaboration and interoperability on cybersecurity matters.

In parallel, the European Cyber Crisis Liaison Organization Network (EU-CyCLONe), is “a cooperation network for the national authorities in charge of cyber crisis management.”¹⁷ EU-CyCLONe focuses on improving preparedness for large-scale cybersecurity incidents, building shared situational awareness, assessing impacts, proposing mitigation measures, coordinating crisis management, supporting political-level decision-making, and, upon request, reviewing national response plans.

In situations where a large-scale cybersecurity incident is anticipated or occurring and could significantly affect these services, the European Commission diligently participates in the activities of EU-CyCLONe. In less critical circumstances, the Commission may join as an observer, ensuring that it remains informed and can contribute to the overall governance of cybersecurity crisis management

¹⁷ European Union Agency for Cybersecurity. EU CyCLONe.

within the EU. In addition, EU-CyCLONe shall submit a report assessing its work to the European Parliament and the Council, reinforcing NIS2 measures to improve incident reporting.

CSIRTs collaborate closely with EU-CyCLONe to enhance European cybersecurity resilience. While CSIRTs focus on technical aspects like monitoring, analyzing, and responding to cyber threats, EU-CyCLONe coordinates cyber crisis management at the operational level among EU MSs. ENISA acts as the secretariat of CSIRTs and EU-CyCLONe, each serving many purposes within the agency. At the initiative of both, ENISA performs analyses of cyber incidents to enhance the learnings from cybersecurity incidents [107].

In short, EU-CyCLONe, in coordination with CSIRTs, strengthens the ability of the EU to manage and deal with complex cybersecurity incidents by uniting MSs and the European Commission to enhance preparedness and coordination [108].

5 Electronic Identification, Authentication and Trust Services

The European Commission published Regulation 910/2014 on electronic identification and trust services (known as eIDAS). This Regulation focuses on enabling “citizens throughout the EU to authenticate to and communicate with online services of MSs while using an electronic Identity (eID) scheme of their own MS” [109]. However, “major privacy shortcomings exist within the eIDAS regulation”, which is the main reason why the EU came up with a revised eIDAS framework (eIDAS 2) in 2024. The amended 2024/1183 regulation will be fully implemented by 2026 and aims to:

- “Provide a highly secure and trustworthy electronic identity solution that enables the user to share only the minimum set of identity data needed by the service provider (SP) to provide the service requested by the user”;
- Introduce the EUDI Wallet from each MS for its residents to use according to approved specifications;
- Enhance privacy by decoupling credential issuance from its presentation to SPs.

The eIDAS2 regulation has become essential to establishing a secure and interoperable digital environment within the EU [110].

5.1 Electronic Identification and Cross-Border Interoperability

Under the eIDAS2 regulation, eID is “the process of using person identification data (PID) in electronic form uniquely representing either a natural or legal person, or a natural person representing another natural person or a legal person” [111]. It’s required to authenticate identities during electronic transactions, “interactions conducted electronically, typically involving data exchanges, that require authentication, trust services or digital signatures”. The authentication process is carried out using electronic identification means, “material and/or immaterial unit containing PID and which is used for authentication for an online service”. This process ensures secure identity verification, builds trust in digital interactions, and enables compliance with the EU legal framework. Complementing PID-based identification, the eIDAS2 regulation also defines electronic attestation of attributes (EAA) as the process of certifying specific information about a natural or legal person, issued in electronic form by a qualified trust service provider (QTSP). Therefore, PID is the core identity, while EAAs are extensions that enrich that identity with verifiable attributes [112].

The main achievement of the regulation regarding eID is the “mutual recognition of electronic identification means for cross border access to public sector services under certain conditions, which require that MSs notify their eID schemes and ensure the availability of authentication services, which shall be provided free of charge for cross-border public sector services” [113]. To address this, the assurance levels of “low”, “substantial”, and “high”, corresponding to the Level of Assurance (LoA), are established. Additionally, it is stated that the notifying MS is responsible for damages resulting from incorrectly issued eID means or failures in the authentication services.

Cross-border interoperability is vital for eID, enabling citizens and businesses to access services in other MSs without barriers. Consequently, it is recognized as essential for the successful implementation of eIDAS2 [114]. In contrast, “the recognition of eID and electronic delivery services is one of the reasons why this regulation is of huge importance for successful cross-border data exchange among public administrations in the EU” [115]. This creates a mutually beneficial arrangement between eIDAS2 and cross-border interoperability.

5.2 Principle of Mutual Recognition

When national law or administrative practice requires eID and authentication using an electronic identification means to access an online service provided by a public sector body in a MS, “the electronic identification means issued in another MS shall be recognized in the first MS for the purposes of cross-border authentication for that service online” [111]. This is Mutual Recognition. Regarding the principle of mutual recognition, authentication for an online service should “concern processing of only those identification data that are adequate, relevant and not excessive to grant access to that service online”.

Mutual recognition of electronic identification means poses several challenges for MSs. Approaching these challenges “can help to improve and increase in efficiency the delivery of eGovernment services so that citizens can identify and authenticate themselves simply, safely and with sufficient confidence” [116].

5.3 Trust Service Providers, Trusted Lists, and Electronic Signatures

According to eIDAS2, trust services are essential for ensuring the security and legal validity of electronic transactions within the EU. The regulation defines trust services as the “creation, verification, and validation” [111] of services such as electronic signatures, electronic time stamps, electronic seals, and electronic certificates for website authentication. Each provides a reliable way to authenticate the identity of individuals or entities and guarantee the integrity of digital documents.

A TSP is an entity that delivers reliable digital trust services. While a service provider could personally provide sufficient guarantees on the trustworthiness of his certificates, a service provider can only be granted the title “qualified” if he complies with the specific requirements set out by eIDAS2. All QTSPs are included in a public trusted list, maintained by the supervisory bodies [117]. The MSs of the EU and European Economic Area “publish trusted lists of QTSPs in accordance with the eIDAS2 regulation”. The European Commission then “publishes a list of these trusted lists, the List of Trusted Lists (LOTL)”. These lists promote transparency and trust across MSs ¹⁸.

Furthermore, QTSPs are subject to conformity assessments carried out by a Conformity Assessment Body (CAB). This accreditation confirms that the QTSP complies with stringent requirements, ensuring

¹⁸ European Commission. (2024). eIDAS Trust Services.

the qualified trust services they provide, such as certification of EUDI Wallets or electronic identification means, meet high standards of security and reliability.

As the most well-known trust service, electronic signatures have revolutionized how digital transactions are authenticated, ensuring trust and legal validity across the EU. eIDAS2 establishes a framework for a range of electronic signatures, from basic to advanced and qualified, with qualified electronic signatures (QES) offering the highest level of trust and being legally equivalent to handwritten signatures. With QES, eIDAS2 can create an ideal environment for electronic documents across MSs. Advanced electronic signatures (AdES) allow for a higher level of security than basic signatures by being uniquely linked to the signer, capable of identifying them, and detecting any tampering with the signed data. However, AdES are not required to be created by or issued through a QTSP, unlike QES, which a QTSP must create to meet the highest legal standards under eIDAS2. This means that AdES provide strong security and integrity guarantees but does not automatically carry the same legal status as QES. Despite these advancements, challenges such as aligning technical solutions with legal requirements remain hurdles to fully realizing the potential of eIDAS2 electronic signatures [118] [119].

In Figure 10, all the mentioned associations can be visualized.

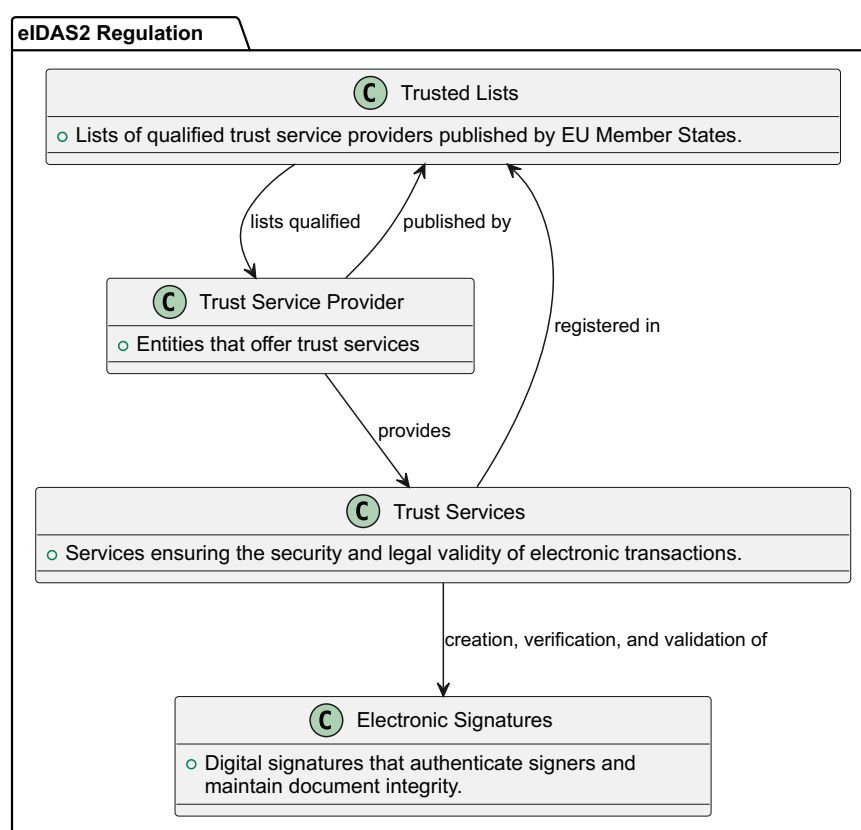


Figure 10: Associations between electronic signatures, trust services, and trusted lists

5.4 Digital Wallet

The EUDI Wallet represents a significant move in how digital identities are managed across the EU. Unlike the previous system, where each country had its own unique eID, the EUDI wallet provides a standardized, cross-border solution for citizens and businesses. Fundamentally, it is an “electronic identification means which allows the user to securely store, manage and validate PID and electronic attestations of attributes” [111].

The wallet aims to simplify cross-border interactions by removing country-specific barriers, offering a unified digital identity for all EU citizens, and integrating both public and private sector services [120]. Its proposed app will include frameworks for Corporate Digital Identity (CDI), addressing unique business needs such as corporate structures, data privacy, and regulatory compliance [121].

EUDI Wallets are the response of the EU to the complexities of digital identification. Every EU MS “will offer its own wallet app, built to the same specifications, to all citizens, residents, and businesses in the next few years. Each version of the wallet will be interoperable and will work across Europe” ¹⁹.

Figure 11 illustrates an EUDI Wallet example.

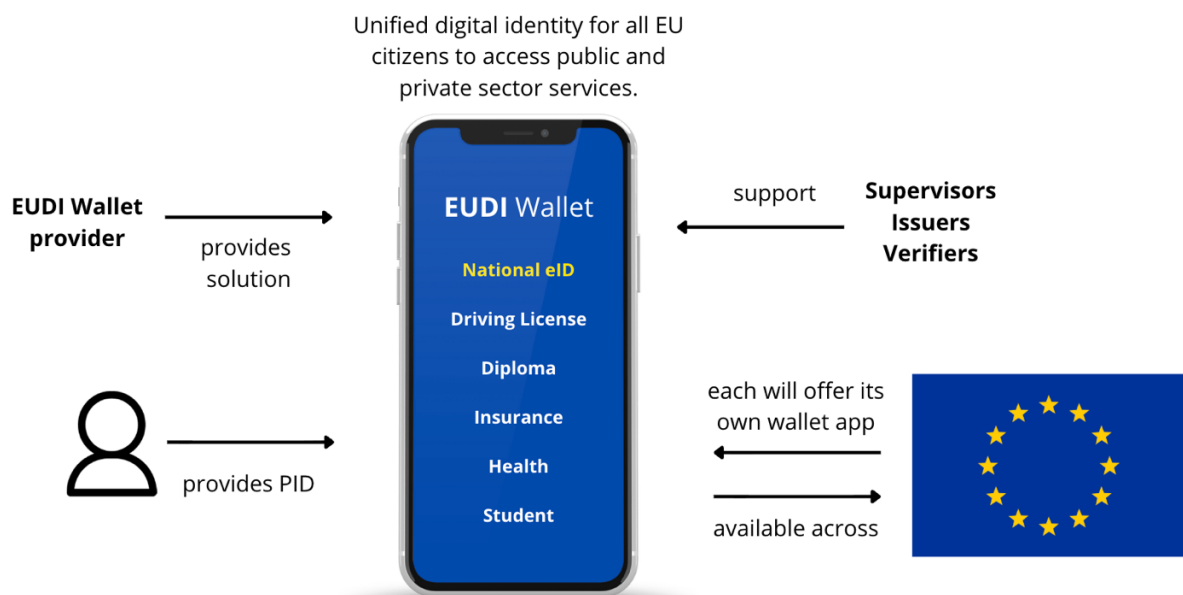


Figure 11: EUDI Wallet representation example

¹⁹ European Commission. EU Digital Identity Wallet.

6 Method

Creating an AD that models complex legislation while addressing diverse legal concerns presents a significant challenge. The guiding question is: How can we simplify and group legal clauses in a way that keeps all important information while making the AD easy to understand? To answer this, a methodical, structured approach presented in this section was followed.

This involved breaking down each piece of legislation into smaller parts, organizing them in a clear format, and then translating those parts into ArchiMate views. It combines legal analysis with conceptual modelling through four main steps: creating a detailed Excel traceability matrix as an external legal reference, developing an ArchiMate Index for internal architectural navigation, connecting both through a hierarchical numbering system, and constructing the ADs in line with defined architectural principles, consistent color coding, and documentation based on official legislative sources.

Nonetheless, all deliverables discussed and presented from this section onward are organized in a dedicated GitHub repository, providing a centralized, structured way to manage project artifacts. This repository ²⁰ ensures that all files are version-controlled and easily accessible.

6.1 Excel Traceability Matrix

The process began with the creation of a traceability matrix in Excel, which provides a structured overview of how the legal text from both NIS2 and eIDAS2 can be broken down into identifiable units. Each unit (row) in the matrix represents a specific legal clause, clearly identifying the chapter, article, paragraph, points, and even sub-points (e.g., 1.1.a or 1.2.e). Samples of these matrixes can be seen in Table 2 and Table 3.

These units are linked to the ArchiMate AD through a numbering system, a consistent reference system used throughout the AD, while also serving as a mapper for navigation. Additionally, colors are used to distinguish between different chapters or types of content, improving visual clarity.

Chapter	Article	ID1	ID2	ID3	ID4	ID5	ID6	ID7	Text
CHAPTER VI		6							CHAPTER VI - INFORMATION SHARING
CHAPTER VI	Article 29	6	29						Article 29 - Cybersecurity information-sharing arrangements
CHAPTER VI	Article 29	6	29	1					1. Member States shall ensure that entities falling within the scope of this Directive and, where relevant, other entities not falling within the scope of this Directive are able to exchange on a (...)
CHAPTER VI	Article 29	6	29	1	1		a		(a) aims to prevent, detect, respond to or recover from incidents or to mitigate their impact;
CHAPTER VI	Article 29								...
CHAPTER VI	Article 30	6	30						Article 30 - Voluntary notification of relevant information
CHAPTER VI	Article 30								...

Table 2: Sample from the traceability matrix of NIS2
** NIS2 Directive (2022/2555) does not contain Sections*

²⁰ Bolinhas, P. (2025). MSc Thesis repository. GitHub. <https://github.com/paulobolinhas/Conceptual-Modelling-of-Legal-Compliance-Requirements-The-cases-of-eIDAS2-and-NIS2>

Chapter	Section	Article	ID1	ID2	ID3	ID4	ID5	ID6	Text
CHAPTER III			3						CHAPTER III - TRUST SERVICES
CHAPTER III	SECTION 8		3	8					SECTION 8 - WEBSITE AUTHENTICATION
CHAPTER III	SECTION 8	Article 45	3	8	45				Article 45 - Requirements for qualified certificates for website authentication
CHAPTER III	SECTION 8	Article 45	3	8	45	1			1. Qualified certificates for website authentication shall meet the requirements laid down in Annex IV. The Evaluation of compliance with those requirements shall be carried out in accordance with the standards, specifications and procedures referred to in paragraph 2 of this Article.
CHAPTER III	SECTION 8	Article 45							...
CHAPTER III	SECTION 8	Article 45a	3	8	45a				Article 45a - Cybersecurity precautionary measures
CHAPTER III	SECTION 8	Article 45a							...
CHAPTER III	SECTION 9		3	9					SECTION 9 - ELECTRONIC ATTESTATION OF ATTRIBUTES
CHAPTER III	SECTION 9	Article 45b	3	9	45b				Article 45b - Legal effects of electronic attestation of attributes
CHAPTER III	SECTION 9	Article 45b	3	9	45b	1			...

Table 3: Sample from the traceability matrix of eIDAS2

6.2 ArchiMate Index

Building on the traceability matrix, the next step was to establish an entry point, in ArchiMate, for each AD: the “Index” view. While the matrix provides an external reference point for navigating the legislation, the “Index” view enables internal navigation between the chapters and articles of the legislation and their corresponding architectural representations from a single location. Together, the traceability matrix, the “Index” view, and the views of the ADs form a “traceability carousel”, linking the legal text to its architecture.

In this “Index” view, each chapter of the legislation is represented as an ArchiMate group, which contains the corresponding ArchiMate views that model each legislation article. While most articles are individually represented, some have been combined into a single view, subdivided across multiple views, or integrated into other related views. All these variations are easy to understand through the numbering system found within the documentation of each view element, which contains an identifier linked to the corresponding legal clause. Nevertheless, the view element was used because it is clickable, which enables navigation and allows for efficient exploration of the AD from a centralized point.

Moreover, extra groups are created to categorize further views that focus on the same topic. For example, within the eIDAS2 section on electronic signatures, the regulation itself does not categorize sub-sections such as advanced electronic signatures or qualified electronic signatures. Therefore, in such cases, additional groups are created here to provide that categorization.

Some views that do not directly correspond to specific articles, but serve a complementary function or represent practical scenarios, are also included. These are marked with distinct colors to differentiate them from article-specific views: **Pink** for practical scenarios and **Yellow** for complementary views. In addition, within the eIDAS2 “Index” view:

- **Blue** was used to represent a mutatis mutandis case, which justifies skipping the modelling of certain views. In detail, the views related to Electronic Seals would have the same structure and logic as those for Electronic Signatures, but with the naming adjustments (mutatis mutandis, as defined in the regulation) to reflect their specific characteristics.
- **Lime** was used in the eIDAS2 "Index" view to avoid redundancy from short views of very similar articles that only differ by the trust service. As a result, it shows two views: "Legal effects of *" and "Requirements for *" (Electronic Time Stamps, Electronic Registered Delivery Services, Electronic Archiving Services, Electronic Ledgers). Each includes clickable references for the specific views of all four trust services. In short, a pragmatic merge to reduce the complexity of the index.

Both index views are presented in Figure 12 and Figure 13.

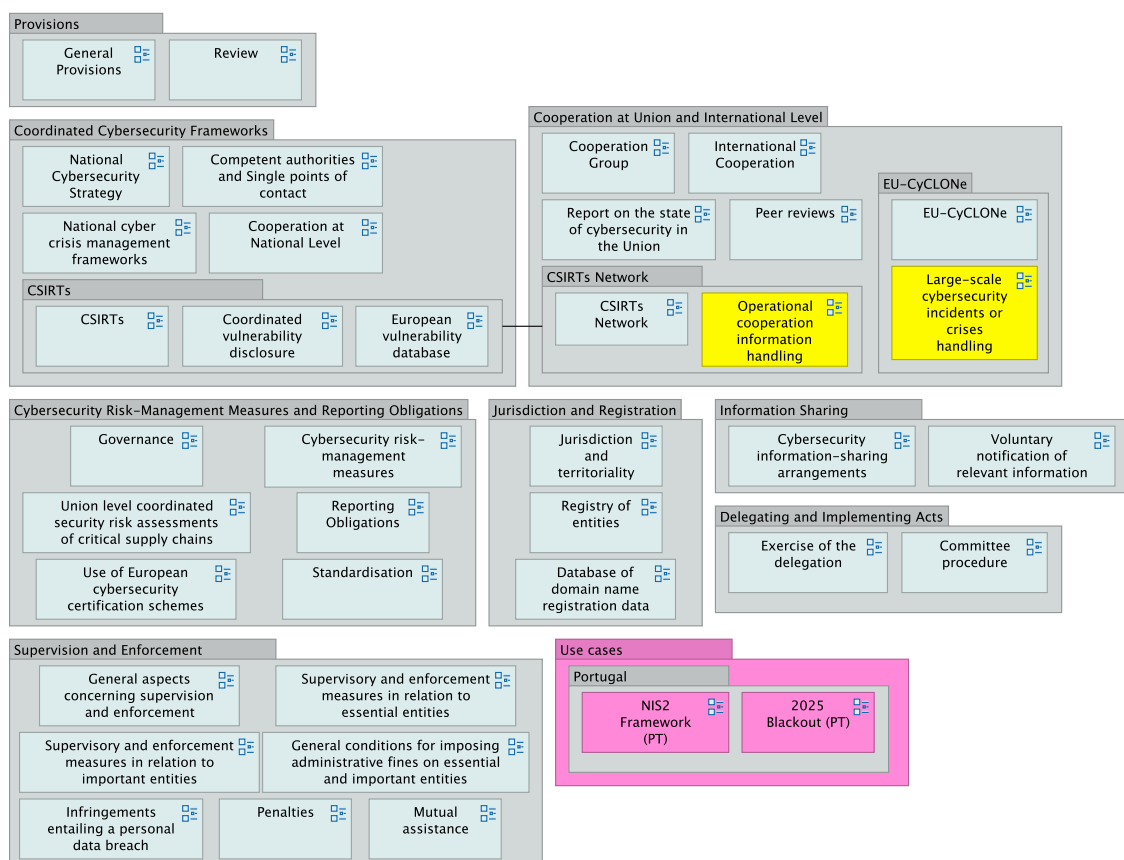


Figure 12: ArchiMate NIS2 Index view

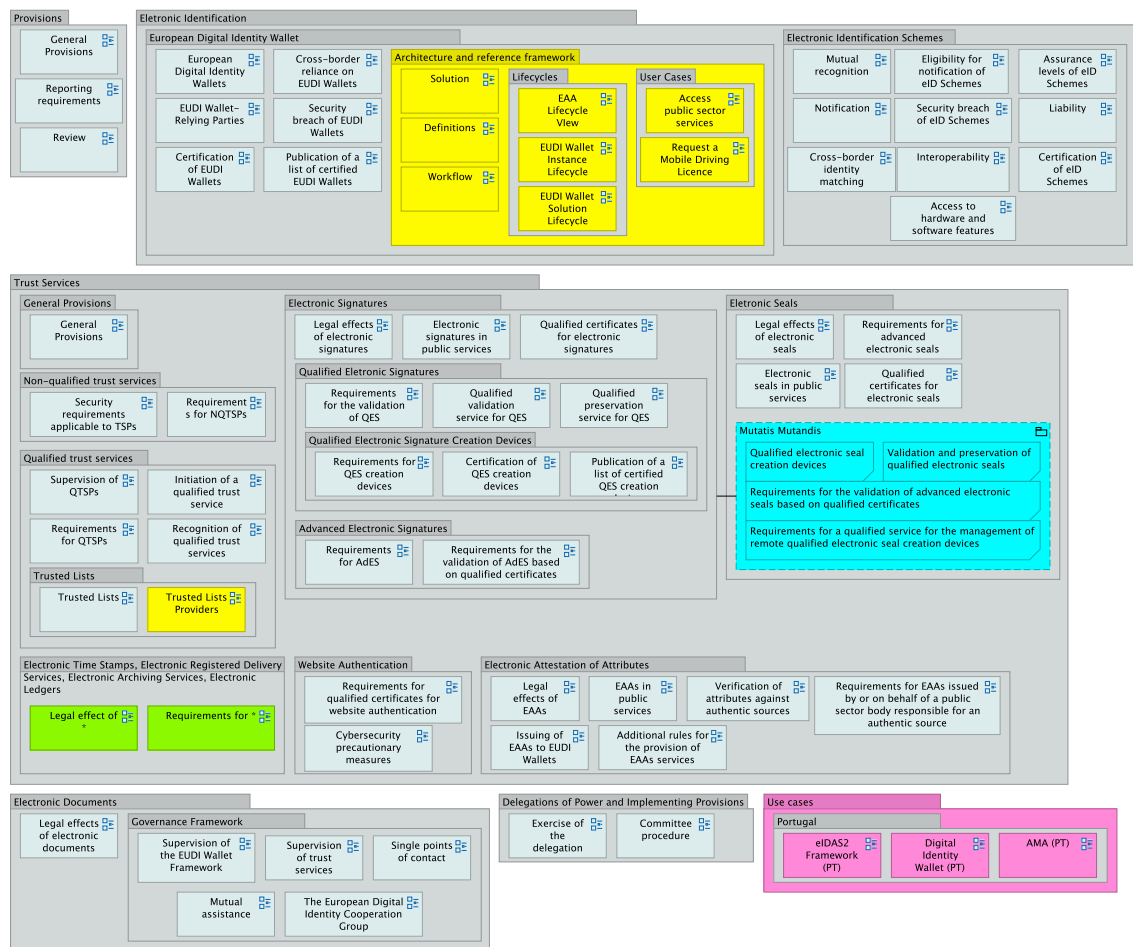


Figure 13: ArchiMate eIDAS2 Index view

6.3 Numbering System

The numbering system, mentioned already, is the key traceability feature, with the Excel matrix serving as its foundation. It provides a simple ID mapping between the legislation clauses listed in the matrix and the elements in the AD. Each ID corresponds to a specific “clause level” in the legislation (e.g., chapter, section, article, paragraph, subparagraph, point, sub-point, etc.). Combining both ADs, there are more than one thousand numbered elements. For example, in Table 4, the sequence “CHAPTER I → Article 2 → 1 → 2 → 2 → 1 → 1 → a → i” corresponds to Chapter I, Article 2, paragraph 2, (a), (i).

Chapter	Article	ID1	ID2	ID3	ID4	ID5	ID6	ID7	Text
CHAPTER I	Article 2	1	2						Article 2 - Scope
CHAPTER I	Article 2								...
CHAPTER I	Article 2	1	2	2	1		a		(a) services are provided by:
CHAPTER I	Article 2	1	2	2	1	1	a	i	(i) providers of public electronic communications networks or of publicly available electronic communications services;
CHAPTER I	Article 2	1	2	2	1	2	a	ii	(ii) trust service providers;

Table 4: Sample 2 from the traceability matrix of NIS2

→ Adopt delegation act (Business Process)		
Main	Specialization:	(none)
Properties	Name:	Adopt delegation act
Analysis	Documentation:	Numbering: 8.38.1, 8.38.2
Appearance		
Label		
Figure		
Image		

→ Implement enforcement measure (Business Process)		
Main	Specialization:	(none)
Properties	Name:	Implement enforcement measure
Analysis	Documentation:	Numbering: 7.32.4 Numbering: 7.33.4
Appearance		
Label		
Figure		
Image		

Figure 14: Numbering examples of business processes

In detail, the traceability matrix assigns a unique hierarchical ID to each clause of the legislation, reflecting its full structural depth and ensuring accurate mapping to elements in the AD. Each ID level corresponds to a structural layer in the legal text. For instance, the clause “Article 2, (2), (a), (i)” might appear straightforward in the legislation, but in the matrix, it is represented as “1.2.2.1.1.a.i” rather than “1.2.2.a.i”. This is because the matrix captures every intermediate level, including numbered lists or grouping structures that are sometimes implicit in the original text. In this case, “1.2.2” represents Chapter I, Article 2, paragraph 2. The additional “1.1” accounts for nested list levels within that paragraph before reaching point (a), and “i” represents the sub-point. For the next clause, “(a), (ii)”, the intermediate levels are now “1.2”, and for a hypothetical “(b), (i)”, they become “2.1”. While these intermediate levels may be disregarded in casual reading, this layered format ensures each clause, no matter how deeply nested, is uniquely identified, avoiding ambiguity. In Figure 14, some numbering examples are shown.

Important to mention that elements and views may be associated with more than one ID, meaning they correspond to multiple clauses or articles. Cases such as “1.-.1” indicate the absence of a Section, while cases like “1.1.*” denote that all sub-clauses within that level are included. Additionally, there are cases where IDs are not assigned to the AD when they correspond to clauses or articles that are either deemed non-relevant or already covered elsewhere within the AD.

6.4 Building the ArchiMate ADs

The process culminates in the development of the ADs for both NIS2 and eIDAS2. It followed a top-down approach, progressing from a high-level legislative perspective to detailed architectural modelling. For each piece of legislation, a **chapter-by-chapter** and **article-by-article** analysis was conducted, critically evaluating the legal text to identify which elements were relevant and should be included in the architecture. This analysis included selecting key actors, roles, processes, and many other elements, with a strong focus on the Business layer of ArchiMate. Emphasizing the business perspective is particularly important for NIS2 and eIDAS2 because both are purpose-driven and affect organizational operations, not just technical systems. This naturally supports the Motivation aspect: once it is clear what is done and by whom, it is also necessary to explain why it is done, which is exactly what this aspect captures. Therefore, special attention was given to this aspect, alongside the others.

Each article was interpreted coherently to reflect the intent and impact of the legislation, in a way that fits within the structure of an ArchiMate model. When a legal clause warranted additional

clarification, complementary views were created. These were only added when they clearly brought value to the interpretation of the legislation within the architectural context.

Moreover, depending on the person reviewing the view, whether an enterprise architect or a legal consultant, each view may be interpreted differently. In designing the views, the perspective of the architect was predominant, but the perspectives of legal actors were not disregarded. Reconciling these viewpoints was often challenging. Therefore, each article was strictly translated into architecture by adhering to core principles while remaining as close as possible to the legal clauses.

6.4.1 Architecture Principles

Given the complexity and granularity of both eIDAS2 and NIS2, careful modelling was required to ensure that the resulting architecture remained clear, interpretable, and faithfully aligned with the legal texts. In the end, each view illustrates a model that facilitates understanding of legal compliance for the stakeholders, and within its design, it adheres to requirements such as the ArchiMate best practices, legal norms, and the ISO/IEC/IEEE 42010 standard. The following architecture principles guided the development of both ADs:

- **Abstraction:** Views are at the right level of legal detail, avoiding unnecessary complexity and focusing on what adds value to the understanding of the legal clauses. This is a must since it allows complex legislation to be represented at different levels of detail, helping stakeholders focus on relevant aspects without being overwhelmed by unnecessary complexity;
- **Reuse:** Once legal elements, style types, and architecture patterns were defined, they were consistently used in all applicable views. For instance: legal elements like data objects were reused where relevant; style types such as labelling, grouping similar cases, arranging relations, and coloring (6.4.2) were applied uniformly; architecture patterns like process-driven and event-driven, that captured legal text with complex behavior, allowed for the usage of their corresponding modelled behavioral design across views. Also, anti-patterns, such as generic association relations and usage of invisible properties, were minimized. This ensures coherence and clarity throughout the model, while reducing duplication of effort;
- **Traceability:** As mentioned before, this is provided by the numbering system between the matrix and the AD, that allows tracking back from elements to the legal clauses, or vice versa. Traceability in legal EA modelling is important because it ensures that every architectural element can be directly linked to its source requirement or legal clause, enabling verification, compliance, and easier maintenance of the architecture;
- **Modularity:** The AD generally models one view per article (legal unit), creating separate views if the content is complementary, and organizing elements using Grouping elements. Views themselves are independent (e.g., the ability of each view to represent a specific architectural area, such as the EUDI Wallet, in isolation). This approach enables stakeholders to understand each view without the need to consult the entire AD. Additionally, the ArchiMate architecture repository is carefully structured into folders for clarity and ease of navigation, as illustrated in Figure 15.



Figure 15: Structure of the folders “Views” and “Business Layer”

6.4.2 Color Table

During the development of views, a color table, showed in Table 5, was followed to guarantee visual consistency and to help distinguish between different types of elements within each view. As mentioned earlier, “colors in ArchiMate have no formal semantics, leaving their use flexible for modelers to highlight specific elements or emphasize certain aspects as needed” [76]. Therefore, they were used only when they were considered relevant to distinguish something.

Color	Representation
Blue (aqua)	EU entity or element
Green (fern)	Portuguese element
Orange (mocha)	Spanish element
Magenta	Other relevant entity or element
Green	State or Positive element
Yellow	State or Complementary element
Orange	State
Red	State or Negative element

Table 5: Color table used throughout the modelling

** this color table does not apply to “Index” views, as their color coding was explained previously*

For instance, in a view, the color Magenta may not cover every relevant entity or element, but only those that are important to consider within the context of that specific view, or to make them stand out for better visual clarity. As an example, Magenta and Blue are not used for NIS2 and eIDAS2 entities, even if those entities are European or relevant to the view. These are always represented in the default color of the ArchiMate element, which means the use of colors is not based on absolute rules. The same principle applies to every color. Nevertheless, the objective is to ensure a clear visual model through the use of a consistent color pattern.

As for Green, Yellow, Orange, and Red, these colors were often used to represent behavioral states within the view, as shown in the view below. However, in other cases, Green and Red represent positive and negative elements, respectively, while Yellow represents a complementary element. Figure 16 presents a view that uses color to represent the states of an EUDI Wallet Solution.

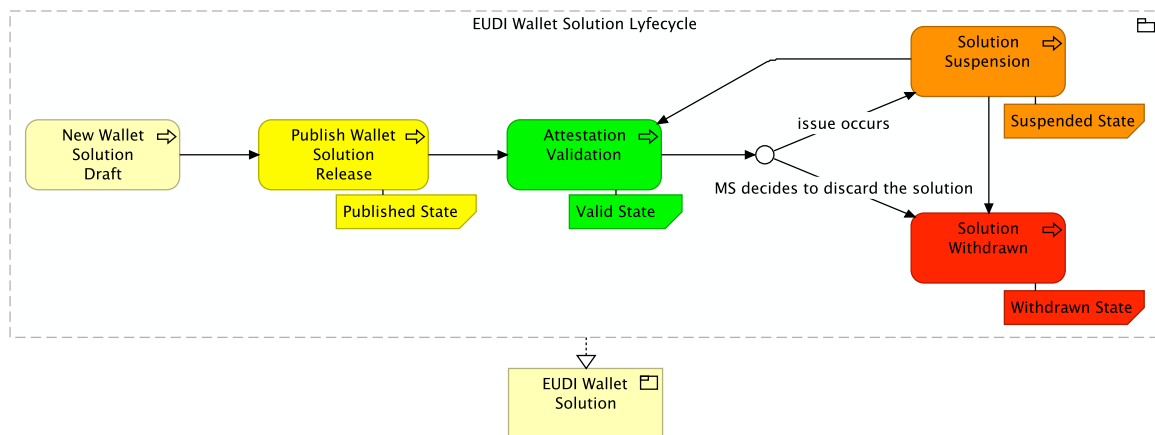


Figure 16: View from the eIDAS2 AD of the EUDI Wallet solution lifecycle

6.4.3 Documentation

Each view is thoroughly documented, with its elements referencing official legislative sources, such as the EU Journal (EUR-Lex), thereby ensuring legal accuracy. It complements the numbering system, which already provides basic information about each element, by including additional details or context when needed. An example is shown in Figure 17.

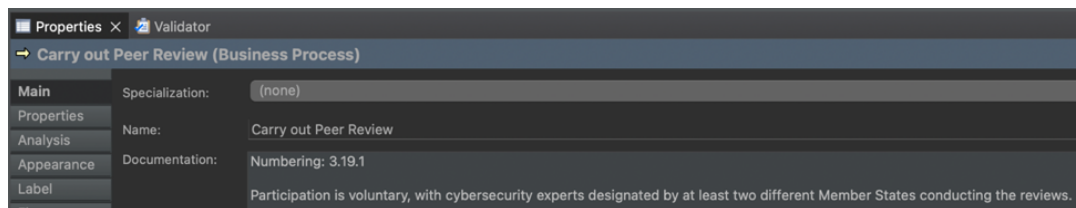


Figure 17: Example documentation of a business process

This helps reduce the need to frequently refer back to the matrix and the risk of missing important nuances, especially since views are independent, and clicking on a single element may require more context than the article or legal clause alone can provide.

A list of the most used sources when documenting the elements in the views:

- Directive 2022/2555 EUR-lex
- Regulation 2024/1183 EUR-lex
- NIS2 Official Website - <https://nis2directive.eu/>
- eIDAS2 Official Website - <https://www.european-digital-identity-regulation.com/>
- GNS Official Website - <https://www.gns.gov.pt/>
- CNCS Official Website - <https://www.cncs.gov.pt/>
- EUDI Wallet ARF [91]
- *Diário da República* - <https://diariodarepublica.pt/>

6.4.4 Versioning with coArchi2 plugin

Versioning is the practice of keeping track of changes made to a project over time, ensuring that each modification is recorded and can be traced back, compared, or reverted if necessary. Versioning was introduced after the initial drafts of both ADs had been produced in order to support the refactoring process.

To enable this functionality in the Archi tool, the coArchi2 plugin was adopted, which integrates Git-based version control into the ArchiMate modelling environment. The installation of coArchi2 was carried out by importing the plugin package provided through the official Archi tool Patreon distribution. Following the installation, the collaboration setup was configured within the Archi tool application settings to define repository access and user parameters. Where required, secure authentication was enabled using SSH keys to ensure reliable communication with the remote Git repository. After these configurations, the collaboration environment was activated by importing the relevant models via the collaboration menu in Archi. The models were then made available in a structured folder system, from which they could be opened and managed directly within the tool.

Through this setup, collaboration with the thesis supervisor was enhanced by enabling him to interact directly with the shared models.

7 Results

As a result of the building process, producing a first consolidated version (**v1**), the ADs provide comprehensive, coherent, and navigable representations of both eIDAS2 and NIS2. However, it is important to acknowledge the inherent difficulty of modelling large and complex legal ADs such as these. With more than a hundred views and over a thousand numbered elements representing different legal clauses, producing perfectly detailed ADs manually is not feasible.

In this section, both ADs are presented. Views were extracted from each AD to represent every subsection from the respective sections 4 and 5. In this way, it is possible to complement the initial discussions, where both legal frameworks were introduced, by directly presenting their architectures together with some relevant considerations that strictly follow the latest ArchiMate 3.2 specification.

7.1 Architecture Description of NIS2

NIS2 AD models the cybersecurity regulatory framework introduced by the directive, focusing on its application to essential and important entities, national CSIRTs, supervisory authorities, and other European-level cooperation institutions.

Using ArchiMate, the AD translates the legal clauses of NIS2 into views that clarify roles, processes, and interactions between national and EU-level cybersecurity actors. This approach helps visualize how cybersecurity obligations are operationalized across different contexts and enables stakeholders to understand better and assess compliance from an architectural perspective.

Based on the previously introduced subsections of NIS2, a view for each was extracted from the AD.

7.1.1 National Cybersecurity Strategy

This subsection introduces the view (Figure 18) representing the NCS as defined by NIS2 in Article 7 [6]. NIS2 requires MSs to establish their NCS, notify the European Commission, and undergo regular assessments by ENISA (upon request) to ensure effectiveness. ENISA supports this assessment by establishing Key Performance Indicators (KPIs) and updating the NCS as needed.

The strategy includes a governance framework, risk assessments, cybersecurity measures, and awareness plans, complemented by policies like ICT supply chain security, cybersecurity education, and SME resilience programs. Aligned with the European Digital Strategy, it ensures consistency across the EU as every country works toward common goals, shared definitions, roles, and practices. This cohesive approach also helps MSs prepare more effectively for large-scale or cross-border cyber incidents, strengthening the European cybersecurity posture.

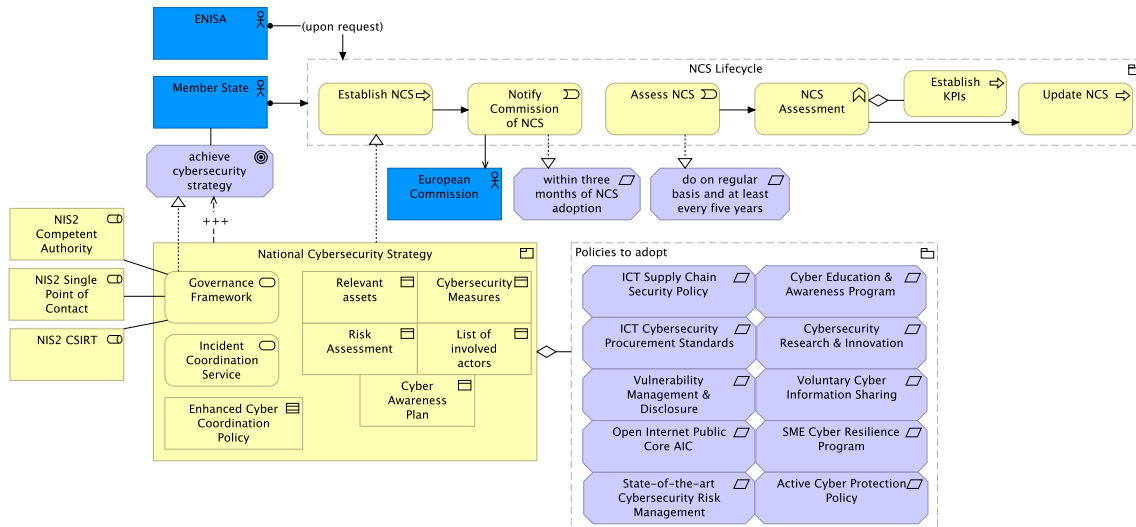


Figure 18: View of the national cybersecurity strategy

ArchiMate architectural comments:

The NCS can be modeled as a business product, as it encapsulates a coherent collection of services and objects, accompanied by a contract. ArchiMate enables a logical representation of this structure. The relevant policies are modeled as requirement elements within the motivation layer, as they express statements of need that justify and guide the development of the NCS. Additionally, the lifecycle of the NCS is represented through business processes, illustrating how its strategy evolves while maintaining alignment with EU plans.

Moreover, grouping in ArchiMate is always a good option when aggregating concepts that belong together based on some common characteristic, which is why it is used a lot on both ADs.

7.1.2 Cybersecurity Measures

This subsection presents the view (Figure 19) containing the cybersecurity risk-management measures imposed by NIS2, in Article 21 [6]. Cybersecurity risk management is a vital aspect of the directive, requiring essential and important entities to manage risks to their network and information systems. Within NIS2, MSs are tasked with managing cybersecurity risks posed to network and information systems, assessing compliance with these measures, and taking appropriate corrective actions if entities do not comply. Essential and important entities must manage risks and prevent or minimize the impact of incidents on their services.

Cybersecurity risk-management measures include policies on risk analysis, business continuity, supply chain security, incident handling, secure communications, and more. These measures ensure a proportional level of security based on risk exposure, entity size, and potential impact, following an all-hazards approach. The diagram emphasizes the importance of continuous risk management and compliance assessment to maintain cybersecurity resilience.

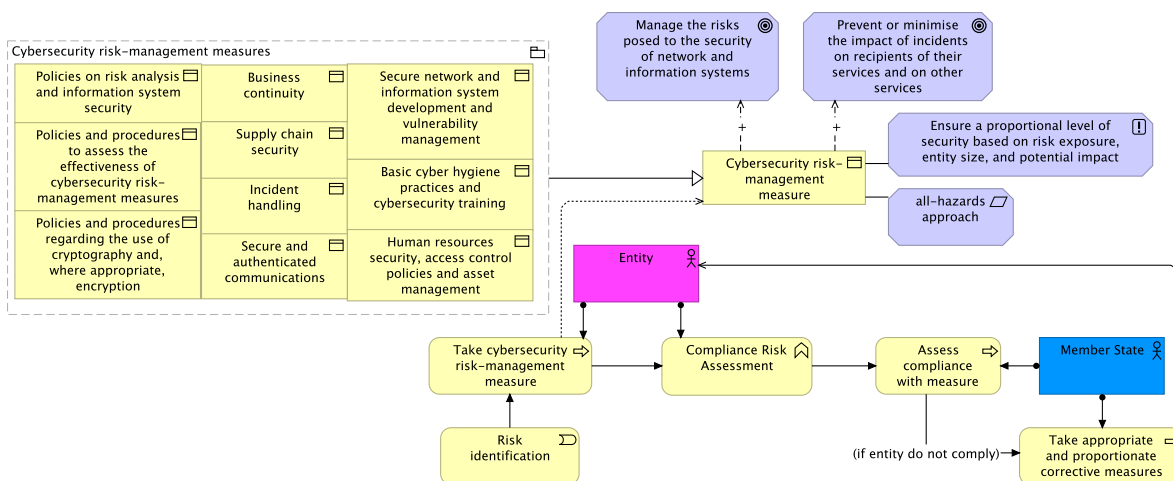


Figure 19: View of the cybersecurity risk-management measures

ArchiMate architectural comments:

This case presents a modelling challenge in ArchiMate, as it revolves around a business object that is referenced across multiple legal paragraphs. Nevertheless, ArchiMate enables a simplified yet structured representation by centralizing the core concept and modelling its variants as distinct business objects. In this approach, the different measures are modeled as a specialization of the main business object, while the conditions that govern them are expressed as motivation elements, linked to the general concept of cybersecurity risk-management measure.

In addition, to reflect the behavioral aspects described in the article, a conditional sequence was introduced, triggered by the identification of a risk.

7.1.3 Incident Reporting and Sector Coverage

This subsection illustrates the workflow (Figure 20) of reporting obligations defined in NIS2, in Article 23 [6]. Incident notification and reporting require entities to promptly notify CSIRTs of significant incidents that cause severe disruption or harm.

When an incident occurs, entities must assess its impact and determine if it qualifies as a significant incident, causing severe service disruption or financial loss. Significant incidents must be reported to the NIS2 CSIRT within 24 hours, which then coordinates the response and notification process. Entities must also notify recipients of service and other MSs if a significant cyber threat occurs. Incident notification updates must be provided within 72 hours, and public awareness of significant threats must be ensured.

Entities are required to submit intermediate and final reports on the incident impact. The CSIRT is responsible for determining the cross-border or cross-sectoral impact of the incident, notifying the SPC if it is cross-border. The SPC should submit a summary report of it, every three months. Both Cooperation Group and the CSIRTs Network should be informed every six months by ENISA on significant incidents that have an impact on more than one MS.

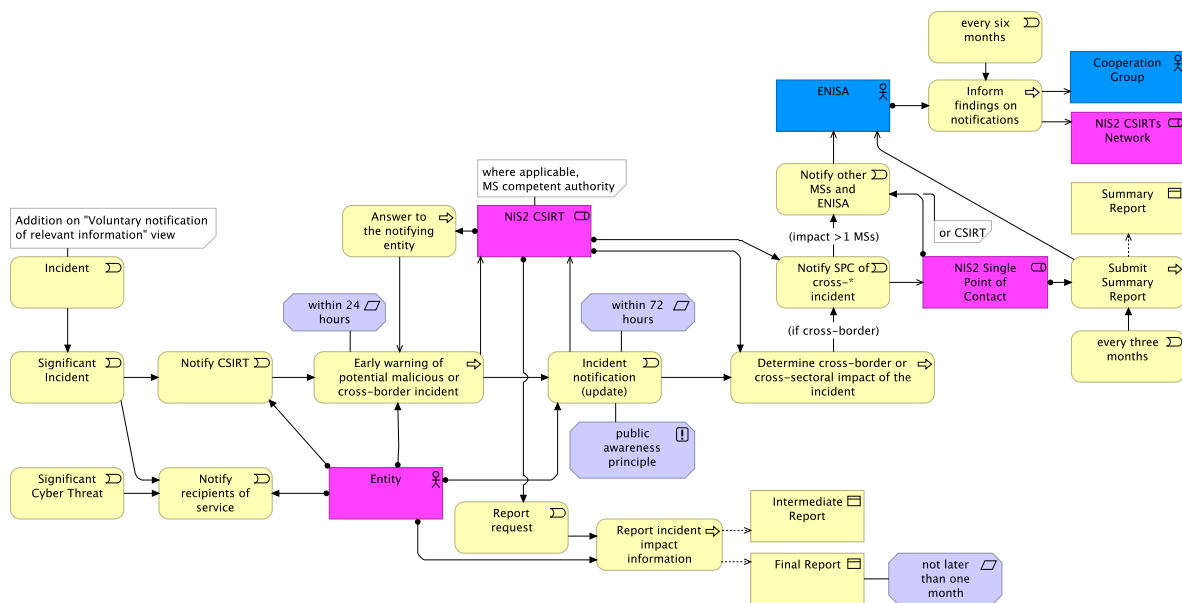


Figure 20: View of the reporting obligations

ArchiMate architectural comments:

Incident reporting follows a sequential flow, which makes it particularly well-suited for representation in ArchiMate through business events and business processes that trigger one another. In these cases, notifications are typically modeled as events, as are incidents and cyber threats.

The Entity element is modeled as an actor rather than a role, since it represents a business entity capable of performing behavior and aggregating both essential and important entities. This contrasts with the CSIRT element, which represents a responsibility for performing specific behavior that can be fulfilled by various teams, such as the national (e.g., PT) or EU-level CSIRTs, therefore represented as a business role.

Moreover, to represent conditional triggers, when needed, these relationships are named with conditions, thereby facilitating the reading of this complex ArchiMate behavioral view.

7.1.4 Cross-Border Cooperation

This subsection highlights a view (Figure 21) representing a clear example of cross-border cooperation, addressing Article 19 of NIS2 [6], the collaborative interaction for conducting a peer review. Assigned to this are the European Commission, ENISA, Cooperation Group, and the NIS2 CSIRTs Network.

A peer review involves multiple MSs working together to evaluate and enhance cybersecurity measures. Participation is voluntary, with cybersecurity experts designated by at least two different MSs conducting the reviews. They must draft reports on the findings and conclusions, with MSs able to comment and potentially publish the reports.

Prior to the commencement of the peer review, MSs may conduct a comprehensive self-assessment covering the aspects to be reviewed, such as their national cybersecurity capabilities, governance structures, and incident management frameworks. This self-assessment serves as a preparatory step,

enabling MSs to evaluate their own compliance with NIS2 requirements and identify potential areas for improvement.

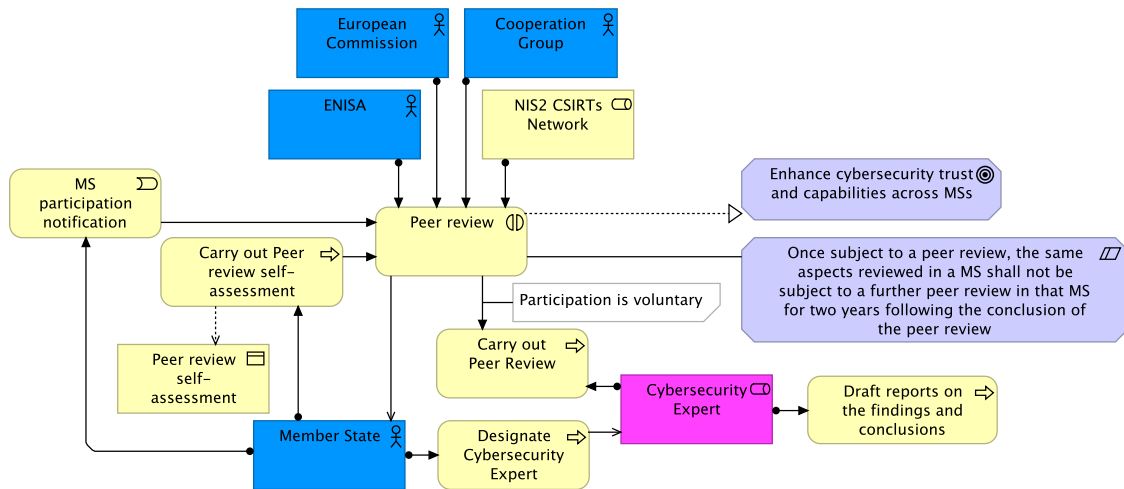


Figure 21: View of the peer reviews

ArchiMate architectural comments:

Peer review, as a unit of collective business behavior involving a collaboration of three actors and one role, is modeled as a business interaction. As such, it is both triggered by and capable of triggering other behavioral elements related to the peer review interaction. This type of collaborations and behavior aligns perfectly with the capabilities of ArchiMate.

7.1.5 CSIRTs and EU-CyCLONe

This subsection describes the establishment and coordination of CSIRTs and the EU-CyCLONe, following Articles 10, 11, and 16 of NIS2 [6]. In Figure 22, NIS2 emphasizes the establishment and coordination of CSIRTs across MSs, facilitated by ENISA. MSs are responsible for establishing CSIRTs and ensuring their effective, efficient, and secure cooperation within the CSIRTs network.

ENISA plays a crucial role by notifying the European Commission about the establishment of CSIRTs and supporting their operational deployment and coordination across MSs. In addition to its facilitating role, ENISA provides technical guidance, promotes best practices, and ensures that CSIRTs operate under consistent standards of effectiveness and security. Each CSIRT is responsible for monitoring cyber threats and incidents within its national scope, conducting incident analysis, and sharing relevant information with other CSIRTs through the CSIRTs Network. This collaborative structure enhances situational awareness, enables rapid detection and response to cross-border cyber incidents, and contributes to the overall resilience of the European cybersecurity ecosystem.

Core CSIRT processes include performing forensic and risk analysis, handling security incidents, participating in peer reviews, providing alerts and incident response, and coordinating international cybersecurity assistance. These processes ensure a robust response to cyber threats and facilitate information sharing among MSs.

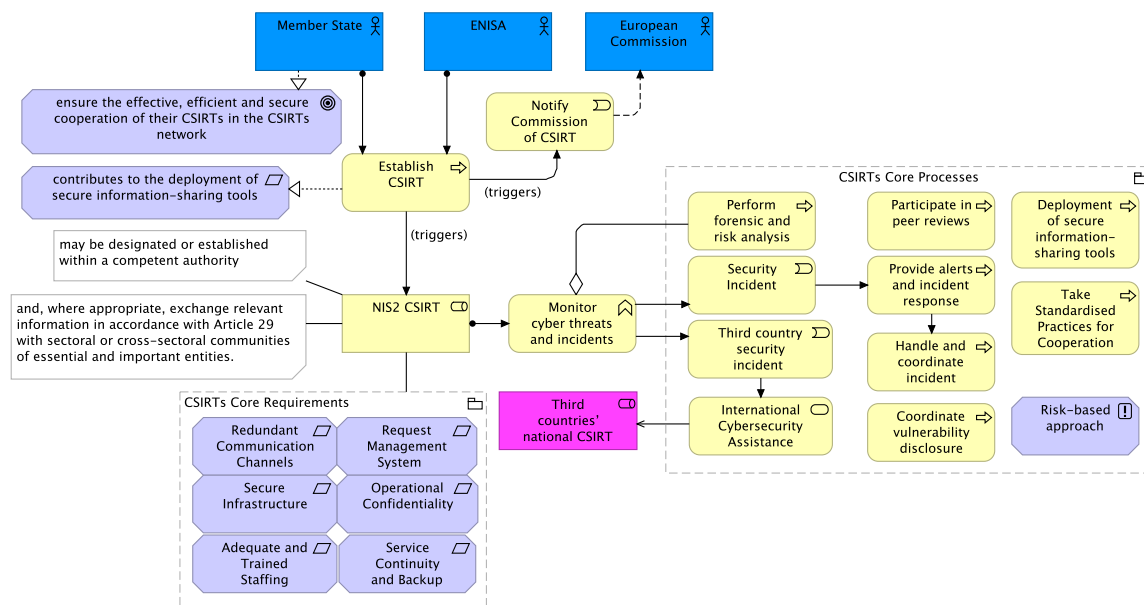


Figure 22: View of the CSIRTs

ArchiMate architectural comments:

The behavior of CSIRTs within NIS2 is extensive, so it was represented through a behavioral grouping and motivation requirements. The objective here was to provide a clear perspective on the activities of CSIRTs rather than detailing each of them, since in other, more detailed views these elements are expanded.

Within Figure 23, the view for the EU-CyCLONE framework that supports coordinated management of large-scale cybersecurity incidents, enabling regular information exchange across MSs and EU bodies.

ENISA facilitates this process by providing the secretariat, ensuring the smooth organization of meetings, documentation, and communication channels. The European Commission actively participates in situations where a potential or ongoing large-scale cybersecurity incident could have a significant impact on services covered by NIS2.

Representatives of relevant stakeholders may participate in the EU-CyCLONE process as observers, providing additional expertise and insight. Regular information exchange, procedural arrangements coordination with the CSIRTs Network, and work assessment reports submissions every 18 months ensure continuous coordination in managing cybersecurity incidents. These reports are submitted to both the Council and the European Parliament, maintaining transparency and accountability at the EU level.

EU-CyCLONE is also responsible for handling large-scale cybersecurity incidents or crises, represented in Figure 23 as a “View” element, with yellow, which refers to a supplementary view detailing the operational procedures and coordination mechanisms involved in such incidents.

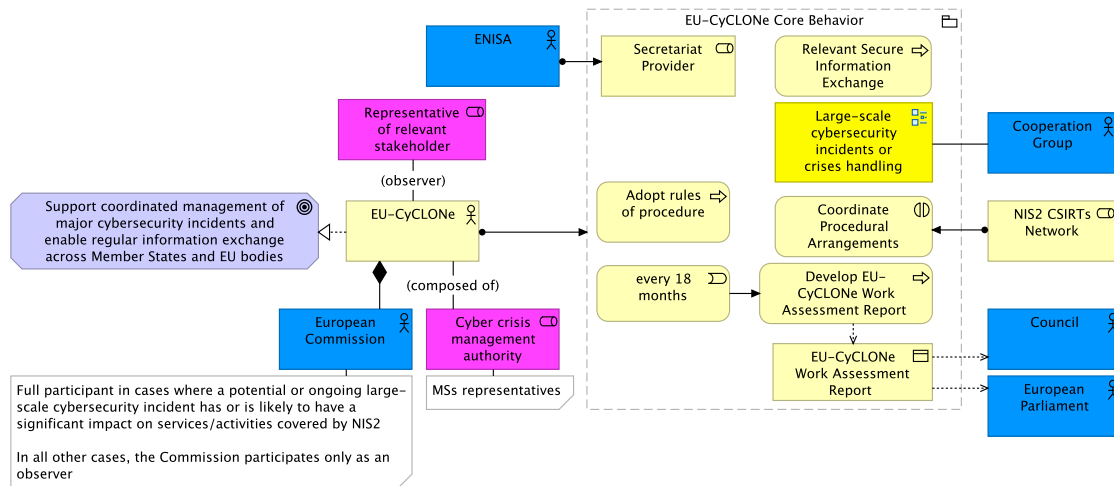


Figure 23: View of the EU-CyCLONe

ArchiMate architectural comments:

Another ArchiMate challenge is presented here: business actors cannot be composed of business roles. The intention was to represent that EU-CyCLONe is composed of both the actor European Commission and other actors holding the role of Cyber Crisis Management Authority, so a named association was used as a workaround, since it is an unspecified case. The decision to use a composition relationship between EU-CyCLONe and the European Commission is further clarified by the explanatory note included in the view.

This view also includes an example of a complementary view, highlighted with yellow, in the most suitable way found to represent this specific EU-CyCLONe activity as a “clickable” element to the complementary view.

Additionally, the Representative of relevant stakeholder is associated with EU-CyCLONe under the designation “observer”, as this was deemed the most appropriate way to model the relationship.

7.2 Architecture Description of eIDAS2

The eIDAS2 AD addresses the expanded legal and technical scope introduced by the revision of the original eIDAS. It covers the legal framework for digital identity, including its means, methods, and mechanisms for secure identification, authentication, and trust services across the EU.

Through ArchiMate, the AD translates the complex regulatory clauses of eIDAS2 into coherent views that represent its complete behavior across different contexts. It encompasses the entire scope by dividing it into smaller, well-defined legal environments, making the AD easier to interpret and navigate.

Following the subsections of eIDAS2 introduced earlier, a view for each can be extracted from the AD.

7.2.1 Electronic Identification and Cross-Border Interoperability

This subsection introduces the view (Figure 24) that represents eID and cross-border interoperability under the Article 11a of eIDAS2 regulation [111]. Cross-border identity matching is a clear example of cross-border interoperability through eID. This view shows the interaction between the user and the relying party (RP), in the context of access to a cross-border service. Users use their notified electronic identification means or EUDI Wallet instances to access cross-border services. The MS acts as an RP, ensuring cross-border identity verification and unequivocal identity matching. This process involves validating the PID of the user, providing access to the cross-border service while implementing data protection measures, such as GDPR. The MS where the eID or Wallet originates is the entity that generates and provides the PID to the RP.

For instance, a person from MS A wants to use an online service in MS B. The service in MS B asks the person to prove their identity using a means from MS A. MS B receives the PID from country A. MS B then checks that the identity data matches the person trying to access the service (this is the identity matching). If the check is successful, the person gets access to the service.

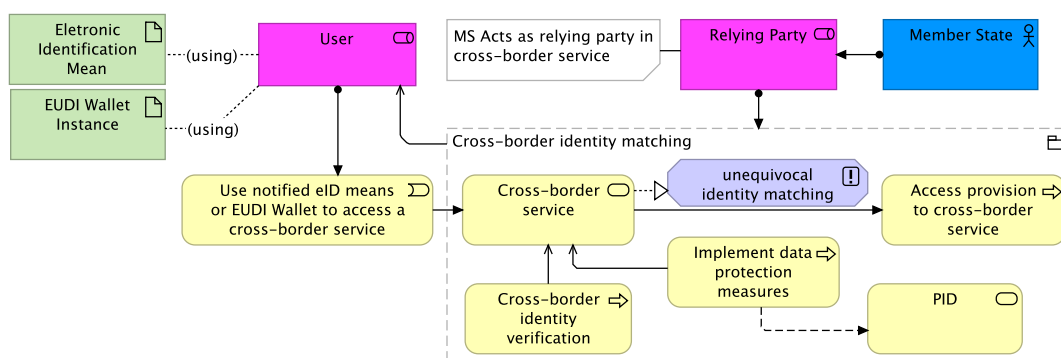


Figure 24: View of the cross-border identity matching case

ArchiMate architectural comments:

A Business Object in ArchiMate represents passive information (data, documents, messages). If PID is modeled as one, it's being reduced to just "data" sitting somewhere. But PID in eIDAS2 is not just static data; it's something that is requested, issued, validated, and consumed across borders, with formal guarantees. A Business Service represents an externally visible unit of functionality provided to customers. The PID is delivered as a service by the MS (issuer) to the user and RPs.

An electronic identification means is a material and/or immaterial unit, such as a smart card or mobile app. The EUDI Wallet instance is a software-based electronic identification means owned and controlled by the user, usually implemented as a mobile app or wallet software. In other words, both represent data used in the eID process and are therefore modeled as Artifacts in ArchiMate.

It is also important to highlight the “unequivocal identity matching” principle in cross-border identity verification, as it expresses a general property that governs this process.

7.2.2 Principle of Mutual Recognition

This subsection presents the view (Figure 25) modelling the principle of Mutual Recognition, following Article 6 of eIDAS2 [111], which underpins trust and interoperability across MSs within the eIDAS2 framework. This directly relates to the previous view, since mutual recognition is a principle behind cross-border identity matching. However, mutual recognition is accepting the ID of another country as valid based on trust, while cross-border identity matching is actively verifying the ID against foreign records.

In short, when a user from MS B attempts to access an online public service in MS A, which requires eID, mutual recognition of cross-border authentication is essential. This process ensures that electronic identification means issued by one MS are recognized and accepted by another, facilitating seamless access to services. If the eID mean is of low assurance level the recognition by MS A is optional.

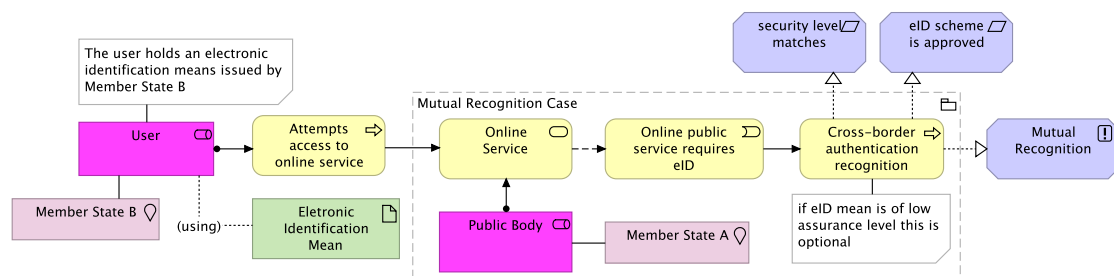


Figure 25: View of the mutual recognition principle

ArchiMate architectural comments:

In this view, it was necessary to represent two different MSs, MS A and MS B. While this could have been done using a note, the location element was used instead, as the mutual recognition process involves two physical contexts: one MS represents the location of the user, and the other MS represents the location of the online service.

7.2.3 Trust Service Providers, Trusted Lists, and Electronic Signatures

This subsection focuses on the TSP ecosystem and its connection to the trusted lists and electronic signatures defined in Articles 21, 22, 23, and 25 of eIDAS2 [111]. A view for each is extracted from the AD.

For instance, regarding the TSPs, the initiation of a qualified trust service (Figure 26) illustrates the intersection between eIDAS2 and NIS2. It begins when a TSP notifies the eIDAS2 supervisory body of its intention to provide a qualified trust service and submits a conformity assessment report issued by a CAB. The supervisory body then performs compliance verification, which includes checks for both eIDAS2 and NIS2 conformity. If required, a NIS2 compliance request is sent to the NIS2 competent authority, which must respond without undue delay and within two months.

Upon successful verification, the eIDAS2 supervisory body decides whether to grant qualified status and, if approved, must notify the European Commission to update the EU trusted lists within three months of the initial notification.

Once the trusted list is updated, the qualified trust service is activated, and the provider gains QTSP status. At this point, the provider may use the EU trust mark to clearly indicate the qualified nature of its services, in accordance with Article 23 of the eIDAS2 regulation.

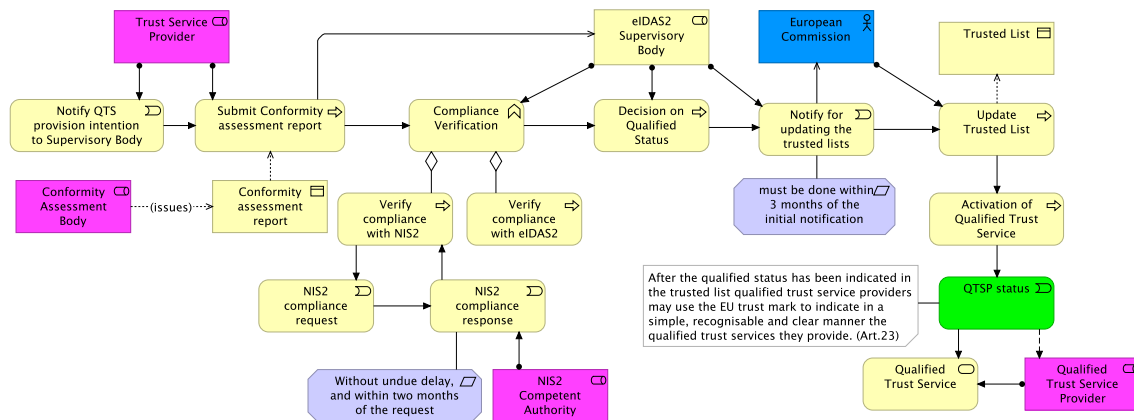


Figure 26: View of the initiation of a qualified trust service

ArchiMate architectural comments:

Here is another case that is easily represented in ArchiMate, a sequential flow of triggers between business events and processes. In this context, compliance verification is modeled as a business function rather than a process, as it represents a collection of business behavior - specifically, ensuring legal compliance with both NIS2 and eIDAS2 - and is performed as a whole. There is no such thing as partial compliance.

About the creation and maintenance of trusted lists by MSs, as highlighted in Figure 27, each MS designates a Trusted List Provider responsible for establishing, maintaining, and publishing trusted lists. These lists must be electronically signed or sealed, published securely, and in a machine-readable format. MSs notify the European Commission, which ensures EU-wide publication of trusted lists, promoting transparency and trust in eID and trust services across the EU.

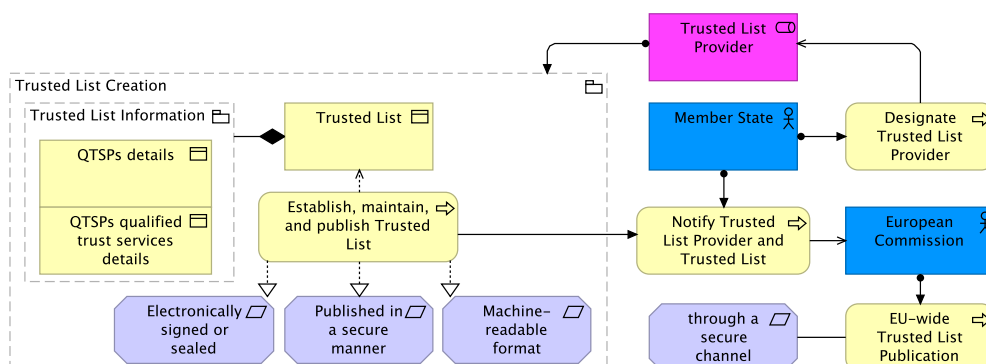


Figure 27: View of the trusted lists

ArchiMate architectural comments:

This is an example of a situation where shorter legal articles with a clear behavioral context are modeled very effectively in ArchiMate.

Furthermore, in Figure 28, eIDAS2 highlights the differentiation between electronic signatures, advanced electronic signatures, and qualified electronic signatures provided by TSPs. Every electronic signature must be non-discriminatory. QES are legally equivalent to handwritten signatures and are emitted by a QTSP. AdES, while not legally equivalent to handwritten signatures, offers a higher level of security by being uniquely linked to the signer and capable of detecting changes to the signed data. Basic electronic signatures, on the other hand, provide a simple digital mark without the stringent security features of advanced or qualified signatures.

This tiered approach ensures that users can choose the appropriate level of security and legal recognition based on their specific needs.

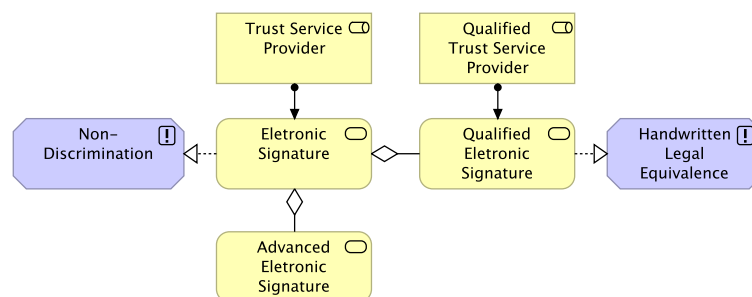


Figure 28: View of the legal effects of electronic signatures

ArchiMate architectural comments:

An electronic signature is data in electronic form used by a signatory to sign. Based on this definition, it can be considered an artifact in ArchiMate, similar to the eID Means or the EUDI Wallet instance. However, given its legal context, as it provides a unit of behavior that is meaningful from the point of view of the legal domain, it was instead modeled as a business service. This applies not only to electronic signatures but also to other trust services such as electronic seals, electronic timestamps, and more. Legally, these are services. Under eIDAS2, they define behaviors that a user, RP, or public service exposes to its environment.

This way it's highlighted how different interpretations can emerge and illustrates the flexibility of EA modelling, particularly in domains, where multiple valid representations may coexist depending on the perspective taken, and how much difficult is to model legal clauses.

7.2.4 Digital Wallet

This subsection explores views on the EUDI Wallet, the central innovation of eIDAS2, and addresses its Articles 5a and 5c [111].

For instance, Figure 29 shows that the provision and use of the EUDI Wallet are voluntary for users, who can request technical support and report incidents to the MS. MSs notify users of security breaches and manage the security of the EUDI Wallet. The wallet solution adheres to GDPR and security-by-design principles, ensuring secure, trusted, and seamless cross-border access to services while

maintaining user control over data and serving each user within their wallet instance. The source code of the wallet is open-source licensed.

MSs designate an EUDI Wallet provider that develops the EUDI Wallet, conducts validity and revocation checks, notifies the European Commission of wallet information, and provides validation mechanisms free of charge and additional functionalities. Ultimately, the Commission shall make available the information notified to the public through a secure channel, in electronically signed or sealed form suitable for automated processing.

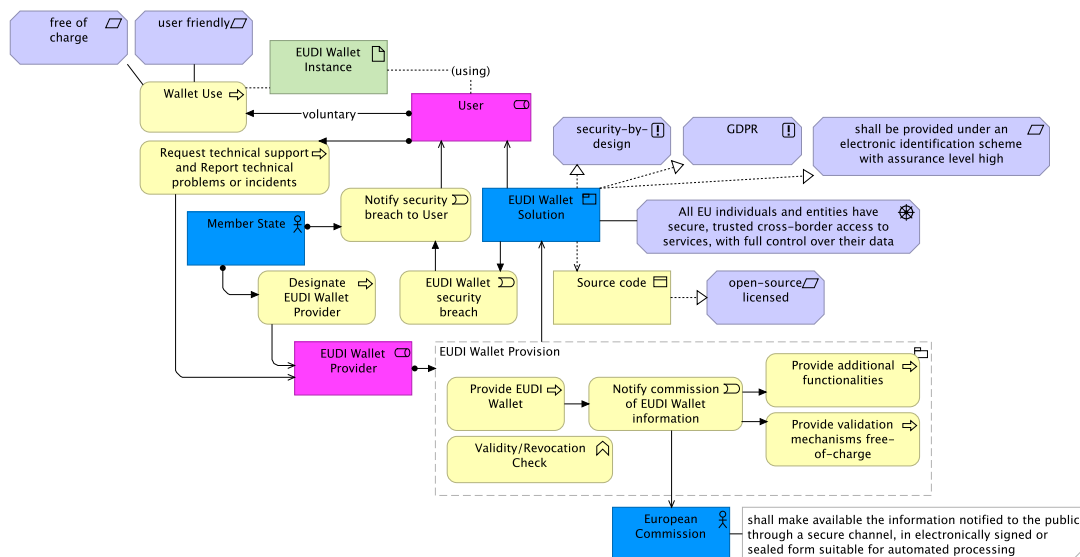


Figure 29: View of the EUDI Wallet

ArchiMate architectural comments:

This is an example of a complex and extensive translation of a legal article into ArchiMate. While it may lack some readability, it remains concise. It effectively shows the major business behaviors associated with the EUDI Wallet Solution, as well as its interactions with key actors, the user, the European Commission, and MSs. It covers scenarios such as wallet usage, provision, support, and provider designation. Although not immediately clear at first glance, the model maintains overall coherence and consolidation.

Moreover, eIDAS2 introduces a comprehensive framework for the certification and use of the EUDI Wallet, as shown in Figure 30. The certification process begins when the EUDI Wallet is ready for certification, under the oversight of a CAB designated by MSs. This certification must comply with GDPR standards. If the certification is cybersecurity-related, it must undergo a cybersecurity certification, served by an EU cybersecurity certification scheme. If this EU scheme doesn't fully cover the certification needs the national MS must establish a national scheme and transmit it to the Cooperation Group. Once the certification is issued, the certification is valid for up to five years, subject to vulnerability assessments every two years. If vulnerabilities are found, the certification can be cancelled.

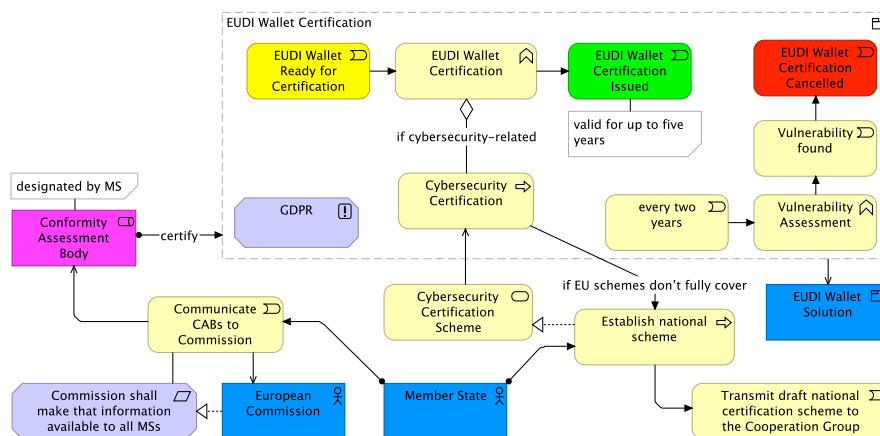


Figure 30: View of the certification of EUDI Wallets

ArchiMate architectural comments:

In ArchiMate, colors are used precisely for situations like this, where they help the reader quickly associate elements with the stages of EUDI Wallet certification - Ready, Issued, and Cancelled.

Beyond the EUDI Wallet views built directly from the eIDAS2 articles, this AD includes a dedicated Group for EUDI Wallet views based on its official ARF [91]. This addition was crucial to provide a realistic and technically grounded view of the EUDI Wallet ecosystem, since it's the primary component of eIDAS2 for secure, cross-border digital identity in the EU.

The “Solution” view, in Figure 31, is almost a general overview of the EUDI Wallet ARF. It illustrates the architecture of the EUDI Wallet. At the center is the wallet solution, which enables users to securely manage and present digital credentials, such as electronic signatures, seals, PID, and attestations of attributes, for use across major scenarios, including online services and national eID schemes. TSPs, including QTSPs, certified by the CAB, supply these digital credentials embedded in the wallet. The wallet is operated by an EUDI Wallet provider and must be certified as eIDAS2-compliant by a CAB. Users control their own wallet instance and present verified credentials to RPs (user cases scenarios). Ultimately, the wallet enables selective disclosure, secure use, identity management, and cross-border recognition, building trust across MSs.

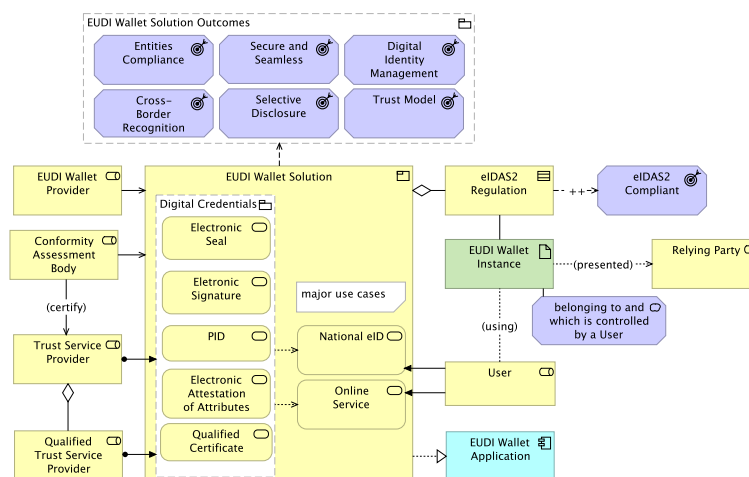


Figure 31: View of the ARF EUDI Wallet solution

ArchiMate architectural comments:

An EUDI Wallet solution refers to the complete product and service offering provided by an EUDI Wallet Provider to all users of that solution. Since it encompasses many user cases and digital credentials (business services), it is modeled as a business product, like the NIS2 NCS. This product is realized by an EUDI Wallet application component, which encapsulates the functionality of the solution in line with its technical implementation.

Nevertheless, it is important to note that, while only two user cases are referenced in this view, the EUDI Wallet solution typically supports more, depending on the specific country context.

Another example, in Figure 32, focuses on the lifecycle of the EUDI Wallet Instance, as described in the ARF, detailing the various states from installation to uninstallation. Initially, the user installs the wallet app, which transitions the wallet instance into the Installed State. Following installation, the wallet instance is activated, moving it into the Operational State where it can perform its intended functions. The wallet provider plays a crucial role in maintaining the attestation of the wallet, which can be revoked or expire, necessitating revalidation. Similarly, the PID provider issues and manages PIDs, which can also be revoked, deleted, or expire, requiring new issuance and validation. If the user decides to uninstall the wallet app, the instance transitions to the Uninstalled State, effectively ending its lifecycle.

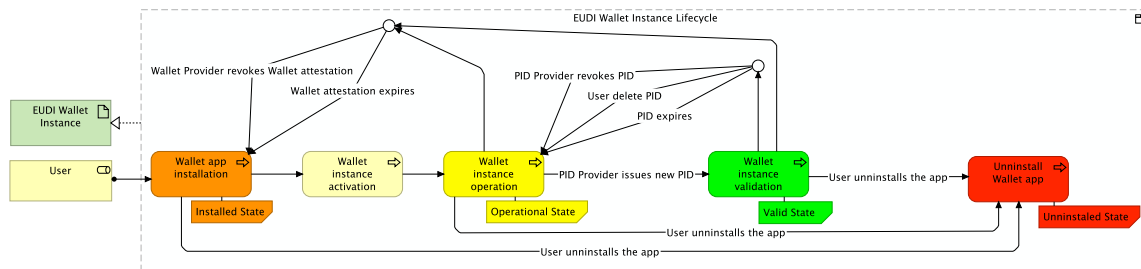


Figure 32: View of the EUDI Wallet instance lifecycle

ArchiMate architectural comments:

ArchiMate allows for modelling a conditional lifecycle. To improve readability, colored notes illustrating the states of the EUDI Wallet Instance were added, along with junctions to support the flow and enhance the overall design.

8 Demonstration

With both ADs now presented, it becomes clear that their primary purpose is to ease, structure, and clarify the business and motivational understanding of both eIDAS2 and NIS2, while also guiding enterprise architects in legal modelling. However, their utility extends further.

This section will introduce a demonstration of the work, showcasing the practicality of the results obtained from the method. Therefore, views describing practical scenarios for hypothetical stakeholders were developed (included in the v1 of the results) to demonstrate the applicability of these views in possible real-world cases, addressing concrete concerns derived from legislative requirements and enabling stakeholders to understand better, implement, and comply with NIS2 and eIDAS2 obligations. These views are built directly within the ADs, reusing some existing elements, representing extensions to the architecture of both legal frameworks. Moreover, they follow the previous [Architecture Principles](#).

8.1 Practical Scenario

The definition and presentation of practical scenarios in this section strictly follow the ISO/IEC/IEEE 42010 standard, regarding the organization of concepts. These scenarios are purely illustrative and demonstrative. They were not submitted to any of the stakeholders mentioned, although they were developed with the awareness that such cases could be relevant and of potential interest for them. Therefore, the process began with identifying hypothetical stakeholders, a critical step in ensuring the relevance and success of these practical scenarios.

Then, efforts were made to address their possible key interests and specific concerns. Afterwards, these were grouped to help define relevant viewpoints through classification. Subsequently, the legal environment was modeled by incorporating elements from the respective legislation AD. Each view was carefully designed to address all identified concerns tied to its viewpoint.

Once the views of practical scenarios were developed, the refinement process started, incorporating necessary changes and addressing any newly identified concerns. This approach ensured that the final views for practical scenarios were fully aligned with stakeholder expectations and requirements. The design of each practical scenario followed a “high-level to detailed” approach, adding elements as needed to capture specific details. Each scenario view is documented, independent, and illustrates a model that facilitates understanding of legal compliance for the stakeholders.

Except for the final practical scenario on cybersecurity maturity, which differs from the others as it was shaped by the interests of a stakeholder without specific national concerns, all the scenarios involve the Portuguese legal environment, since they were shaped by the interests of Portuguese stakeholders with national concerns. The developed views reflect the national context, addressing how key actors interpret and respond to the requirements of NIS2 and eIDAS2, with particular attention given to authorities and bodies actively engaged in digital identity and cybersecurity implementation.

Important to underscore that when viewpoints are not well defined, or not used: stakeholders get confused, some will demand too much detail and others will complain about missing info; regulatory obligations might be misinterpreted; important concerns may be ignored; effort is wasted; is harder to

scale and maintain compliance, because every new entity or change needs reinterpreting without a coherent lens. With well-established viewpoints, views are easier to design and align, leading to a consolidated AD that supports every perspective from the stakeholders.

8.2 NIS2 Framework in Portugal

To provide a clearer understanding of how cybersecurity governance is organized in Portugal under the NIS2 Directive, the following scenario, illustrated in Figure 33 and Figure 34, outlines the Portuguese structure of cybersecurity entities. This representation builds upon the context introduced in 2.3.3.

CNCS, which operates under GNS, serves as the central authority for cybersecurity, therefore being the major stakeholder. CNCS plays a pivotal role in implementing NIS2 at the national level, ensuring the security and resilience of critical sectors, and fostering cooperation between public and private stakeholders. It oversees the main CSIRT for Portugal, CERT.PT, which is part of both the national CSIRTs network (RNCSIRT) and the European CSIRTs Network (CERT-EU), ensuring effective information sharing and joint response across national and European cybersecurity communities.

Major Stakeholder: CNCS

Viewpoint:

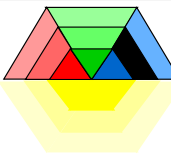
NIS2 Framework Viewpoint		
Stakeholders	1. CNCS	
Concerns	1. Compliance with the NIS2 directive at the national level 2. NIS2 Portugal environmental context 3. Major Portuguese actors and roles for NIS2	
Purpose	Informing	
Abstraction Level	Coherence	
Layer	Business	
Aspects	Active structure, Motivation	

Figure 33: Viewpoint of the NIS2 framework in Portugal

View:

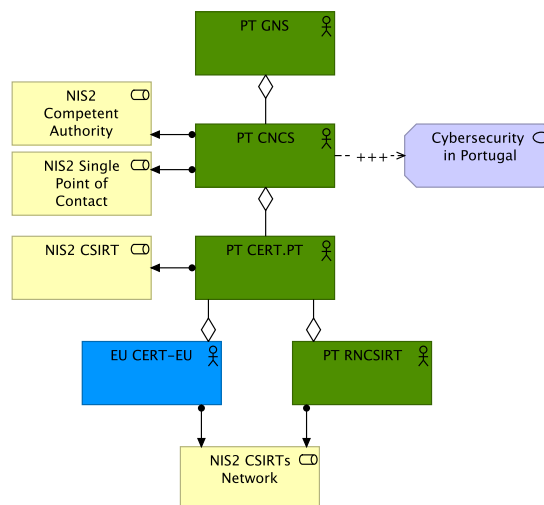


Figure 34: View of the NIS2 framework in Portugal

ArchiMate architectural comments:

Green is used for the Portuguese elements. The structured hierarchy is represented using the aggregation relationship, as in all cases, the source element aggregates the entire (whole of) target concept. The composition relationship was not used, since the elements on the aggregating side are not strictly composed of the aggregated ones.

8.3 Cybersecurity Perspective of the 2025 Blackout

The 2025 blackout event scenario, in Figure 35 and Figure 36, offers a practical context for examining how the Portuguese compliance with NIS can strengthen national resilience and compliance with EU cybersecurity standards, since the directive has been approved by the Portuguese Parliament but is not yet fully in force, awaiting official publication in the *Diário da República*. Therefore, this scenario represents an example of how the blackout events can be aligned with NIS2 and be useful for an entity like CNCS.

Portugal experienced a historic blackout that began on April 28 at 11:30 AM, causing disruption across transport, telecommunications, and healthcare. The CNCS quickly ruled out a cyberattack and coordinated with Spain, France, the European Commission, and ENTSO-E to investigate. On April 29, the civil group Citizens for Cybersecurity (CpC) reported the alleged sale of access to energy infrastructure on the dark web, prompting a formal complaint and requests for telecom accountability. By April 30, CNCS reiterated the absence of malicious activity, while INCIBE, from Spain, began a technical investigation with Portuguese authorities. On May 1, Anacom issued recommendations to improve emergency communications, and the government ordered an urgent review of the SIRESP system ^{21 22 23 24 25}.

On May 9, the report from ENTSO-E confirmed that frequency instabilities in Spain triggered the blackout, with no evidence of cyber compromise. In parallel, Portugal advanced the transposition of the NIS2 Directive, reinforcing obligations for critical infrastructure, and on June 25, the government unveiled new energy security measures, including expanded "black start" capabilities and battery installations in key facilities.

Furthermore, representing the blackout scenario in an ArchiMate view is valuable because it demonstrates how consolidated EA ADs can extend real "compliance in practice". The visual model consolidates and adapts NIS2 requirements and the event timeline of the blackout into a clear and single report, showing interactions according to the legislation. This approach allows for a single visual reference of a large incident handling in a national context and its understanding, as well as supports

²¹ ENTSO-E & National Authorities. (2025, April 30). Grid Incident in Spain and Portugal on 28 April 2025 - Incident Report. Brussels: ENTSO-E.

²² Reuters. (2025, April 28). Portugal's REN says no sign blackout caused by cyberattack. London: Reuters.

²³ CNCS. (2025). Relatório sobre a coordenação nacional no contexto do apagão de abril de 2025. Lisboa: Centro Nacional de Cibersegurança.

²⁴ Anacom. (2025). Recomendações para comunicações de emergência após o apagão de abril de 2025. Lisboa: Autoridade Nacional de Comunicações.

²⁵ Governo de Portugal. (2025, Junho 25). Medidas de reforço da segurança energética nacional. Lisboa: Ministério do Ambiente e da Energia.

compliance analysis, and helps identify gaps or improvements in national resilience and cybersecurity preparedness.

Major Stakeholder: CNCS

Viewpoint:

Cybersecurity Perspective of 2025 Blackout Viewpoint	
Stakeholders	1. CNCS
Concerns	1. Overall response of portuguese entities to the Blackout 2. Core behavior and timeline of events 3. Systemic risk of large-scale power outage and its cascading effects on critical sectors. 4. Incident response coordination and communication between national and EU authorities
Purpose	Designing
Abstraction Level	Coherence
Layer	Business
Aspects	Active structure, Behavior, Passive structure

Figure 35: Viewpoint of the cybersecurity perspective of the 2025 blackout

View:

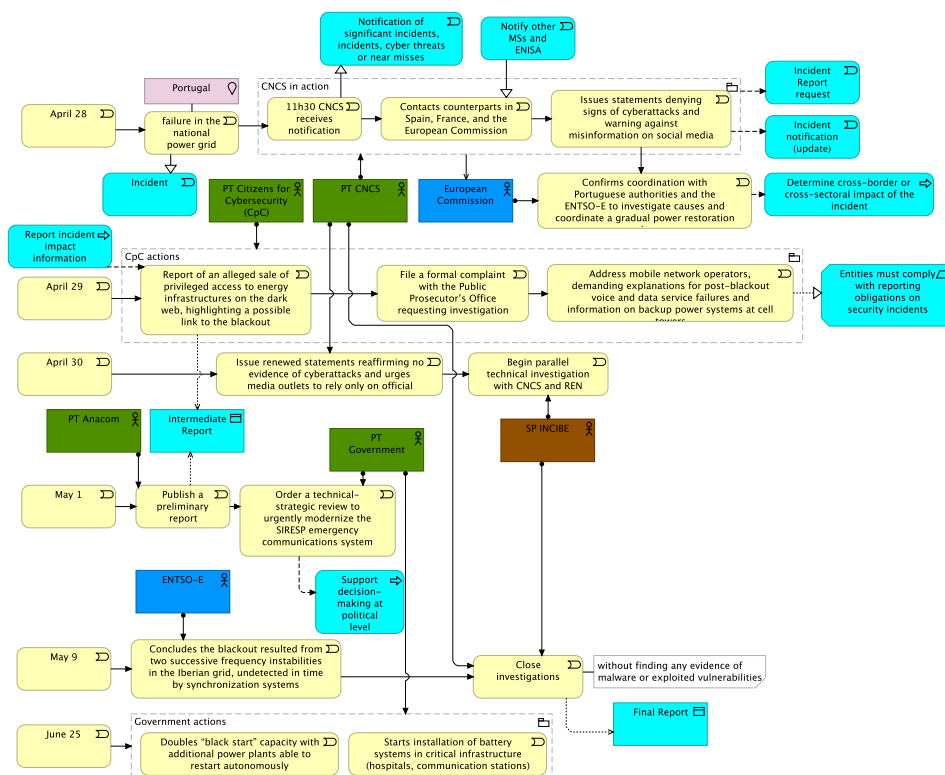


Figure 36: View of the cybersecurity perspective of the 2025 blackout

ArchiMate architectural comments:

The Blackout behavior represents a long sequence of events over multiple days, so separate flows were created for each day. This separation supports a more accurate and simple representation of the operational timeline. NIS2 elements are highlighted in **cyan** to clearly illustrate the elements from the original AD, serving as a guiding framework for these cases. All non-cyan elements represent business events, as they are not specific processes described in the directive.

Moreover, implementing this complex case shows how ArchiMate can model multi-day, multi-actor operational events, capturing both the temporal sequence and interactions between national authorities, international bodies, and civil groups, while maintaining clarity in the representation of cascading effects and responses, although the layout lacks some readability.

8.4 eIDAS2 Framework in Portugal

The scenario represented in Figure 37 and Figure 38 presents an architectural overview of the eIDAS2 framework in Portugal, similar to what was done for NIS2. Portugal is actively implementing the eIDAS 2.0 framework. The country has deployed its eIDAS-Node, facilitating interoperability with other EU MSs systems. Additionally, the Portuguese National Digital Identity, *Chave Móvel Digital* (CMD), aligns with eIDAS 2.0 standards, offering citizens a secure means of authentication and digital signature. The government has also certified several QTSPs, ensuring the availability of trusted electronic services ²⁶ ²⁷ ²⁸.

As depicted in the view, *Agência para a Modernização Administrativa* (AMA) is the eIDAS2 National Authority and SPC, reason for being the major stakeholder of this view. AMA coordinates with GNS, which has the eIDAS2 supervisory role. It is also responsible for both the supervision and development of the Portuguese Digital Identity Wallet and the Portuguese electronic identification scheme, CMD.

Relevant relationships to the Portuguese eIDAS2 scenario include: TSP is supervised under the GNS and certified by a CAB, which is accredited and monitored by the IPQ, the National Accreditation Body (NAB) of Portugal. GNS create and send a report summary of trust services notifications, from TSPs, to both ENISA and CNPD. CABs perform compliance assessments to the Portuguese Digital Identity Wallet. Furthermore, AMA is connected to broader European oversight through the European Commission and other MSs, aligning with eIDAS2 interoperability.

Major Stakeholder: AMA

Viewpoint:

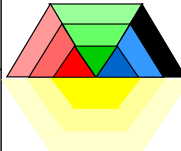
eIDAS2 Framework Viewpoint			
Stakeholders	1. AMA		
Concerns	1. Compliance with the eIDAS2 regulation at the national level 2. eIDAS2 Portugal environmental context 3. Major Portuguese actors and roles for eIDAS2		
Purpose	Informing		
Abstraction Level	Coherence		
Layer	Business		
Aspects	Active structure, Behavior, Passive structure		

Figure 37: Viewpoint of the eIDAS2 framework in Portugal

²⁶ Gabinete Nacional de Segurança. (2023). Regulamento eIDAS. Lisboa: GNS.

²⁷ Diário da República. (2021, September 10). Decreto-Lei n.º 88/2021. Lisboa: Imprensa Nacional.

²⁸ European Commission. eIDAS Dashboard.

View:

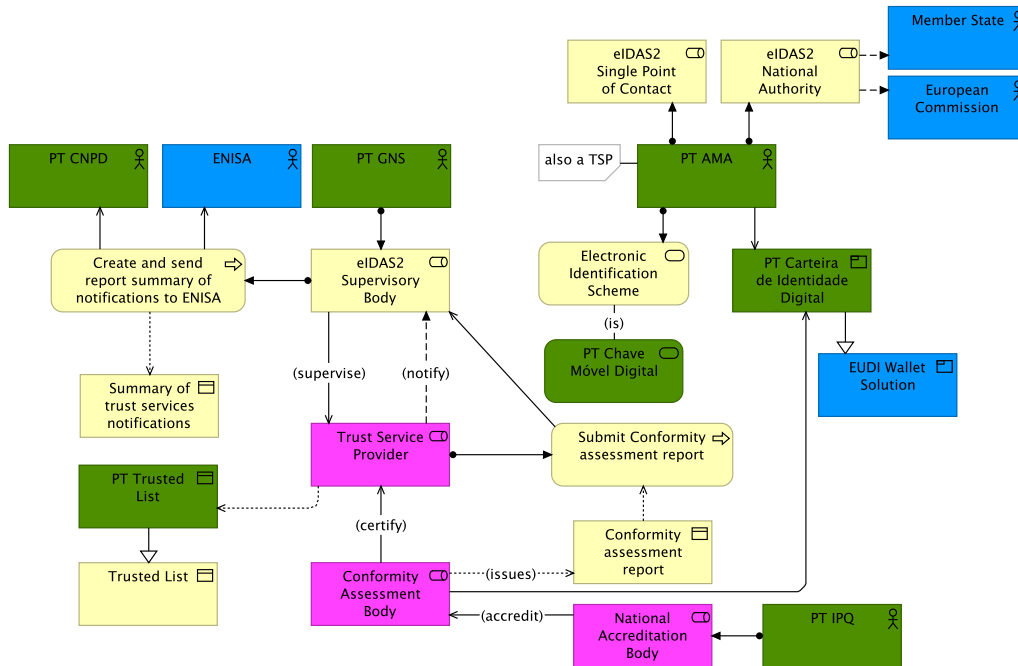


Figure 38: View of the eIDAS2 framework in Portugal

ArchiMate architectural comments:

The flow relation from AMA to MS and the European Commission indicates an information flow or interaction (not control), showing data/report flow.

Although cases like the accreditation of the CAB by the NAB could have been modeled as business processes, a more generic representation was used to maintain a high-level perspective in this view.

8.5 Strategic view of AMA in the context of eIDAS2

As previously mentioned, AMA serves as the central authority for digital identity and authentication in Portugal, fulfilling the roles of the eIDAS2 National Authority and Single Point of Contact. Therefore, a specific scenario, illustrated in Figure 39 and Figure 40, was developed to demonstrate its main responsibilities within eIDAS2 and the EUDI Wallet.

AMA is shown as the key actor driving the modernization of public administration. It manages critical Portuguese digital systems such as *Autenticação.gov*, CMD, and the Digital Identity Wallet. The roadmap highlights a phased approach: participation in EU pilots and requirement definition (2023 - 2024), prototype development and pilot testing with real users (2024 - 2025), and the public launch of the Portuguese Digital Identity Wallet in late 2025 or early 2026, demonstrating progressive alignment with eIDAS2 ²⁹.

Major Stakeholder: AMA

²⁹ Agência para a Modernização Administrativa (n.d.). AMA.

Viewpoint:

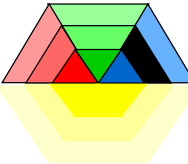
AMA Viewpoint		
Stakeholders	1. AMA	
Concerns	1. AMA general overview and major motivations 2. Alignment of AMA's mission with eIDAS2 and Digital Wallet adoption in Portugal 3. AMA's digital identity key systems (e.g., Autenticação.Gov, Chave Móvel Digital, etc.)	
Purpose	Informing	
Abstraction Level	Overview	
Layer	Business, Application	
Aspects	Active structure, Behavior, Passive structure, Motivation	

Figure 39: Strategic viewpoint of AMA in the context of eIDAS2

View:

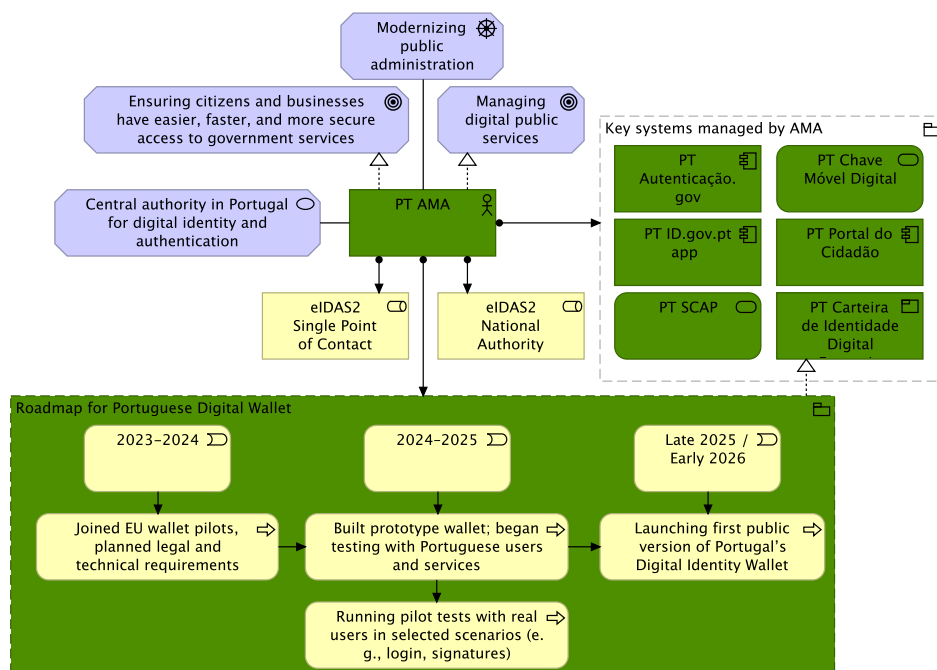


Figure 40: Strategic view of AMA in the context of eIDAS2

ArchiMate architectural comments:

AMA is the central element of the view, so all the other elements were built around it. A simple behavior roadmap was added to represent the latest events in the timeline of the Portuguese Digital Identity Wallet, the most relevant upcoming eIDAS2 implementation from AMA.

Autenticação.gov, the id.gov.pt app, and *Portal do Cidadão* are application components because they are self-contained, modular units that provide user-facing functionality, while *Sistema de Certificação de Atributos Profissionais* (SCAP) and CMD are application services because they offer reusable capabilities that these application components rely on or expose through interfaces.

8.6 Portuguese Digital Identity Wallet

The following scenario, shown in Figure 41 and Figure 42, presents the Portuguese Digital Identity Wallet, jointly supervised by AMA and INCM, as a central component of the broader EUDI Wallet Solution. The wallet will support key data objects such as the national eID, mobile driving license, diplomas, etc. ^{30 31}

The European Commission funded several large-scale pilot projects to test real-world uses of the EUDI Wallet. Portugal is involved in two main pilots: POTENTIAL and EWC. Aligned with them, the solution focuses on enabling strong authentication, qualified electronic signatures, cross-border interoperability, and personal data control, meeting eIDAS2 requirements. The view also shows potential integration with national systems like *Autenticação.gov* and CMD, with pilot tests scaling to public release by late 2025 or early 2026. This positions Portugal to meet the mid-2026 deadline from the EU for launching a compliant EUDI Wallet.

Major Stakeholder: AMA

Viewpoint:

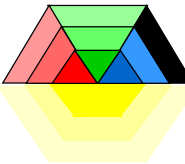
Portuguese Digital Identity Wallet Viewpoint			
Stakeholders	1. AMA		
Concerns	1. Portuguese implementation of eIDAS2 Digital Identity Wallet primary elements 2. Current status 3. Major services/user cases		
Purpose	Informing		
Abstraction Level	Overview		
Layer	Business, Application		
Aspects	Active structure, Behavior, Passive structure, Motivation		

Figure 41: Viewpoint of the Portuguese digital identity wallet

View:

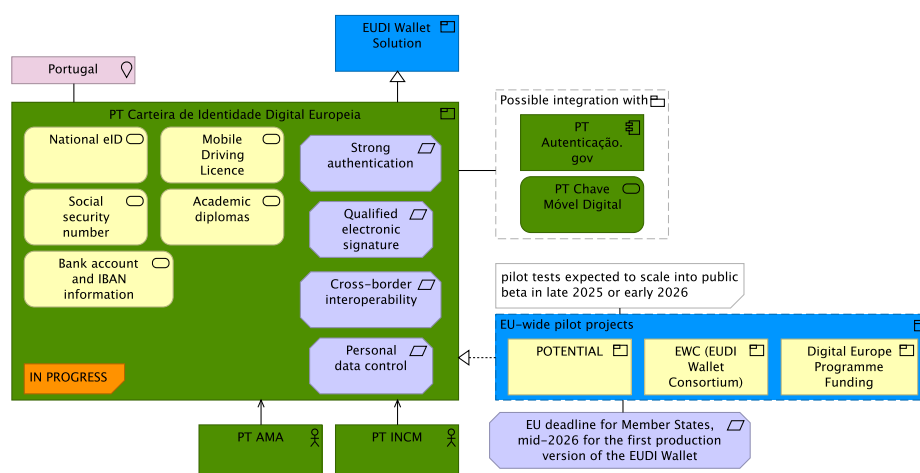


Figure 42: View of the Portuguese digital identity wallet

³⁰ European Commission. (2025). EU Digital Identity Wallet Pilot implementation.

³¹ Agência para a Modernização Administrativa. (n.d.). Carteira de Identidade Digital Portuguesa.

ArchiMate architectural comments:

The large-scale pilot projects are carried out to test real-world use cases of the Portuguese Digital Identity Wallet. Therefore, the realization relationship appropriately indicates that the wallet implements the results derived from these projects.

A possible integration with other Portuguese digital identity services is currently unspecified. Thus, an association relationship is used. This represents another modelling challenge in ArchiMate, where certain design decisions require abstraction due to a lack of specification detail.

8.7 Maturity Assessment Integration with NIS2

The scenario represented in Figure 43 and Figure 44 illustrates a cybersecurity maturity self-assessment model aligned with the principles of the NIS2 Directive, developed for a possible MSc thesis of a student from *Instituto Superior Técnico* (IST) wanting to integrate this concept within the directive. This directly relates to the view from the NIS2 AD on the report on the state of cybersecurity in the Union (Article 18), illustrated in Figure 45, representing an extension example of how legal compliance and detail can be improved.

Official documents, such as the NIS2 Directive itself, guidelines, or the domain-specific outputs of ENISA, do not define or impose a standardized maturity model. Therefore, the maturity levels (1–5) are only an example of the progressive sophistication of cybersecurity capabilities across the Union. At the national level, MSs, via their NIS2 Competent Authorities, oversee entities in carrying out cybersecurity maturity self-assessments. These entities, classified as either Important or Essential, evaluate themselves against defined the maturity levels. These self-assessments are aggregated within a report submitted by the ENISA to the European Parliament.

ENISA, together with the European Commission, the Cooperation Group, and the CSIRTs Network, aggregates these national results into a Union-level assessment. Every two years, ENISA presents its report on the state of cybersecurity to the European Parliament, using a common methodology and indicators to track progress and guide policy improvements.

Major Stakeholder: MSc Student from IST

Viewpoint:

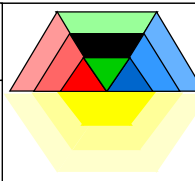
Maturity Viewpoint			
Stakeholders	1. MSc Student from Tecnico		
Concerns	1. NIS2 possible integration with maturity levels 2. Understanding of cybersecurity maturity self-assessment behavior 3. EA diagram for MSc Thesis		
Purpose	Deciding		
Abstraction Level	Coherence		
Layer	Business		
Aspects	Active structure, Behavior, Passive structure, Motivation		

Figure 43: Viewpoint of the maturity assessment integration with NIS2

View:

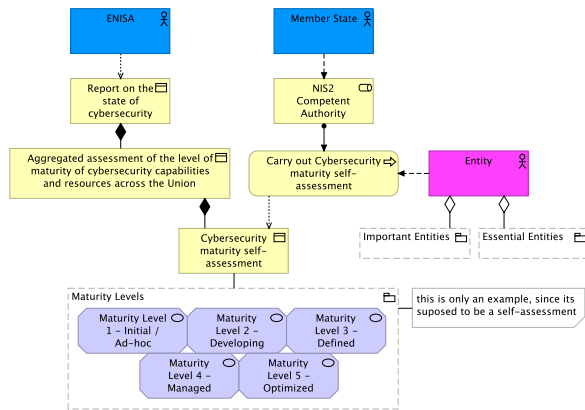


Figure 44: View of the maturity assessment integration with NIS2

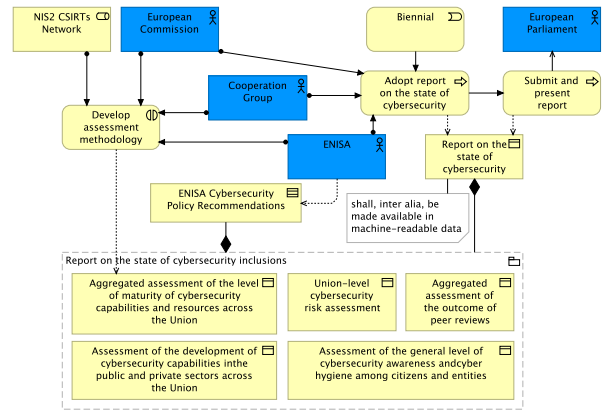


Figure 45: View of the report on the state of cybersecurity in the Union

ArchiMate architectural comments:

The Maturity Levels are modeled almost as a “catalog” or “value set,” but in ArchiMate terms they could be represented as assessment or goal elements, showing that entities aim to progress toward higher maturity. However, since they constitute value of information and knowledge representing advantage, benefit, and desirability for both the Entities and European actors, they are modeled as motivation values.

Overall, the demonstration confirms that the ADs are not only theoretical constructs but also practical tools for clarifying legislative requirements, aligning stakeholders, and supporting compliance with NIS2 and eIDAS2.

Modelling real-world contexts within national frameworks for hypothetical legal scenarios shows how different stakeholders can extend these ADs into their “own practical scenarios”, adapting the legislation to their business environment and representing it clearly in a view, facilitating legal compliance.

9 Validation and Documentation

Exploratory work with AI was conducted on the validation and documentation of ArchiMate ADs. Two preliminary and complementing approaches were explored: a unit-based review with ChatGPT and an AI-enhanced JArchi integration with LLaMA and Gemini. These experiments did not aim to replace formal validation but to test experimental methods that point towards possible future directions of EA.

At this point, AI is still not advanced enough to be fully integrated with complex EA models, especially in ArchiMate. Even when provided with significant context, AI can diverge significantly, and in difficult scenarios with multiple views, it can easily get lost. However, building on small, simpler cases could make it possible to validate and document large ADs.

Therefore, to address this question, this section presents the attempts and resources examined, which led to new versions of the consolidated ADs.

9.1 Unit-Based Review with ChatGPT

First, an exploration with ChatGPT 5, from OpenAI, was conducted through unit-based reviews. In this approach, small “units” of the AD, such as views linked to a single article, were selected and provided to the AI together with their traceability information and context. The goal was to test whether ChatGPT could analyze these units step by step and suggest improvements, rather than attempting to process the full model at once.

To represent this technique in the context of ISO 42010, an extension diagram (using ArchiMate elements) was created based on its original conceptual model in Figure 5. The illustration in Figure 46, represents how AI is integrated into the ISO 42010 conceptual model to support unit-based validation of the AD. At the top, the diagram shows the process flow: starting from a specific given context (legal text, traceability information, and explanations), the information is sent and processed by ChatGPT 5. This feeds into a review response, which provides the actor with the basis to carry out unit validation, checking the quality and consistency of the given architectural view or “unit” and updating if necessary. Once it is done for every unit, the whole AD is AI-validated.

Moreover, this represents a way of making the best use of AI: by concentrating on small, well-defined contexts where all relevant information is consolidated, it maximizes the quality and relevance of AI feedback. The iterative, view-by-view review ensures that improvements accumulate systematically, resulting in a significant enhancement in the overall validation, completeness, and coherence of the AD. Beyond validation, this method also applies to documentation, as each AI-assisted unit review adds an additional dimension to the elements, establishing a clear record of architectural decisions and adjustments.

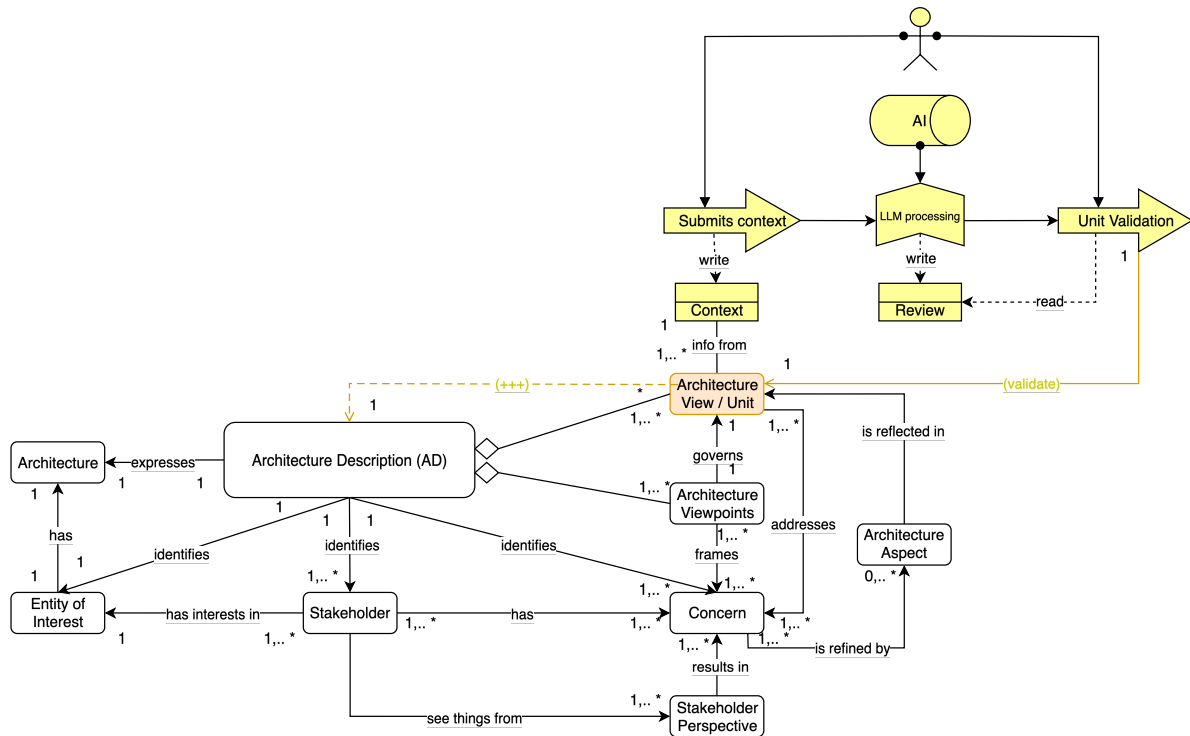


Figure 46: Extension to the ISO 42010 conceptual model of an AD

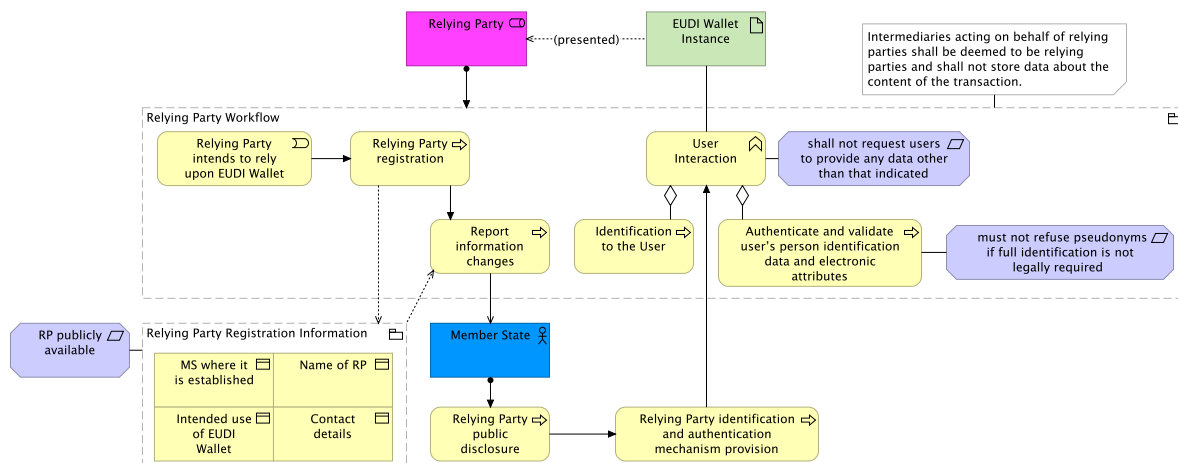


Figure 47: View of EUDI Wallet relying parties (v1)

From this point, unit-based validation with AI is exemplified to show how it was used to refine certain views of the first consolidated version of the AD (v1). Initially, a simple view representing a small legal article was selected. The chosen view, Figure 47, comes directly from the eIDAS2 AD: “EUDI Wallet-Relying Parties” (Numbering 2.1.5b, representing Chapter 2, Section 1, Article 5b).

The XML (.archimate file) of this view was provided to ChatGPT 5, along with the corresponding entries from the traceability matrix, the original legal text, and a contextual explanation of how the numbering system works. Then, it was written what the AI should analyze. The instructions were for the AI first to generate a brief overview paragraph, then produce a small table linking the legal clauses in the matrix to the respective elements. Next, it should assess the validity of the view and whether it strictly follows the legal text. Finally, it was asked to provide ArchiMate-specific comments on what

aspects of the view are well-modeled and what should be updated. In Figure 48, one of the used prompts is shown.

You are given a XML of an ArchiMate view from an Architecture Description (AD) of eIDAS2 regulation, representing the Article "EUDI Wallet Relying Parties" and an excel traceability matrix sample linking the elements in the view to the article legal clauses in the eIDAS2 regulation. The numbering system in the matrix assigns a unique hierarchical ID to each clause (e.g., chapter, section, article, paragraph, subparagraph, point, sub-point). Each AD element may map to one or more IDs.

Tasks:

- Generate a brief overview paragraph of the view.
- Produce a small table linking each legal clause in the matrix to the respective AD element.
- Assess the validity of the view: does it follow the legal text strictly?
- Provide ArchiMate-specific comments on what is well-modeled and what should be updated.

Excel traceability matrix sample: <...>

eIDAS2 "EUDI Wallet-Relying Parties" article legal text: <...>

Annexes: XML and PNG of "EUDI Wallet-Relying Parties" view

Figure 48: ChatGPT 5 provided context example

The response to this prompt can be seen in Annex B. On the first task, requiring a short description of the view, as expected, ChatGPT 5 was able to capture the essence of the view well. Attention should be drawn to its reference to constraints modeled as requirements. This is a typical case where legal text can be difficult to model in ArchiMate, since the constraint it mentions - "shall not request users to provide any data other than indicated" - does not represent a limitation on aspects of the "User Interaction" business function, but rather a statement of need defining a property that applies to this function, therefore, a requirement.

Regarding the second task, the generated clause-to-element table was very accurate. It correctly divides the content into three main columns: legal clause ID, clause text, and mapped AD element. There is only one minor misalignment. For instance, it does not include 5b (2a), the information necessary to authenticate to EUDI Wallets, mapped to the "Relying Party registration information" element. This represents only a small deviation, as the overall alignment was very good. Notably, the analysis also correctly identified the absence of 5b (4) and 5b (11) clauses from the view, which had not been represented due to their limited relevance in the architectural perspective.

The validity assessment also provided useful insights. Since the view was already a consolidated representation, only a few contrary observations were made, which confirmed its correctness. As highlighted in the second task, the missing clauses 5b (4) and 5b (11) from the view, and, in addition, the missing link to 5b (2) "proportionality" requirement were noted. Moreover, ChatGPT 5 mentions that some aspects of the duties of MSs are only partially captured. This is not particularly true since the view captures most of the key obligations in Article 5b: registration, required information, data request limitations, user identification, pseudonym handling, public availability, and intermediary responsibilities, and MSs duties are not further elaborated in the article. When AI refers to "partial", this is, for example, the case with the "publicly available" requirement, which does not mention that the availability must be in electronically signed or sealed form suitable for automated processing, as specified in 5b (5). This represents a relevant point that could have been overlooked by the architect. Nevertheless, the numbering system adequately covers most of these cases.

As for the ArchiMate-specific feedback, the comments of the AI focused on the improvements: modelling the omitted legal provisions mentioned on the third task, clarifying certain elements (e.g.,

representing RP information as a Business Object), and representing intermediaries more formally instead of using a note. While the level of detail in the ArchiMate critique was not very deep, this is likely due to the prompt design. With more precise, detailed instructions, the analysis could yield stronger modelling recommendations. Nevertheless, the results were positive, as these are indeed valid points to take into consideration and may not be immediately clear to someone reviewing the view. However, their implementation ultimately depends on architectural decisions, since they represent optional considerations. That said, the ones presented by the AI were considered a plus and correct; therefore, the view was updated.

About the positive points, they were also well observed. For instance, the comment “the separation of registration information into attributes is well-structured and traceable to the legislation” is valuable because it highlights the good practice of logically separating data into distinct attributes. Complementing this, the suggested improvement to model the same object as a data object rather than a grouping shows very promising potential for the AI.

Ultimately, this shows how, by using AI within legal and architectural “units” that represent small environments, it is possible to gradually improve a large and complex model. Therefore, this approach with ChatGPT was applied after the first complete draft of the ADs (v1). The AI was used to review and refine some selected views of this draft, following a similar procedure as the one described, which contributed to the development of an improved version **v2** (Figure 49).

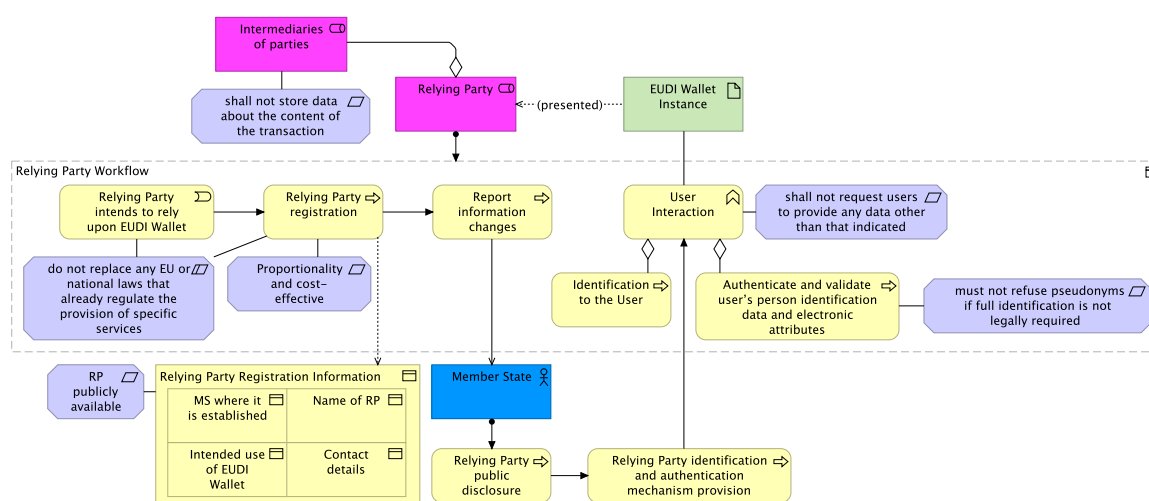


Figure 49: Updated view of EUDI Wallet relying parties (v2)

As a bonus in this version, some AI considerations were also requested on layout validation. For any enterprise architect, it is important that a view is clear, concise, and well-centralized. The recommendations from AI proved valuable in addressing spatial organization and readability issues, successfully guiding improvements in problematic line crossings, element sizing, and visual grouping. However, the analysis also revealed a critical limitation: the AI struggled to interpret some enterprise architecture practices, particularly with respect to ArchiMate notation standards. The outcome was the version **2.1** of this view (Figure 50).

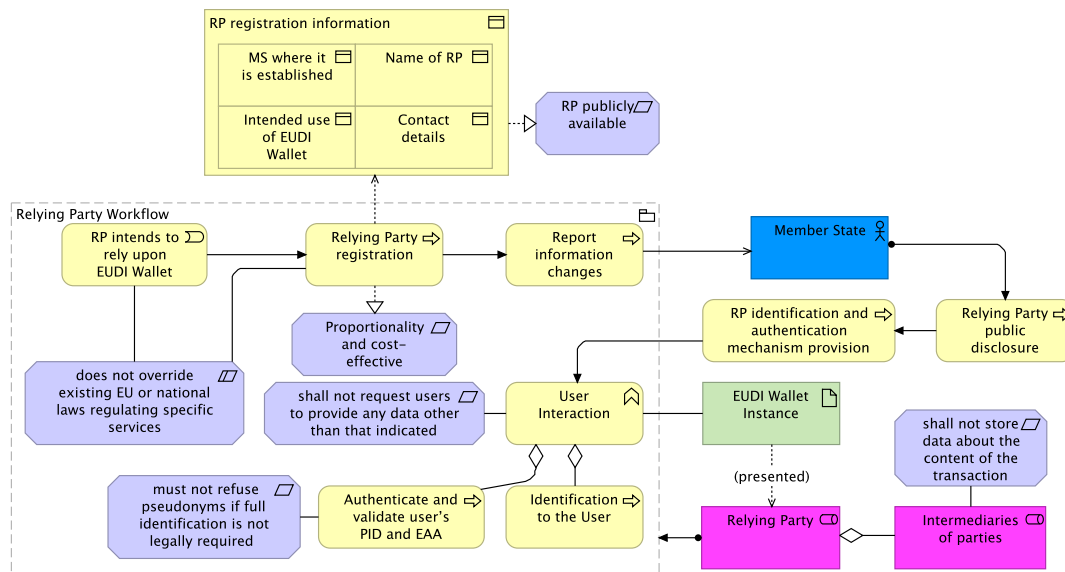


Figure 50: Updated view of EUDI Wallet relying parties (v2.1)

For the version 2 of this view, ChatGPT was asked to produce documentation to assess the viability and possibilities of adding comments to the views automatically, using the unit approach, based on the legal text and its knowledge of the traceability matrix and numbering system. Some numbering omissions were detected, but minor and not critical. Overall, the documentation provided by ChatGPT was on point - concise and assertive (see the documentation task in Annex B).

Interestingly, in another test run with similar inputs but a more extensive context, ChatGPT succeeded in recognizing more elements and producing documentation with greater detail and accuracy, without losing precision, highlighting the value of iterative and expert use of AI and the importance of maintaining a good traceability system when designing legal ADs. Nonetheless, it must be emphasized that documentation with AI through unit-based review is an excellent approach, but it requires a well-established AD and a traceability system linking views to the legal clauses. Providing views without this system, unconsolidated, or not extremely legally aligned does not produce good results nor does it give a bad context or a lot of views.

After these results, some of the initial v1 views, along with the validated v2 views, were submitted to this process to refine elements whose documentation was missing or unclear, whether because the legal clause alone was insufficient, or it represented an extra element, ultimately resulting in the **v3** of the ADs.

9.2 AI-enhanced JArchi

JArchi is a JavaScript-based scripting extension for the Archi tool that enables the automation, extension, and customization of ArchiMate modelling tasks through JavaScript. To exploit its potential with AI, two complementary approaches, with their different scripts, were integrated. These scripts were tested to validate and improve the clarity and consistency of some views. The results of this process were analyzed, identifying where the AI scripts were most and least helpful.

First, the “archi-scripts” GitHub library³² was added to the Archi environment, providing an extensive set of pre-built scripts that significantly enhance the modelling and AI-assisted capabilities of JArchi. After cloning the repository, the scripting preferences of Archi tool were configured to point directly to the scripts folder and the scripts editor. This setup provided access to the collection of JArchi scripts in the library, which could be accessed on Archi through the “Scripts Manager” to run, delete, or edit them. When a script starts running, it opens the “Scripts Console”.

The key features of the library include model analysis and bulk operations, which allow users to perform tasks like running a model health check to get improvement recommendations or using smart style sync to update the appearance of elements based on their properties. It also offers powerful AI-assisted scripts that leverage AI for complex tasks. For example, a user can provide a textual description to generate an entire ArchiMate model or expand an existing one. Other AI features include creating documentation for ArchiMate views and elements model, layout views, and even generating a meta-model schema from an image. Nevertheless, the library provides many reporting and integration utilities, such as publishing views to Confluence, exporting diagrams to Draw.io, and a “Where Used” script to locate elements within the model. The scripts are highly customizable to fit specific modelling needs, making them valuable tools.

Second, based on the GitHub Gists gathered from the work entitled “AI support for Archi Models using Gemini and jArchi”³³, a more direct integration of AI was implemented using the Gemini API, from Google, with JArchi. This involved configuring: an API key from Gemini, library of jars, and the scripts together with the Gemini Java client libraries. The integration allows selected ArchiMate views or components to be automatically critiqued, analyzed, or document by Gemini. The AI feedback is then displayed in Archi and can be appended to, or replace, the documentation of the view, or even used to generate an “Assessment” element directly inside the model. This workflow provides enterprise architects with AI-generated insights that are contextually embedded in their ArchiMate diagrams.

However, both approaches come with limitations. The extensive functionality of the library can be overwhelming for new users, requiring a steep learning curve to understand all scripts and certain AI-assisted features may produce inconsistent results depending on: the complexity of the model, the clarity of the input description, the size and depth of views, overlapping or nested views, the number of elements displayed per view, or when views contain cross-cutting relationships that span multiple layers of the ArchiMate model. Likewise, the Gemini integration, while powerful for critique and documentation, relies on proper configuration of external dependencies (such as the Google GenAI Java SDK and Apache HttpClient), and AI outputs also require manual interpretation or adjustment. Moreover, automatically generating or updating views may lead to missing or misaligned elements, incorrect relationships, or loss of intended layout and readability.

Finally, since both solutions rely heavily on community contributions and external APIs, the level of ongoing support and compatibility can be inconsistent. Automated generation or updating of views may result in missing or misaligned elements, incorrect relationships, or unintended changes to the layout

³² ThomasRohde, 2023. “archi-scripts” GitHub repo. <https://github.com/ThomasRohde/archi-scripts#>

³³ Mileham, Steven. 2025. AI support for Archi Models using Gemini and jArchi. General Geekery blog, July 2, 2025. <https://smileham.co.uk/2025/07/02/ai-support-for-archi-models-using-gemini-and-jarchi/>

and readability. Some scripts may also require manual adjustments after execution, and integration with external tools can occasionally encounter compatibility issues.

To show some technical features explored within these approaches, the “EUDI Wallet-Relying Parties” view was again picked, this time its v2 (Figure 49), and submitted to some of the scripts. Even so, it is relevant to point out that these scripts were used only for investigation purposes and were not used to directly improve the ADs. Therefore, one script usage from each approach will be presented below.

9.2.1 Documentation Augmenter with LLaMA AI

Starting with the documentation part, a script from the “archi-scripts” library that enables the automatic generation of descriptive text for a selected view, was used. It’s called “Documentation Augmenter” and produces concise explanations for every element on a view.

The chosen AI model was LLaMA by Meta. This AI was required to generate extensive documentation according to the defined settings in Figure 51.

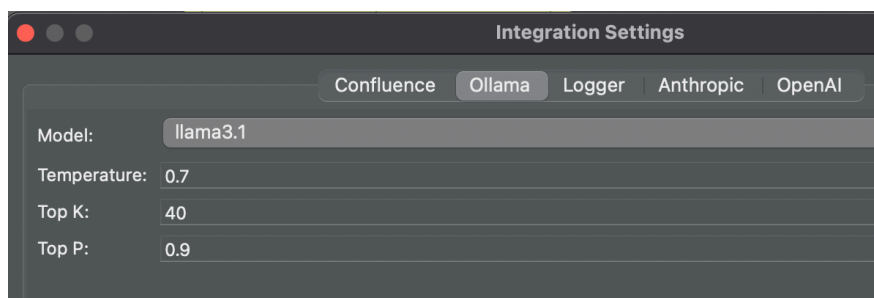


Figure 51: LLaMA AI integration settings within Archi tool

It is important to note that this AI was not provided with any legal context, it only considers the current view for its response. This makes the task more challenging, but it was intentional to evaluate how well the model could infer insights from the consolidated elements, given that the view is already well-established.

The script was then executed. While the generated output would require subsequent post-editing to clearly align with the legal provisions, filtering unnecessary information, and correcting misalignments or digressions, it effectively demonstrated its value as a supportive assistant rather than a substitute for the expertise of the architect. LLaMA can provide useful documentation information but still requires expert oversight.

For instance, an analysis made on the generated documentation for the “Relying Party registration” business process is presented in Figure 52.

AI response without any change	Analysis
# “Relying Party registration” (business-process) Relying Party registration is a business-process within the European Digital Identity Wallet (EUDI Wallet) architecture. ## Purpose The purpose of Relying Party registration is to facilitate secure and efficient identity verification processes for relying parties (RPs). ## Key Attributes Key attributes include RP’s unique identifier, contact information, and intended use of EUDI Wallet. ## Relationships Interpretation: The Relying Party registration process aggregates multiple activities, including User Interaction and Identification to the User. It also triggers the Report information changes process. ## Business Impact The business impact is a secure identity verification framework that ensures compliance with regulatory requirements (e.g., eIDAS2 Regulation). ## Technical Details Technical details involve a secure communication protocol between RP and EUDI Wallet Instance, ensuring confidentiality and integrity of user data. It also incorporates authentication mechanisms to prevent unauthorized access. ## Risks and Issues N/A ## Future Considerations Future considerations may include scalability and integration with existing systems.	## Purpose Legal alignment: The text says: “facilitate secure and efficient identity verification processes for relying parties.” Article 5b defines the purpose more narrowly. Relying parties must register in the Member State where they are established in order to use the EUDI Wallet. The legal focus is on registration as a prerequisite to use, not on identity verification efficiency. The AI generalized too much here, but it’s correct. ## Key Attributes Legal alignment: The documentation mentions unique identifier, contact information, and intended use. This aligns well with Article 5b(2)(a–c), which specifies the MS establishment, and RP’s name (and registration number if applicable), contact details, and intended use of EUDI Wallet (including data requested). ## Relationships Legal alignment: AI analyzed well the trigger of the “Report Information Changes” business process, and in some way, it is also related to user activities. Legal misalignment: Although the registration process in Article 5b (paragraphs 1–7) is separate from the user interaction process described in paragraphs 8–9: Registration (1–7): For the Member State to officially recognize the relying party and make their information publicly available; User interaction (8–9): When the relying party actually interacts with the wallet user. Given that the AI did not have access to the full legal text, this analysis is satisfactory. ## Business Impact Legal alignment: The text mentions regulatory compliance with eIDAS2. Legal misalignment: While generally correct, Article 5b emphasizes proportionality, cost-effectiveness, and legal safeguards. These aspects are missing, while the AI narrowed the focus only to “secure identity verification.” ## Technical Details Legal misalignment: Article 5b does not specify protocols at this stage. Instead, it emphasizes registration, disclosure, pseudonyms, and intermediaries. The AI added technical details not grounded in the regulation or diagram. ## Risks and Issues Legal misalignment: Left as “N/A.” Risks are central. Article 5b highlights: RPs must not request more data than declared (risk of data overreach), pseudonyms must not be refused if not legally required (risk of unlawful exclusion), and intermediaries must not store transaction content (risk of misuse). The AI missed all of these. ## Future Considerations To consider: The documentation mentions scalability and integration. These points are speculative and not part of Article 5b. The regulation focuses instead on obligations for continuous updating (Article 5b(6)) and Member State mechanisms (Article 5b(7)). The AI drifted into generic IT foresight.

Figure 52: Analysis of the generated documentation from LLaMA AI 3.1

9.2.2 Validation Critique with Gemini

Next, it was used one of the key AI-assisted features from the second approach, with the ability to generate a validation “critique” of a selected view. This process uses a script that exports the view to markdown and then uses the Gemini AI model to produce a few paragraphs highlighting areas for improvement, potential opportunities, and gaps. Once the critique is generated, the user is offered three options for integrating it back into the model: “Replace Documentation”, “Append Documentation”, or “Create Assessment”. The last option automatically creates an “Assessment” ArchiMate element

attached to the selected view, embedding AI insights directly within the architecture. The generated critique text is shown in Figure 53.

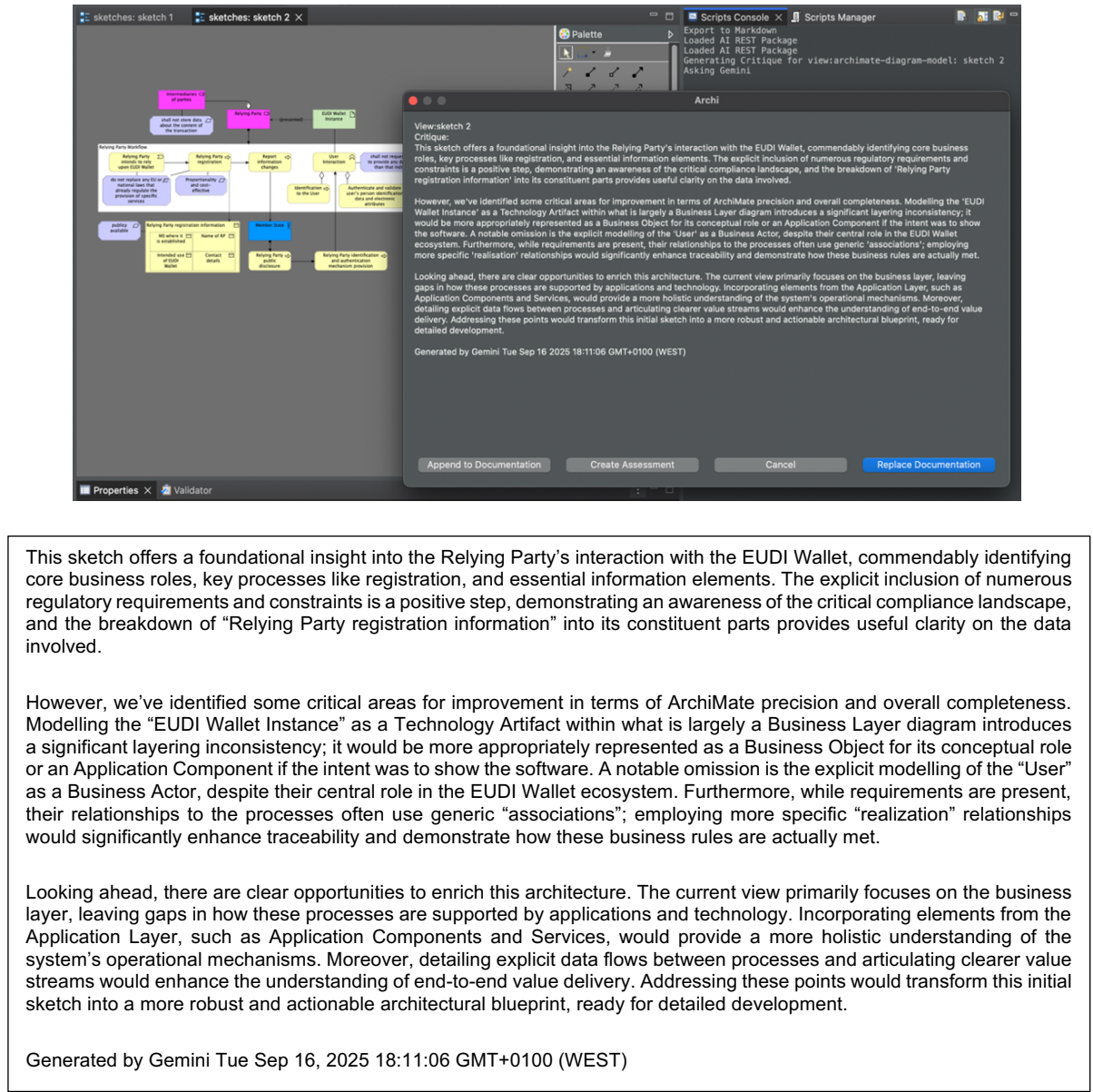


Figure 53: Gemini AI critique to the view of EUDI Wallet-Relying Parties (v2)

Again, no context was given to the AI, only the view. Regarding the critique, it is divided into three paragraphs. The first one provides a small overview, which was expected to be general.

The second paragraph highlights the aspects it didn't consider valid, suggesting more critical improvements, which are also relevant. For example, when it says "modelling EUDI Wallet Instance as a Technology Artifact within what is largely a Business Layer diagram introduces a significant layering inconsistency", that is indeed true in ArchiMate. With no context, this is a reasonable criticism from an architectural perspective at first glance. However, it does not apply here since it is a legal implementation, a piece of data that represents a specific operational deployment, belonging to and controlled by a User, rather than being an object. Nevertheless, it was initially represented as a Business Object, so it is indeed an excellent comment, given the absence of legal information.

Another case is when it points out that the “User”, an important role, is omitted. This is true and should be explicitly included if it was not implied by the legal context within the AD. It’s a decision of the architect. It also discusses the generic associations with the requirements, which is a good observation, though in some cases it may be debatable whether realization or association should be used. Realization is more appropriate when an element plays a critical role in the creation, achievement, sustenance, or operation of a requirement, whereas association could be more suitable in unspecified cases. In this view, however, realization is indeed the correct choice in some of the elements. This adjustment was applied to the v2.1 of this view (Figure 50), demonstrating that this method serves as a way to validate additional views from the ADs, potentially leading to a hypothetical version 4.

The third paragraph gives comments on possible improvements. These were not as strong, for example suggesting the inclusion of more layers, which in this case does not apply. It also recommends “detailing explicit data flows between processes and articulating clearer value streams to enhance the understanding of end-to-end value delivery”. Although this again overlooks the context, from a pure ArchiMate validation perspective, the overall critique provides valuable insights.

It is worth highlighting that there are several other libraries and JArchi validations available that do not rely on AI and are specifically designed to ensure model correctness, consistency, and compliance with ArchiMate standards. These non-AI scripts are often more stable and predictable for routine validations and model checking. However, these two approaches were chosen in this context to explore the potential of AI-assisted validation, offering an experimental approach that goes beyond traditional JArchi capabilities. This allowed testing limited-context AI validation (only the view) within ArchiMate, in contrast with unit-based review validation, and opened space for a deeper investigation into how AI can complement standard conceptual modelling practices, even if some features remain less reliable or require manual refinement, highlighting possibilities for the future.

10 Conclusion

Early in the research process, a significant gap was identified: the absence of structured conceptual models that fully structure eIDAS2 and NIS2. Building on it, a research question was internally formulated: “How can we simplify and group legal clauses in a way that keeps all important information while making an AD easy to understand?”.

Through an in-depth analysis of both legal frameworks and the ArchiMate language, it was possible to develop a method that allows for the traceability and visualization of legal clauses as architectural elements, simplifying the understanding of complex legal compliance requirements within EA conceptual modelling, filling the research gap.

This final chapter presents the outcomes and considerations of the work described in this dissertation. It also highlights potential supplementary work that could further enhance this study in the future.

10.1 Outcomes and Considerations

This work proved effective in systematically transforming the legal requirements of eIDAS2 and NIS2 into ArchiMate elements. As outcomes and considerations:

- **Use of a traceability matrix in Excel, combined with a solid numbering system, a standardized color scheme, and clear architecture principles:** Represents a structured and standardized methodology, and should be followed and improved when modelling other legislation. It is important to acknowledge, however, that this is extensive work and not automated. All ArchiMate views were implemented manually, with each element carefully translated from a legal clause. At present, there are no “easy” ways of achieving this, even with other EA languages. Such remains very challenging. Some parts of the documentation can be supported by scripts, but the modelling effort itself requires significant manual input.
- **ArchiMate ADs of NIS2 and eIDAS2:** Revealed the inherent complexity of modelling legal compliance, particularly the difficulty of reconciling perspectives for different possible stakeholders, such as enterprise architects and legal consultants. Nevertheless, the results are promising, as legislation is often ambiguous and difficult for organizations to interpret and comply with. Practices such as those followed in this research offer a structured way of reducing this ambiguity and creating a shared legal understanding of both NIS2 and eIDAS2. Moreover, being able to extract independent and documented views and compare them with each other and with the legal text covering each clause, directly on the architecture and following the same architecture principles, indicates that the major objective was achieved.
- **Inclusion of practical scenarios:** Demonstrated that the developed views can serve as a foundation for allowing organizations to adapt and extend representations of NIS2 and eIDAS2 to their specific needs. The views not only illustrate real entities but also simulate compliance situations with both legal frameworks, showing how these entities can align processes and

responsibilities with the legal expectations. Ultimately, it confirmed the applicability of ArchiMate in bridging legal models with real world practice.

- **Exploratory validation and documentation with AI:** Highlighted the potential for partial automation in compliance modelling, although this remains an area for future refinement, with some areas yet to be addressed. At present, AI tends to play a role in the consolidation phases of EA rather than in its implementation. However, it shows promise as a supportive tool for simplifying legal texts during the modelling process and warrants consideration for future use. Nevertheless, it cannot fully automate the task, as the involvement of an architect will always remain necessary.

Overall, the outcomes of this research provide valuable insights for entities seeking to understand and comply with eIDAS2 and NIS2 (from an EA perspective), demonstrate the practical application of ArchiMate in legal modelling, and open the door to future applications with AI.

10.2 Future work

Primarily, any improvements to this work - whether by enhancing the methodology, demonstration, or validation approaches and applying them to other legal frameworks, or by refining the current ADs - represent a consolidated step forward.

Looking ahead, the integration of AI-powered scripts offers a wide range of opportunities to advance architectural modelling. This evolution suggests a paradigm shift in which enterprise architecture modelers may increasingly rely on intelligent assistants to accelerate model creation, improve reporting, enhance analytical depth, and support systematic validation and documentation. Such an approach opens the way for richer and more adaptive modelling practices in both academic and professional contexts.

Further work should focus on evolving the “unit” size of the unit-based review approach presented in this study. This would enable more effective comparison of different views across the ADs and their escalation into broader legal contexts, allowing interdependencies between regulations to be better captured. Another key direction is the pursuit of complete legal validation, ensuring that the models not only represent compliance requirements accurately but also withstand rigorous legal interpretation and practical application.

New research could also explore alternative methods for improving traceability and the numbering system used in legislative modelling. For instance, experimenting with hierarchical or semantic numbering schemes, dynamic linking between clauses and architectural elements, or integrating more advanced cross-referencing techniques could enhance navigability and automation. Additionally, approaches that systematically compare and reconcile different views across multiple legislative instruments could provide richer insights into interdependencies and overlaps between legal clauses, supporting more robust compliance modelling. Such refinements would further strengthen the practical applicability and accuracy of ADs in legal and enterprise contexts.

Bibliography

- [1] Blanco-Lainé, Gaëlle, Jean-Sébastien Sottet, and Sophie Dupuy-Chessa. "Using an enterprise architecture model for GDPR compliance principles." IFIP Working Conference on The Practice of Enterprise Modeling. Cham: Springer International Publishing, 2019.
- [2] Organization for Economic Co-operation and Development. (2021). OECD Regulatory Policy Outlook 2021. OECD Publishing.
- [3] Singhal, A., & Breaux, T. (2025). Legal Requirements Translation from Law. *arXiv preprint arXiv:2507.02846*.
- [4] Schwalm, S. (2023). The possible impacts of the eIDAS 2.0 digital identity approach in Germany and Europe. In *Open Identity Summit 2023* (pp. 109-120). Gesellschaft für Informatik eV.
- [5] GSMA Europe. (2024). Architecture considerations for eIDAS 2.0 (Version 1.1). GSMA Europe.
- [6] European Parliament & Council of the European Union. (2022). Directive (EU) 2022/2555 (NIS 2 Directive). Official Journal of the European Union, L 333, 80–152. <https://eur-lex.europa.eu/eli/dir/2022/2555>
- [7] Foorthuis, R., van Steenberghe, M., Mushkudiani, N., Bruls, W., Brinkkemper, S., & Bos, R. (2020). On course, but not there yet: Enterprise architecture conformance and benefits in systems development. *arXiv preprint arXiv:2008.11026*.
- [8] European Commission. (2024). eArchiving Initiative: Version 2.0 of the reference architecture. <https://digital-strategy.ec.europa.eu/en/news/earchiving-initiative-announces-version-20-earchiving-reference-architecture>
- [9] Cruz, M., Guerreiro, S., Borbinha, J., & Proença, D. (2024). Enhancing the Assessment of Digital Archive Maturity using Enterprise Architecture. In *CEUR Workshop Proceedings* (Vol. 3855). CEUR-WS.
- [10] Krimmer, R., Prentza, A., & Mamrot, S. (2021). The Once-Only Principle: The TOOP Project (p. 241). Springer Nature.
- [11] Campmas, A., Iacob, N., & Simonelli, F. (2022). How can interoperability stimulate the use of digital public services? An analysis of national interoperability frameworks and e-Government in the European Union. *Data & Policy*, 4, e19.
- [12] Syrpis, P. (2015). The relationship between primary and secondary law in the EU. *Common Market Law Review*, 52(2).
- [13] Dutt, T., Holzinger, K., Malang, T., Schäubli, T., Schimmelfennig, F., & Winzen, T. (2017). Opting out from European Union legislation: the differentiation of secondary law. *Journal of European Public Policy*, 24(3), 406-428.
- [14] DeVries, W. T. (2003). Protecting privacy in the digital age. *Berkeley Tech. LJ*, 18, 283.
- [15] Beck, E. N. (2015). The invisible digital identity: Assemblages in digital networks. *Computers and Composition*, 35, 125-140.
- [16] Románský, R. (2017). A survey on digital world opportunities and challenges for user's privacy. *Proceedings of the 2017 International Conference on Information Technology and Applications*, 1–6. <https://doi.org/10.1109/ICITA.2017.00011>

- [17] Zaeem, R. N., & Barber, K. S. (2020). The effect of the GDPR on privacy policies: Recent progress and future promise. *ACM Transactions on Management Information Systems (TMIS)*, 12(1), 1-20.
- [18] Sharma, S. (2019). *Data privacy and GDPR handbook*. John Wiley & Sons.
- [19] European Parliament & Council of the European Union. (2016). Regulation (EU) 2016/679 (General Data Protection Regulation). *Official Journal of the European Union*, L 119, 1–88. <https://eur-lex.europa.eu/eli/reg/2016/679/oj>
- [20] Gobeo, A., Fowler, C., & Buchanan, W.J. (2018). *GDPR and Cyber Security for Business Information Systems* (1st ed.). River Publishers. <https://doi.org/10.1201/9781003338253>
- [21] Bay, M. (2016). What is cybersecurity. *French Journal for Media Research*, 6, 1-28.
- [22] Corallo, A., Lazoi, M., & Lezzi, M. (2020). Cybersecurity in the context of industry 4.0: A structured classification of critical assets and business impacts. *Computers in industry*, 114, 103165.
- [23] Christen, M., Gordijn, B., & Loi, M. (2020). *The ethics of cybersecurity* (p. 384). Springer Nature.
- [24] Sievers, T. (2021). Proposal for a NIS directive 2.0: companies covered by the extended scope of application and their obligations. *International Cybersecurity Law Review*, 2(2), 223-231. <https://doi.org/10.1365/s43439-021-00033-8>
- [25] Mulligan, D. K., & Schneider, F. B. (2011). Doctrine for cybersecurity. *Daedalus*, 140(4), 70-92.
- [26] Sawano, D., Johnsson, D. B., & Deogun, D. (2019). *Secure by Design*. Simon and Schuster.
- [27] Srinivas, J., Das, A. K., & Kumar, N. (2019). Government regulations in cyber security: Framework, standards and recommendations. *Future generation computer systems*, 92, 178-188.
- [28] Syafrizal, M., Selamat, S. R., & Zakaria, N. A. (2020). Analysis of cybersecurity standard and framework components. *International Journal of Communication Networks and Information Security*, 12(3), 417-432.
- [29] Dunn Cavelty, M., & Smeets, M. (2023). Regulatory cybersecurity governance in the making: The formation of ENISA and its struggle for epistemic authority. *Journal of European Public Policy*, 30(7), 1330-1352.
- [30] Shalabayev, I. (2022). Towards coherent cybersecurity in the EU? The impact of the NIS Directive implementation on the European cyberspace.
- [31] Drivas, G., Chatzopoulou, A., Maglaras, L., Lambrinoudakis, C., Cook, A., & Janicke, H. (2020, July). A nis directive compliant cybersecurity maturity assessment framework. In *2020 IEEE 44th Annual Computers, Software, and Applications Conference (COMPSAC)* (pp. 1641-1646). IEEE.
- [32] Calder, A., & Watkins, S. (2024). *IT governance: an international guide to data security and ISO 27001/ISO 27002*.
- [33] European Parliament & Council of the European Union. (2019). Regulation (EU) 2019/881 on the European Cybersecurity Competence Centre. *Official Journal of the European Union*, L 151, 15–30. <https://eur-lex.europa.eu/eli/reg/2019/881/oj/eng>

- [34] Shaffique, M. R. (2024). Cyber Resilience Act 2022: A silver bullet for cybersecurity of IoT devices or a shot in the dark?. *Computer Law & Security Review*, 54, 106009.
- [35] Plan, N. P. A. National Institute of Standards and Technology (NIST).
- [36] National Institute of Standards and Technology. (2024). The NIST Cybersecurity Framework (CSF) 2.0. <https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.29.pdf>
- [37] Ibrahim, A., Valli, C., McAteer, I., & Chaudhry, J. (2018). A security review of local government using NIST CSF: a case study. *The Journal of Supercomputing*, 74, 5171-5186.
- [38] Centro Nacional de Cibersegurança. Quadro Nacional de Referência para a Cibersegurança. <https://www.cncs.gov.pt/pt/quadro-nacional/>
- [39] Gabinete Nacional de Segurança. (2017). Decreto-Lei n.º 136/2017: Alteração à Lei Orgânica do Gabinete Nacional de Segurança. <https://www.cncs.gov.pt/docs/dl-136-2017-alt-lo-gns.pdf>
- [40] Ghadge, N. (2024). Challenges with securing digital identity. *International Journal on Cybernetics & Informatics (IJCI)*, 13(13), 1.
- [41] Abelson, H., Lessig, L., Covell, P., Gordon, S., Hochberger, A., & Kovacs, J. (1998). Digital identity in cyberspace. White Paper Submitted for 6.805/Law of Cyberspace: Social Protocols.
- [42] Sullivan, C. (2011). Digital identity: An emergent legal concept (p. 178). University of Adelaide Press.
- [43] Ruiu, P., Saiu, S., & Grosso, E. (2024). Digital Identity in the EU: Promoting eIDAS Solutions Based on Biometrics. *Future Internet*, 16(7), 228.
- [44] De Marco, G., Marino, F. A., & De Maria, A. (2024). Bridging eIDAS 2.0 Legal Requirements and Technical Solutions. *Regulation*, 2.
- [45] Van Kessel, R., Haig, M., & Mossialos, E. (2023). Strengthening cybersecurity for patient data protection in Europe. *Journal of Medical Internet Research*, 25, e48824.
- [46] Organization for Economic Co-operation and Development. (2024). G7 mapping exercise of digital identity approaches.
- [47] Robles-Carrillo, M. (2024). Digital identity: an approach to its nature, concept, and functionalities. *International Journal of Law and Information Technology*, 32(1), eaae019.
- [48] Tobin, A., & Reed, D. (2016). The inevitable rise of self-sovereign identity. *The Sovrin Foundation*, 29(2016), 18.
- [49] Takemiya, M., & Vanieiev, B. (2018). Sora identity: Secure, digital identity on the blockchain. In 2018 IEEE 42nd annual computer software and applications conference (compsac) (Vol. 2, pp. 582-587). IEEE.
- [50] Kimmo, M., Pappel, I., & Draheim, D. (2018, April). E-residency as a nation branding case. In *Proceedings of the 11th International Conference on Theory and Practice of Electronic Governance* (pp. 419-428).
- [51] Gopichandran, V., Ganeshkumar, P., Dash, S., & Ramasamy, A. (2019). Ethical challenges of digital health technologies: Aadhaar, India. *Bulletin of the World Health Organization*, 98(4), 277.
- [52] Dahl, A., Mattila, R., Olkkonen, L., Saarinen, H., Sandell, T., & Törnroos, T. (2021). Baseline study of cross-border data exchange in the Nordic and Baltic countries. *Nordic Council of Ministers*.

- [53] Norwegian Digitalisation Agency. The Nordic-Baltic eID Project (NOBID).
- [54] Musoni, M., Domingo, E., & Ogah, E. (2023). Digital ID systems in Africa: Challenges, risks and opportunities. European Centre for Development Policy Management.
- [55] Foster, C. (2024). Intellectual property rights and control in the digital economy: Examining the expansion of M-Pesa. *The Information Society*, 40(1), 1-17.
- [56] World Bank Group & Global Partnership for Financial Inclusion. (2018). G20 digital identity onboarding.
- [57] Anand, N., & Brass, I. (2021). Responsible innovation for digital identity systems. *Data & Policy*, 3, e35.
- [58] Organization for Economic Co-operation and Development. (2024). G7 mapping exercise of digital identity approaches.
- [59] Organisation for Economic Co-operation and Development. (2023). Recommendation of the Council on the Governance of Digital Identity.
- [60] Organisation for Economic Co-operation and Development. (2023). Draft Recommendation on the Governance of Digital Identity.
- [61] Romanyuk, A., Otchik, Y., Sukhaver, E., & Lukashevich, K. (2021). The Importance of Standardization.
- [62] Dumitru, F. (2008). THE ORGANISMS OF EUROPEAN STANDARDIZING. *Annals of the University of Oradea, Economic Science Series*, 17(1).
- [63] Calder, A., & Watkins, S. (2024). IT governance: an international guide to data security and ISO 27001/ISO 27002
- [64] Bago, E. (2020). Review–ISO/IEC 24760-3: 2016. IDPro Body of Knowledge, 1(2).
- [65] Kanevskaia, O., Delimatsis, P., & Bijlmakers, S. (2024). The European Telecommunication Standards Institute (ETSI): in search for legitimacy and resilience of European standardisation. *Innovation: The European Journal of Social Science Research*, 1-20.
- [66] Rannenbergh, K. A framework for identity management (ISO/IEC 24760).
- [67] European Telecommunications Standards Institute. (2023). ETSI EN 303 645: Cyber Security for Consumer Internet of Things: Baseline Requirements (V2.1.1).
- [68] Oliveira, M. R. S. D. (2024). eIDAS Qualified services: serviço de validação (Doctoral dissertation).
- [69] Nelson, H. J., Poels, G., Genero, M., & Piattini, M. (2005). Quality in conceptual modelling: five examples of the state of the art. *Data & Knowledge Engineering*, 55(3), 237-242.
- [70] International Organization for Standardization. (2022). ISO/IEC/IEEE 42010:2022. <http://www.iso-architecture.org/ieee-1471/index.html>
- [71] Architects” Council of Europe. (2024). Architecture policies Europe. <https://ace-cae.eu/publication/architecture-policies-europe/>
- [72] Perez-Castillo, R., Ruiz, F., Piattini, M., & Ebert, C. (2019). Enterprise architecture. *IEEE software*, 36(4), 12-19.

- [73] Ellerm, A., & Morales-Trujillo, M. E. (2020). Modelling security aspects with archimate: a systematic mapping study. In 2020 46th Euromicro Conference on Software Engineering and Advanced Applications (SEAA) (pp. 577-584). IEEE.
- [74] Kitsios, F., Kyriakopoulou, M., & Kamariotou, M. (2022). Exploring business strategy modelling with ArchiMate: A case study approach. *Information*, 13(1), 31.
- [75] Storey, V. C., Lukyanenko, R., & Castellanos, A. (2023). Conceptual modelling: topics, themes, and technology trends. *ACM Computing Surveys*, 55(14s), 1-38.
- [76] The Open Group. (2021). ArchiMate® 3.2 specification. <https://pubs.opengroup.org/architecture/archimate32-doc.singlepage/>
- [77] Chaves Costa, J. E., & Soares, M. D. S. (2023). A Review on the Capacities of the ISO/IEC/IEEE 42020: 2019 Standard for Architecture Elaboration of Software and Systems. In *Proceedings of the XIX Brazilian Symposium on Information Systems* (pp. 412-418).
- [78] Santos, V., & Soares, M. S. (2023). Design and Evaluation using Technology Acceptance Model of an Architecture Conceptualization Framework System based on the ISO/IEC/IEEE 42020. *Journal of Universal Computer Science*, 29(12), 1510.
- [79] Emery, D., & Hilliard, R. (2009). Every architecture description needs a framework: Expressing architecture frameworks using ISO/IEC 42010. In 2009 Joint Working IEEE/IFIP Conference on Software Architecture & European Conference on Software Architecture (pp. 31-40). IEEE.
- [80] Institute of Electrical and Electronics Engineers. (2022). ISO/IEC/IEEE 42010:2022: Systems and software engineering - Architecture description.
- [81] Bork, D., Ali, S. J., & Roelens, B. (2023). Conceptual modelling and artificial intelligence: A systematic mapping study. *arXiv preprint arXiv:2303.06758*.
- [82] Fill, H. G., Fettke, P., & Köpke, J. (2023). Conceptual modelling and large language models: impressions from first experiments with ChatGPT. *Enterprise Modelling and Information Systems Architectures (EMISAJ)*, 18, 1-15.
- [83] Antunes, J., Barata, J., da Cunha, P. R., Estima, J., & Tavares, J. (2024, May). A reference architecture for dry port digital twins: preliminary assessment using ArchiMate. In *International Conference on Research Challenges in Information Science* (pp. 131-145). Cham: Springer Nature Switzerland.
- [84] Takeuchi, H. (2022). Enterprise Architecture-Based Project Model for AI Service System Development. In *Handbook on Artificial Intelligence-Empowered Applied Software Engineering: VOL. 1: Novel Methodologies to Engineering Smart Software Systems* (pp. 101-121). Cham: Springer International Publishing.
- [85] Rozehnal, P., & Novák, V. (2018). The Core of Enterprise Architecture as a Management Tool: GDPR Implementation Case Study. *26th Interdisciplinary Information Management Talks*, 359-366.
- [86] Hoogenboom, C. (2019). An enterprise architecture approach to implementing the NIST Cyber Security Framework (Master's thesis, Leiden University)

- [87] Strîmbei, C. (2023). EXPLORING EUROPEAN INTEROPERABILITY REFERENCE ARCHITECTURE (EIRA). In European Financial Resilience and Regulation. Proceedings of the International Conference EUFIRE 2023 (pp. 316-326).
- [88] European Commission. (2019). European Interoperability Reference Architecture (EIRA) v3.0.0 Overview. Interoperable Europe.
- [89] Pöhn, D., & Hommel, W. (2022). Reference Service Model Framework for Identity Management. IEEE Access, 10, 120984-121009.
- [90] Häberle, S., Lenk, J., & Hwang, J. (2022). Reference service model for federated identity management. ResearchGate.
- [91] European Commission. EU digital identity wallet. GitHub. <https://eu-digital-identity-wallet.github.io/eudi-doc-architecture-and-reference-framework/1.1.0/arf/>
- [92] Terry, M., & Bagnasco, A. (2019). Electronic identification for universities: Building cross-border services based on the eIDAS infrastructure. ResearchGate.
- [93] Bella, G., Castiglione, G., & Santamaria, D. F. (2023). An Ontological Approach to Compliance Verification of the NIS 2 Directive. In JOWO.
- [94] Seeba, M., Valgre, M., & Matulevičius, R. (2025). Evaluating organization security: User stories of European Union NIS2 Directive (Section 4). arXiv preprint arXiv:2504.19222v1.
- [95] National Institute of Standards and Technology. (2020). Digital identity guidelines: NIST special publication 800-63-4. <https://pages.nist.gov/800-63-4/sp800-63/model/>
- [96] Markopoulou, D., Papakonstantinou, V., & De Hert, P. (2019). The new EU cybersecurity framework: The NIS Directive, ENISA's role and the General Data Protection Regulation. Computer Law & Security Review, 35(6), 105336.
- [97] Ruohonen, J. (2024). A Systematic Literature Review on the NIS2 Directive. arXiv preprint arXiv:2412.08084.
- [98] Galvão, R. J. V. (2024). NIS2 Directive Analysis and Preparation for its Implementation (Master's thesis). Repositório científico da Universidade de Coimbra.
- [99] Lemnitzer, J. M., & Prockl, G. (2023). The NIS 2 Directive—Where Cyber Risk meets Supply Chain Management. Effektivitet, (3), 26-27.
- [100] Van Welie, A. (2024). Legislation within cybersecurity: Preparing for NIS2 - A detailed framework in the healthcare sector in the Netherlands (Master's thesis). University of Turku.
- [101] Eckhardt, P., & Kotovskaia, A. (2023). The EU's cybersecurity framework: the interplay between the Cyber Resilience Act and the NIS 2 Directive. International Cybersecurity Law Review, 4(2), 147-164.
- [102] Ferrari, L. B. (2023). The relationship between the mandatory reporting obligations of NIS2 and GDPR in light of premature public disclosure.
- [103] Vandezande, N. (2024). Cybersecurity in the EU: How the NIS2-directive stacks up against its predecessor. Computer Law & Security Review, 52, 105890.
- [104] Contreras, P. (2023). The Transnational Dimension of Cybersecurity: The NIS Directive and Its Jurisdictional Challenges. In Proceedings of the International Conference on Cybersecurity,

- Situational Awareness and Social Media: Cyber Science 2022; 20–21 June; Wales (pp. 327-341). Singapore: Springer Nature Singapore.
- [105] Božić, V. (2023). Strengthening cybersecurity resilience: Collaborative efforts in the NIS2 framework. *International Journal of Cybersecurity*, 1(1), 1-10. <https://doi.org/10.1000/journal.2023.001>
- [106] Kadioglu Kumtepe, C., & Gonzales, W. (2023). The NIS 2 Directive on Cybersecurity and International Arbitration. *LexisNexis Uk Blog: Research and legal analysis*, 17.
- [107] Falch, M., Henten, A. H., Kaloudis, A., & Windekjælde, I. (2024). Economics and regulation of cybersecurity-the case of EU.
- [108] Boban, M. (2024). CYBERSECURITY IN THE DIGITAL AGE: REGULATORY FRAMEWORK BASED ON THE IMPLEMENTATION OF THE NIS2 DIRECTIVE. In *Economic and Social Development (Book of Proceedings)*, 112th International Scientific Conference on Economic and Social Development (p. 592).
- [109] Sharif, A., Ranzi, M., Carbone, R., Sciarretta, G., Marino, F. A., & Ranise, S. (2022). The eIDAS regulation: a survey of technological trends for European electronic identity schemes. *Applied Sciences*, 12(24), 12679.
- [110] Kurg, R. EU Cross-border Service Delivery Analysis in the Context of eIDAS 2.0.
- [111] European Parliament & Council of the European Union. (2024). Regulation (EU) 2024/1183 of the European Parliament and of the Council of 30 April 2024 on the European Digital Identity Framework. *Official Journal of the European Union*, L 333, 1–80. <https://eur-lex.europa.eu/eli/reg/2024/1183>
- [112] Roelofs, F., Verheul, E., & Jacobs, B. (2019). Analysis and comparison of identification and authentication systems under the eIDAS regulation. European Commission: Brussels, Belgium.
- [113] Hühnlein, D. (2014). Towards eIDAS as a Service. In *ISSE 2014 Securing Electronic Business Processes: Highlights of the Information Security Solutions Europe 2014 Conference* (pp. 241-248). Wiesbaden: Springer Fachmedien Wiesbaden.
- [114] Hölbl, M., Kežmah, B., & Kompara, M. (2023). eIDAS Interoperability and Cross-Border Compliance Issues. *Mathematics*, 11(2), 430.
- [115] Krimmer, R., Dedovic, S., Schmidt, C., & Corici, A. A. (2021). Developing cross-border e-Governance: Exploring interoperability and cross-border integration. In *International Conference on Electronic Participation* (pp. 107-124). Cham: Springer International Publishing.
- [116] Andraško, J. (2017). Mutual recognition of electronic identification means under the eIDAS Regulation and its application issues. *AD ALTA: Journal of Interdisciplinary Research*, 7(2), 1-6.
- [117] Alamillo, I., Schwalm, S., Stoecker, C., & Thiermann, R. (2024). Qualified Ledgers—Breakthrough for proven security and legal trust in DLT through eIDAS2 Regulation?. In *Open Identity Summit 2024* (pp. 195-200). Gesellschaft für Informatik eV.
- [118] Marti Llull, X. (2024). Qualified electronic signatures for european digital identity wallets transition (Bachelor's thesis, Universitat Politècnica de Catalunya).

- [119] Kutylowski, M., & Błażkiewicz, P. (2023). Advanced electronic signatures and eIDAS—analysis of the concept. *Computer Standards & Interfaces*, 83, 103644.
- [120] Sjöholm, M. (2023). Designing a trustworthy EU digital identity wallet: A study of user needs and preferences (Independent thesis, Uppsala University). DiVA portal.
- [121] Balani, H. (2024). Securing the Future of the EU Digital Identity Wallet: Why we need Corporate Digital Identity standards now. Available at SSRN 4910859.
- [122] Strîmbei, C. (2023). EXPLORING EUROPEAN INTEROPERABILITY REFERENCE ARCHITECTURE (EIRA). In *European Financial Resilience and Regulation. Proceedings of the International Conference EUFIRE 2023* (pp. 316-326). Editura Universităţii» Alexandru Ioan Cuza «din Iaşi.

Annexes

A. Glossary

European Union	"Group of 27 countries in Europe. They agreed to work together and help each other.", (European Union)
Member States	"A candidate country that meets the accession criteria and signs the accession Treaty with the individual EU Member States becomes a Member State of the European Union.", (EU Glossary of summaries)
Legislation	"EU law.", (European Commission)
Regulation	"Binding legislative act. It must be applied in its entirety across the EU.", (European Union)
Directive	"Legislative act that sets out a goal that EU countries must achieve.", (European Union)
Legal Compliance	"Property of (national/local) legislation to be coherent or to implement European legislation.", (Interoperable Europe Portal)
Interoperability	"The ability of systems to exchange and make use of information.", (Interoperable Europe Portal)
Cross-border	"between different countries, or involving people from different countries", (Cambridge Dictionary)
GDPR	"EU regulation 2016/679 that governs how the personal data of individuals in the EU may be processed and transferred.", (GDPR)
NIS2 Directive	"EU directive 2022/2555 on network and information systems. It provides legal measures to boost the overall level of cybersecurity in the EU.", (European Commission)
Network and information system	"(a) an electronic communications network as defined in Article 2, point (1), of Directive (EU) 2018/1972; (b) any device or group of interconnected or related devices, one or more of which, pursuant to a programme, carry out automatic processing of digital data; or (c) digital data stored, processed, retrieved or transmitted by elements covered under points (a) and (b) for the purposes of their operation, use, protection and maintenance.", (EUR-lex, 2022)
Security of network and information systems	"Ability of network and information systems to resist, at a given level of confidence, any event that may compromise the availability, authenticity, integrity or confidentiality of stored, transmitted or processed data or of the services offered by, or accessible via, those network and information systems.", (EUR-lex, 2022)
Cybersecurity	"Practice of defending computers, servers, mobile devices, electronic systems, networks, and data from malicious attacks.", (EU4Digital Glossary)
National Cybersecurity Strategy	"A coherent framework of a Member State providing strategic objectives and priorities in the area of cybersecurity and the governance to achieve them in that Member State.", (EUR-lex, 2022)
Cybersecurity Measure	"Achieve a high common level of cybersecurity across the EU, with a view to improving the functioning of the internal market." (EUR-lex, 2022)
Mutual assistance	"Where an entity provides services in more than one Member State or provides services in one or more Member States and its network and information systems are located in one or more other Member States, the competent authorities of the Member States concerned shall cooperate with and assist each other as necessary.",
Cyber threat	"Based on its technical characteristics, can be assumed to have the potential to have a severe impact on the network and information systems of an entity or the users of the entity's services by causing considerable material or non-material damage.", (EUR-lex, 2022)
Vulnerability	"Weakness, susceptibility or flaw of ICT products or ICT services that can be exploited by a cyber threat.", (EUR-lex, 2022)
Incident	"An event compromising the availability, authenticity, integrity or confidentiality of stored, transmitted or processed data or of the services offered by, or accessible via, network and information systems.", (EUR-lex, 2022)

Incident handling	"Any actions and procedures aiming to prevent, detect, analyse, and contain or to respond to and recover from an incident.", (EUR-lex, 2022)
Risk	"Potential for loss or disruption caused by an incident and is to be expressed as a combination of the magnitude of such loss or disruption and the likelihood of occurrence of the incident.", (EUR-lex, 2022)
ENISA	"EU agency dedicated to achieving a high common level of cybersecurity across Europe.", (ENISA)
CSIRTs	"Expert groups that handle computer security incidents.", (EU4Digital Glossary)
CSIRTs Network	"The CSIRTs network shall be composed of representatives of the CSIRTs designated or established pursuant to Article 10 and the computer emergency response team for the Union's institutions, bodies and agencies (CERT-EU).", (EUR-lex, 2022)
CERTs	"CERTs are expert groups that handle computer security incidents. Alternatively known as Computer Security Incidents Response Teams (CSIRTs).", (EU4Digital Glossary)
EU-CyCLONe	"Cooperation network for Member States national authorities in charge of cyber crisis management.", (ENISA. EU CyCLONe)
Cooperation Group	"In order to support and facilitate strategic cooperation and the exchange of information among Member States, as well as to strengthen trust and confidence, a Cooperation Group is established. The Cooperation Group shall be composed of representatives of Member States, the Commission and ENISA.", (EUR-lex, 2022)
Essential Entity	Essential entities are organizations that provide services of high criticality and whose disruption would have significant impact on the functioning of society or the economy (EUR-lex, 2022).
Important Entity	Important entities are organizations that are also significant for the economy or society but have a lower criticality than essential entities (EUR-lex, 2022).
Competent Authority	"The competent authorities shall monitor the implementation of NIS2 at national level.", (EUR-lex, 2022).
Single Point of Contact	"Each Member State shall designate or establish a single point of contact. Where a Member State designates or establishes only one competent authority, that competent authority shall also be the single point of contact for that Member State. Each single point of contact shall exercise a liaison function to ensure cross-border cooperation of its Member State's authorities with the relevant authorities of other Member States, and, where appropriate, with the Commission and ENISA, as well as to ensure cross-sectoral cooperation with other competent authorities within its Member State.", (EUR-lex, 2022).
eIDAS2 Regulation	"EU regulation 2024/1183 on electronic identification, authentication, and trust services for electronic transactions in the internal market, ensuring the cross-border mutual recognition of electronic identification.", (EU4Digital Glossary)
Digital Identity	"Unique characteristics representing an individual or entity online. It encompasses the distinct traits and attributes that define an individual or entity in the online environment.", [40]
Digital Identity Services	"Services that enable the management of a person's identity using digital means. Digital identity services are designed to serve as the primary source of information to identify a citizen. Digital identity services include identity creation, identity update, and identity preservation services.", (EU4Digital Glossary)
Person Identification Data	"Set of data that is issued in accordance with Union or national law and that enables the establishment of the identity of a natural or legal person, or of a natural person representing another natural person or a legal person.", (EUR-lex, 2024)
Electronic Attestation of Attributes (EAAs)	"An attestation in electronic form that allows the authentication of attributes.", [91]
Attestation	"A signed set of attributes, in either the mdoc format specified in [ISO18013-5] or the SD-JWT format specified in [SD-JWT]. This may be a PID, QEAA or EAA.", [91]
Issuer	"A Person Identification Data Provider issuing PID or a (Qualified) Trust Service Provider issuing (Q)EAA. In the case of the EUDI Wallet there may be multiple Issuers for PID and (Q)EAA.", [91]
Electronic Identification	"Process of using person identification data in electronic form uniquely representing either a natural or legal person, or a natural person representing another natural person or a legal person.", (EUR-lex, 2024)

Electronic Identification Means	"Material and/or immaterial unit containing person identification data and which is used for authentication for an online service or, where appropriate, for an offline service", (EUR-lex, 2024)
Electronic Identification Scheme	"means a system for electronic identification under which electronic identification means are issued to natural or legal persons or natural persons representing other natural persons or legal persons;", (EUR-lex, 2024)
EUDI Wallet	"Electronic identification means which allows the user to securely store, manage and validate person identification data and electronic attestations of attributes for the purpose of providing them to relying parties and other users of European Digital Identity Wallets, and to sign by means of qualified electronic signatures or to seal by means of qualified electronic seals", (EUR-lex, 2024)
EUDI Wallet Instance	"Instance of an EUDI Wallet Solution belonging to and which is controlled by a User.", [91] "A software-based electronic identification means issued under an electronic identification scheme and recognised pursuant to Article 6a of Regulation (EU) No 910/2014 [eIDAS], enabling a natural or legal person to store, manage and share identification data and electronic attestations of attributes.", (EUR-Lex, 2024)
EUDI Wallet Solution	"An EUDI Wallet Solution is the entire product and service owned by an EUDI Wallet Provider, offered to all Users of that solution. An EUDI Wallet solution can be certified as being EUDI-compliant by a CAB.", [91]
Electronic Transaction	"Interactions conducted electronically, typically involving data exchanges, that require authentication, trust services or digital signatures.", (EUR-lex, 2014)
Electronic Signature	"Data in electronic form which is attached to or logically associated with other data in electronic form and which is used by the signatory to sign.", (EUR-lex, 2014)
Trust Service	"Electronic service normally provided for remuneration.", (EUR-lex, 2024) "An electronic service normally provided against payment which consists of: (a) the creation, verification, and validation of electronic signatures, electronic seals or electronic time stamps, electronic registered delivery services, electronic attestation of attributes and certificates related to those services; (b) the creation, verification and validation of certificates for website authentication; (c) the preservation of electronic signatures, seals or certificates related to those services; (d) the electronic archiving of electronic documents; (e) the management of remote electronic signature and seal creation devices; (f) the recording of electronic data into an electronic ledger.", [91]
Trust Service Provider	"Natural or a legal person who provides one or more trust services either as a qualified or as a non-qualified trust service provider.", (EUR-lex, 2014)
Trusted List	"List of recognised trust service providers", (EUR-lex, 2024)
Mutual Recognition	"The electronic identification means issued in another MS shall be recognized in the first MS for the purposes of cross-border authentication for that service online.", (EUR-lex, 2014)
Relying Party	"Natural or legal person that relies upon electronic identification, European Digital Identity Wallets or other electronic identification means, or upon a trust service.", (EUR-lex, 2024)
Conformity Assessment Body	"A body as defined in Article 2, point 13, of Regulation (EC) No 765/2008, which is accredited in accordance with that Regulation as competent to carry out conformity assessment of a qualified trust service provider and the qualified trust services it provides, or as competent to carry out certification of European Digital Identity Wallets or electronic identification means.", (EUR-lex, 2024)
Identity and Access Management	"Service that manages users and their access to resources. Often, these services are used to implement a single sign-on.", (European Alternatives)
Enterprise Architecture	"Allows organizations to align their business goals and processes to information technology.", [97]
Reference Architecture	"Ensembles the components, principles, and guidelines that define an architectural solution (an EA solution).", [122]
EUDI Wallet ARF	"A set of common standards and technical specifications and a set of common guidelines and best practices. (...) The purpose of the document is to provide all the specifications needed to develop an interoperable EUDI Wallet Solution based on common standards and practices.", [91]
TOGAF	"Framework for EA.", (TOGAF Standard)

EIRA	"Represents a metamodel defining the architectural building blocks to build interoperable eGovernment systems.", [87]
Conceptual Modelling	"Important part of information systems development and use that involves identifying and representing relevant aspects of reality", [75]
ArchiMate	"Provides instruments to enable enterprise architects to describe, analyze, and visualize the relationships among business domains in an unambiguous way." Emphasizes "modelling with concepts.", (ArchiMate Specification) "As an Architecture Description Language (ADL), ArchiMate is any form of expression for use in architecture descriptions", [70]
ISO/IEC/IEEE 42010	"Specifies a set of processes for architecting which can be employed in support of creating one or more ADs.", (ISO/IEC/IEEE 42010:2022)
Architecture Description	"Work product used to express an architecture. It identifies the system of interest whose architecture is being expressed.", [70]
Entity of Interest	"System of Interest.", [70]
Stakeholder	"Individuals, groups or organizations holding concerns for the system of interest.", [70]
Concern	"Any interest in the system.", [70]
View	From ISO/IEC/IEEE 42010 perspective: "Expresses the Architecture of the System of Interest from the perspective of one or more Stakeholders to address specific Concerns, using the conventions established by its viewpoint.", [70] From ArchiMate perspective: "A representation of a system from the perspective of a related set of concerns.", [76]
Viewpoint	From ISO/IEC/IEEE 42010 perspective: "Set of conventions for constructing, interpreting, using and analyzing one type of architecture view.", [70] From ArchiMate perspective: "A specification of the conventions for a particular kind of architecture view.", [76]
Layer	"An abstraction of the ArchiMate framework at which an enterprise can be modeled.", [76]
Aspect	"Classification of elements based on layer-independent characteristics related to the concerns of different stakeholders.", [76]
Artificial Intelligence	"Artificial intelligence is technology that enables computers and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy." (IBM)
Machine Learning	"Machine learning is a branch of artificial intelligence (AI) focused on enabling computers and machines to imitate the way that humans learn, to perform tasks autonomously, and to improve their performance and accuracy through experience and exposure to more data." (IBM)
Large Language Model (LLM)	"Large language models are a category of foundation models trained on immense amounts of data making them capable of understanding and generating natural language and other types of content to perform a wide range of tasks." (IBM)

B. Unit Review from ChatGPT 5

1. Overview of the View

The ArchiMate view models the workflow of *Relying Parties* that intend to use the European Digital Identity (EUDI) Wallet in line with **Article 5b of eIDAS2**. It depicts their lifecycle from registration in the Member State to public disclosure, user identification, authentication, and interaction rules. Constraints from the legislation are modeled as requirements/constraints (e.g., “shall not request users to provide any data other than indicated”). The diagram also highlights the responsibilities of relying parties, member states, and intermediaries.

2. Traceability Table

Legal Clause ID	Legal Text (shortened)	AD Element(s)
5b(1)	RP intends to rely → must register in MS	Relying Party intends to rely upon EUDI Wallet
5b(2)	Registration process proportionate/cost-effective	Relying Party registration
5b(2)(a)(i)	MS in which RP is established	MS where it is established (attribute)
5b(2)(a)(ii)	Name & registration details of RP	Name of RP (attribute)
5b(2)(b)	Contact details of RP	Contact details (attribute)
5b(2)(c)	Intended use of Wallet incl. data requested	Intended use of EUDI Wallet (attribute)
5b(3)	RP shall not request data other than in 2(c)	shall not request users... (requirement)
5b(4)	Without prejudice to EU/national law	(Not explicitly represented in view)
5b(5)	MS shall make info publicly available online	publicly available (requirement)
5b(6)	RP must inform MS of changes	Report information changes
5b(7)	MS shall provide identification/authentication mechanism	Relying Party identification and authentication mechanism provision
5b(8)	RP must identify itself to the user	Identification to the User
5b(9)	RP validates data and shall not refuse pseudonyms	Authenticate and validate user’s person identification data...; must not refuse pseudonyms...
5b(10)	Intermediaries deemed RPs, shall not store data	Note
5b(11)	Commission establishes technical specifications	(Not explicitly represented in view)

(cont.)

3. Validity Assessment

The view **aligns well with Article 5b** of eIDAS2, as it captures all main requirements (registration, disclosure, user identification, authentication, pseudonyms, intermediaries). The constraints are explicitly modeled as requirements, and the workflow represents the legal sequence.

However:

- **Art. 5b(2) proportionality principle** (registration must be cost-effective and proportionate-to-risk) is not modeled.
- **Art. 5b(4)** (without prejudice to Union/national law) is missing.
- **Art. 5b(11)** (Commission implementing acts) is not represented.
- Some aspects of Member States' duties (common mechanism for authentication, electronically signed public info) are only partially captured.

So, the model is faithful but **not exhaustive** — it covers operational obligations but omits meta-legal provisions and implementation timelines.

4. ArchiMate-Specific Comments

Strengths:

- Good use of **Business Roles/Actors** (Relying Party, Member State, User).
- **Requirements and constraints** are modeled with the correct ArchiMate notation (purple requirements).
- Workflow is logically structured with proper **business processes/functions**.
- Separation of *registration information* into attributes is well-structured and traceable to the legislation.

Improvements:

- The "**proportionality**" principle (Art. 5b(2)) could be modeled as a **requirement** linked to *Relying Party registration*.
- **Art. 5b(4)** "**without prejudice**" could be modeled as a **constraint** influencing all RP processes.
- **Art. 5b(11)** could be added as a **Business Process** or **Outcome** assigned to *Commission*, showing the implementing acts.
- "Publicly available online in electronically signed/sealed form" (Art. 5b(5)) should be clarified by an additional **business object (signed/sealed data)**.
- Intermediaries could be represented as a distinct **Business Role** realizing the "Relying Party" role instead of just a note

(documentation task)

Actors / Roles

- **Relying Party (Role):** Entity intending to rely on the EUDI Wallet. Central role linked to registration, user interaction, and authentication.
- **Member State (Actor):** Responsible for registering RPs, publishing data, and providing authentication mechanisms.
- **Intermediaries (Role):** Entities acting on behalf of RPs; legally treated as RPs and subject to same obligations.

Business Processes / Functions

- **Relying Party intends to rely upon EUDI Wallet:** Trigger for registration obligation (5b(1)).
- **Relying Party registration:** Core process of registering with the MS (5b(1)).
- **Report information changes:** Obligation to update information (5b(6)).
- **Relying Party public disclosure:** MS publishes RP registration information (5b(5)).
- **Relying Party identification and authentication mechanism provision:** MS ensures authentication mechanisms (5b(7)).
- **Identification to the User:** RPs must identify themselves when interacting (5b(8)).
- **User Interaction:** Generic process for wallet uses between RP and user.
- **Authenticate and validate user's person identification data and electronic attributes:** RP responsibility (5b(9)).

Business Objects

- **Relying Party Registration Information:** Aggregates required attributes.
 - **MS where it is established** (5b(2)(a)(i))
 - **Name of RP** (5b(2)(a)(ii))
 - **Contact details** (5b(2)(b))
 - **Intended use of EUDI Wallet** (5b(2)(c))

Requirements / Constraints

- **Proportionality and cost-effective** (5b(2))
- **Shall not request users to provide any data other than indicated** (5b(3))
- **Do not replace any EU or national laws** (5b(4))
- **Publicly available** (5b(5), partial)
- **Must not refuse pseudonyms if full identification is not legally required** (5b(9))
- **Shall not store data about the content of the transaction** (5b(10))